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(54) **ENZYMATIC METHOD FOR THE
PRODUCTION OF L-GLUFOSINATE
P-ESTERS**

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C12N 15/52 (2006.01)

C12N 15/70 (2006.01)

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(58) **Field of Classification Search**

None

See application file for complete search history.

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ABSTRACT

A first enzymatic method for the production of an L-glufosinate P-ester includes reacting an L-glufosinate P-ester carbamoylate to give the corresponding L-glufosinate P-ester in the presence of a carbamoylase as a catalyzer. A second enzymatic method includes the enantioselective production of an L-glufosinate P-ester from a mixture $M_{III,A}$ of L- and D-glufosinate P-ester hydantoins. The L-glufosinate P-ester obtained in the two methods may be saponified to give L-glufosinate.

13 Claims, 2 Drawing Sheets

Specification includes a Sequence Listing.

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FIG. 1

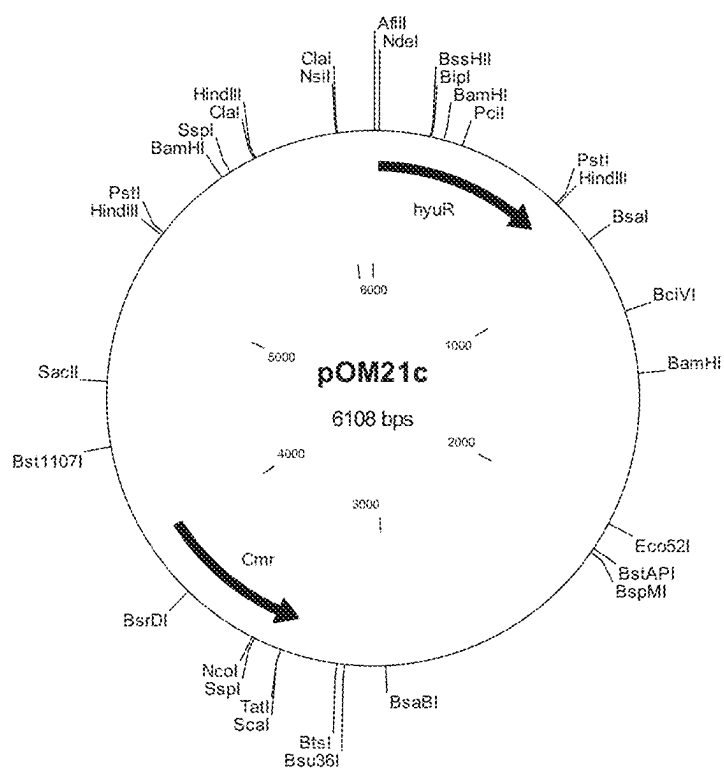
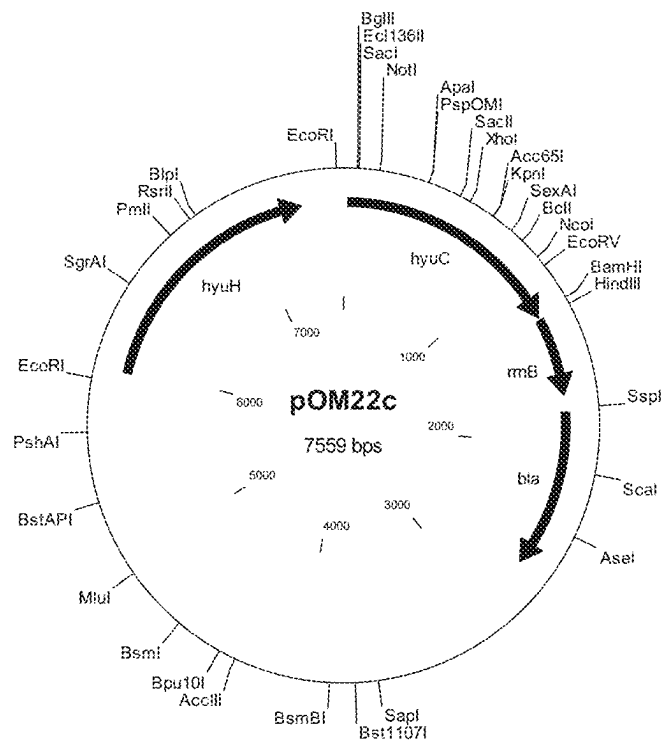


FIG.2



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ENZYMATIC METHOD FOR THE PRODUCTION OF L-GLUFOSINATE P-ESTERS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a National Stage entry under § 371 of International Application No. PCT/EP2022/056539, filed on Mar. 14, 2022. The content of this application is hereby incorporated by reference in its entirety.

REFERENCE TO A SEQUENCE LISTING

The present application is accompanied by an XML file as a computer readable form containing the sequence listing entitled, "006128USPCT-SeqList.txt", created on Sep. 11, 2024, with a file size of 86,979 bytes, the content of which is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates in a first aspect to an enzymatic method for the production of an L-glufosinate P-ester. This method is characterized by a step (c) in which an L-glufosinate P-ester carbamoylate is reacted to give the corresponding L-glufosinate P-ester. This step (c) is catalyzed by a carbamoylase, preferably by an L-enantioselective carbamoylase.

The L-glufosinate P-ester carbamoylate employed in step (c) may be obtained by reaction from the corresponding L-glufosinate hydantoin P-ester. This optional, antecedent step (b) is catalyzed by a hydantoinase, preferably an L-enantioselective hydantoinase. Even more preferred, the L-glufosinate hydantoin P-ester employed in step (b) may be obtained by reaction from the corresponding D-glufosinate hydantoin P-ester. This optional, antecedent step (a) is preferably catalyzed by a hydantoin racemase.

In a second aspect, the present invention relates to an enzymatic method for enantioselective production of an L-glufosinate P-ester from a mixture $M_{III A}$ of L- and D-glufosinate P-ester hydantoins. In the method according to the second aspect of the invention, the mixture $M_{III A}$ is provided in a step (i-A) and employed in steps (ii) and (iii). Steps (ii) and (iii) correspond to steps (b) and (c) described above, respectively, wherein in step (c), an L-carbamoylase is employed.

In an optional, preferred step (i-B) of the method according to the second aspect of the invention, corresponding to preferred step (a) described above, a hydantoin racemase is preferably used to convert at least a part of the D-glufosinate P-ester hydantoin in the mixture $M_{III A}$ into its corresponding L-isomer, giving a mixture $M_{III B}$, which is then employed in steps (ii) and (iii).

The L-glufosinate P-ester obtained in the methods according to the first or second aspect of the invention may be saponified to give L-glufosinate.

Description of Related Art

Organic phosphorous compounds, i.e. chemical agents comprising a carbon-phosphor bond, are widely applied as herbicides in the area of plant protection. Agents such as the herbicides glyphosate (Roundup®, Touchdown®) and glufosinate (Basta®, Liberty®) as well as the growth regu-

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lator glyphosine (Polaris®) are used for this purpose (as described for example by G. Hörlein, Rev. Environ. Contam. Toxicol. 1994, 138, 73-145).

The esters of P-methyl phosphinic acid (for example, P-methyl phosphinic acid butyl ester; "MPBE"; CAS-No: 6172-80-1) have a key role as synthetic building blocks in the synthesis of the non-selective herbicide glufosinate. These esters are accessible via two fundamental synthetic pathways (summarized in FIGS. 3 a and 3 b, page 130, of the article of K. Haack, Chem. Unserer Zeit 2003, 37, 128-138):

a. Reacting diethyl chlorophosphite $[CIP(OC_2H_5)_2]$ with CH_3MgCl provides methyl diethoxy phosphine $[H_3CP(OC_2H_5)_2]$; "DEMP"; CAS-No. 15715-41-0], which is partially hydrolyzed to give the corresponding methylphosphinic acid ethyl ester (MPPE; CAS-Nr: 16391-07-4).

b. Alternatively, methane can be reacted with phosphor trichloride at 500° C. to give methyl dichloro phosphane H_3CPCl_2 . The latter can then be solvolyzed in alcohols to give the corresponding methyl phosphinic acid esters.

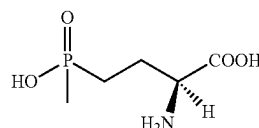
The esters of P-methyl phosphinic acid add to carbon-carbon double bonds regioselectively. This property is used in the synthesis of glufosinate for the formation of the second phosphor-carbon bond. For example, $H_3CPH(O)OR$ ($R=Alkyl$) reacts with 1-cyano allyl acetate in an addition reaction to provide an intermediate. Subsequent exchange of the acetate substituent with ammonia and hydrolysis of the cyano group and the ester group of the phosphinic acid moiety give glufosinate.

Acrylic acid ester is a cheaper alternative starting material. It can react with the ester of P-methyl phosphinic acid to 3-[alkoxy(methyl)phosphinyl]propionic acid alkyl ester. Claisen reaction of this diester with diethyl oxalate, hydrolysis and decarboxylation provide the corresponding α -keto acid, which can be reductively aminated to give glufosinate.

These and further synthetic routes towards glufosinate are also described in the art, e.g. in WO 1999/009039 A1, EP 0 508 296 A1.

A general disadvantage of all synthetic routes to glufosinate is that the obtained glufosinate is a racemic mixture. However, as there is no herbicidal activity of the D-enantiomer, L-glufosinate [hereinafter "LGA"; CAS-Nr. 35597-44-5; other names "(S)-glufosinate", "(-)-glufosinate"] is the enantiomer of economical interest.

LGA



For example, CN 111662325 A discloses a synthetic pathway in which L-homoserine is reacted to the respective hydantoin, followed by addition of methane phosphor dichloride, which results in a methane phosphane which is disubstituted with L-homoserine hydantoin. After a final Arbuzov reaction and hydrolysis, LGA is obtained. Although the disclosure describes a high enantiomeric excess ("ee"), this ee could not be reproduced by the inventors of the present invention. A reason could be that during this synthesis pathway, strong acidic (HCl for saponification of the hydantoin phosphor bond) and alkaline con-

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ditions (NaOH, 100° C. for ring opening) are applied. Such conditions usually lead to racemization, as described by M. Bovarnick & H. T. Clarke, Journal of the American Chemical Society 1938, 60, 2426-2430, by R. A. Lazarus, J. Org. Chem. 1990, 55, 4755-4757, and by A. S. Bommarius, M. Kottenhahn, H. Klenk, K. Drauz: "A direct route from hydantoins to D-amino acids employing a resting cell biocatalyst with D-hydantoinase and D-carbamoylase activity" on page 164 and 167 in "Microbial Reagents in Organic Synthesis" Series C: Mathematical and Physical Sciences-Vol. 381, S. Servi (Ed.), 1992, Springer Science+Business Media, B. V., Dordrecht.

For enantioselective syntheses of LGA, chemical and enzymatic pathways are described in the art.

WO 2020/145513 A1 and WO 2020/145514 A1 describe a chemical route to LGA. In this route, an L-homoserine derivative such as O-acetyl L-homoserine or O-succinyl L-homoserine is used as starting material and L-glufosinate is obtained by a sequence of reactions including lactonization and halogenation.

WO 2020/145627 A1 describes a similar route, wherein during halogenation, a bromo derivative is obtained.

The route disclosed by CN 106083922 A resembles these synthetic pathways but starts off from L-methionine.

CN 108516991 A describes another synthetic pathway to LGA, starting from the azeotropic dehydration of L-homoserine to give L-3,6-bis(2-haloethyl)-2,5-diketopiperazine, followed by introduction of a methylphosphonate diester group, and hydrolysis.

WO 2017/151573 A1 discloses a two-step enzymatic synthesis of LGA from D-glufosinate. In the first step, D-glufosinate is oxidatively deaminated to give 2-oxo-4-[hydroxy(methyl)phosphinoyl]-butyric acid ("PPO"), followed by the specific amination of PPO to LGA as the second step. The first step is carried out by catalysis of a D-amino acid oxidase, the second step is catalyzed by a transaminase.

WO 2020/051188 A1 discloses a similar method of converting racemic glufosinate to the L-enantiomer. In addition, it discloses a step in which the α -ketoacid or ketone byproduct formed during amination of PPO with an amine donor is converted by ketoglutarate decarboxylase to further shift the equilibrium to LGA.

WO 2019/018406 A1 discloses a method of purifying LGA from a mixture comprising LGA and glutamate. Glutamate is converted to pyroglutamate enzymatically by glutaminy-peptidyl cyclotransferase, and LGA is then purified from the resulting mixture with an ion-exchange resin.

WO 2013/072486 A1 disclose hydantoinase mutants which have a greater activity towards D-amino acids.

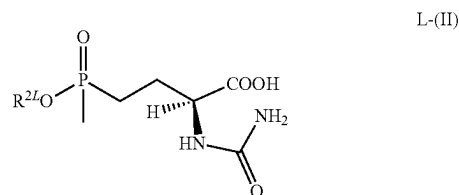
WO 00/58449 A1 disclose hydantoinase mutants which have a greater activity towards L-amino acids.

The object of the present invention is to provide a further enzymatic process for producing glufosinate or glufosinate P-esters. This process should provide products with a high excess of the L-enantiomer over the D-enantiomer. Moreover, this process should allow to use new substrates which heretofore were not used in the enzymatic synthesis of glufosinate or glufosinate esters. Moreover, there is a need in the art for an enantioselective method for the production of L-glufosinate or L-glufosinate esters from starting materials comprising D- and L-enantiomers, such as racemic mixtures, in a minimal number of synthetic steps.

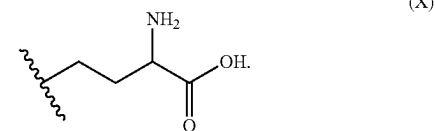
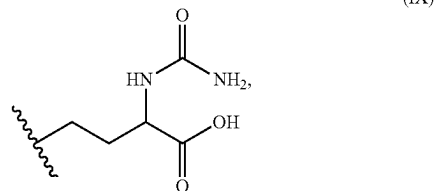
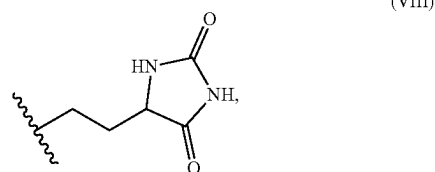
SUMMARY OF THE INVENTION

Namely, it was surprisingly found that the desired L-glufosinate P-esters can be produced by an enzymatically catalyzed reaction from the corresponding carbamoylates of formula L-(II):

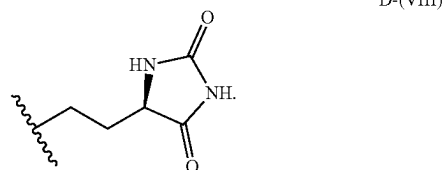
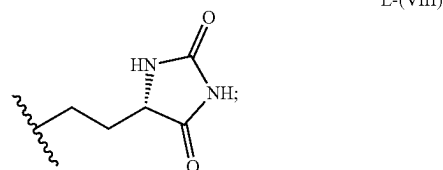
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wherein R^{2L} is a radical selected from the following formulae (VIII), (IX), (X), preferably selected from the following formulae (IX), (X), more preferably R^{2L} has the following formula (IX):

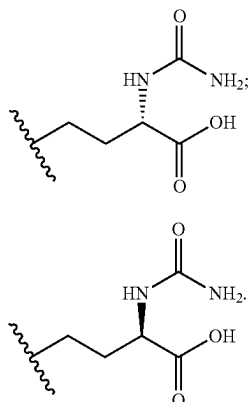


When R^{2L} is a radical of the formula (VIII), the radical R^{2L} may be selected from the formulae L-(VIII) and D-(VIII), and preferably has a formula according to L-(VIII):

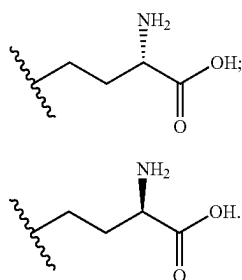


When R^{2L} is a radical of the formula (IX), the radical R^{2L} may be selected from the formulae L-(IX) and D-(IX), and preferably has a formula according to L-(IX):

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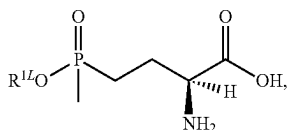


When R^{2L} is a radical of the formula (X), the radical R^{2L} may be selected from the formulae L-(X) and D-(X), and preferably has a formula according to L-(X):



Such carbamoylates were not used in the enzymatically catalyzed production of L-glufosinate or its P-esters before. This finding was even more surprising, as it turned out that the presence of a radical selected from the following formulae (VIII), (IX), (X) is mandatory, as there is no corresponding reaction of compounds according to formula L-(II) in which $R^{2L}=H$.

$\alpha.1$) The present invention hence solves the problems mentioned above by providing in a first aspect a method for the production of a glufosinate P-ester according to formula L-(I), comprising a step (c) in which a compound according to the above-mentioned formula L-(II) is reacted to give a compound according to formula L-(I):



wherein R^{1L} is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the above-mentioned formulae (IX), (X), more preferably R^{1L} has formula (X).

When R^{1L} is a radical of the formula (VIII), the radical R^{1L} may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

When R^{1L} is a radical of the formula (IX), the radical R^{1L} may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

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When R^{1L} is a radical of the formula (X), the radical R^{1L} may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

The reaction according step (c) is catalyzed by a carbamoylase E_1 , preferably by an L-enantioselective carbamoylase E_1 ("L-enantioselective carbamoylase"="L-carbamoylase").

The L-carbamoylase E_1 is preferably categorized in the EC class 3.5.1.87.

The polypeptide sequence of the L-carbamoylase E_1 is preferably selected from SEQ ID NO: 1 and variants thereof, SEQ ID NO: 2 and variants thereof, SEQ ID NO: 3 and variants thereof, SEQ ID NO: 4 and variants thereof, SEQ ID NO: 5 and variants thereof, SEQ ID NO: 6 and variants thereof, SEQ ID NO: 7 and variants thereof, SEQ ID NO: 8 and variants thereof, SEQ ID NO: 9 and variants thereof.

$\alpha.2$) In a preferred embodiment of the method according to the first aspect of the invention, the compound according to formula L-(II) that is employed in step (c) is obtained by a step (b) in which a compound according to formula L-(III) is reacted to give the compound according to formula L-(II):

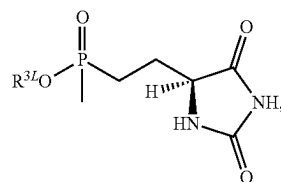
L-(X)

25

D-(X)

30

L-(III)



wherein the reaction according to step (b) is catalyzed by a hydantoinase E_2 , preferably an L-enantioselective hydantoinase E_2 ("L-enantioselective hydantoinase"="L-hydantoinase").

R^{3L} is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (VIII), (IX), more preferably R^{3L} has formula (VIII).

When R^{3L} is a radical of the formula (VIII), the radical R^{3L} may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

When R^{3L} is a radical of the formula (IX), the radical R^{3L} may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

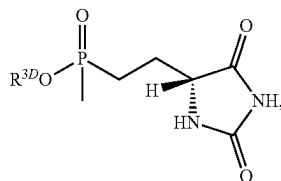
When R^{3L} is a radical of the formula (X), the radical R^{3L} may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

The hydantoinase E_2 , in particular the L-hydantoinase E_2 , is preferably categorized in the EC class 3.5.2.2.

The polypeptide sequence of the hydantoinase E_2 , in particular the L-hydantoinase E_2 , is preferably selected from SEQ ID NO: 10 and variants thereof, SEQ ID NO: 11 and variants thereof, SEQ ID NO: 12 and variants thereof, SEQ ID NO: 13 and variants thereof, SEQ ID NO: 14 and variants thereof, SEQ ID NO: 15 and variants thereof, SEQ ID NO: 16 and variants thereof.

$\alpha.3$) In an even more preferred embodiment of the method according to the first aspect of the invention, the compound according to formula L-(III) that is employed in step (b) is obtained by a step (a) in which a compound according to formula D-(III) is reacted to give the compound according to formula L-(III):

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wherein the reaction according to step (a) is preferably catalyzed by a hydantoin racemase E₃.

R^{3D} is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (VIII), (IX), more preferably R^{3D} has formula (VIII).

When R^{3D} is a radical of the formula (VIII), the radical R^{3D} may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

When R^{3D} is a radical of the formula (IX), the radical R^{3D} may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

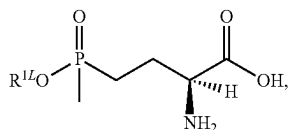
When R^{3D} is a radical of the formula (X), the radical R^{3D} may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

The hydantoin racemase E₃ is preferably categorized in the EC class 5.1.99.5.

The polypeptide sequence of the hydantoin racemase E₃ in step (a) is preferably selected from SEQ ID NO: 17 and variants thereof, SEQ ID NO: 18 and variants thereof, SEQ ID NO: 19 and variants thereof, SEQ ID NO: 20 and variants thereof, SEQ ID NO: 21 and variants thereof, SEQ ID NO: 22 and variants thereof, SEQ ID NO: 23 and variants thereof, SEQ ID NO: 24 and variants thereof, SEQ ID NO: 25 and variants thereof, SEQ ID NO: 26 and variants thereof.

It was further surprisingly found that the above-mentioned finding can be advantageously used for a method for enantioselective production of an L-glufosinate P-ester, in particular from mixtures of stereoisomers, preferably from mixtures of enantiomers, such as racemic mixtures.

β.1) Hence, the present invention relates in a second aspect to a method for the production of an L-glufosinate P-ester according to formula L-(I):



wherein R^{1L} is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the above-mentioned formulae (IX), (X), more preferably R^{1L} has formula (X).

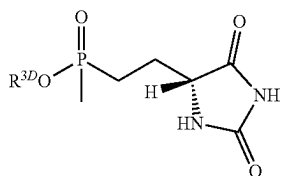
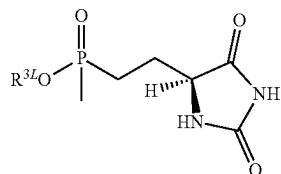
When R^{1L} is a radical of the formula (VIII), the radical R^{1L} may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

When R^{1L} is a radical of the formula (IX), the radical R^{1L} may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

When R^{1L} is a radical of the formula (X), the radical R^{1L} may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

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In this method, a mixture M_{III A} comprising at least one compound L-(III) and at least one compound D-(III) is provided [step (i)-A]:



R^{3L} is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (VIII), (IX), more preferably R^{3L} has formula (VIII).

When R^{3L} is a radical of the formula (VIII), the radical R^{3L} may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

When R^{3L} is a radical of the formula (IX), the radical R^{3L} may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

When R^{3L} is a radical of the formula (X), the radical R^{3L} may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

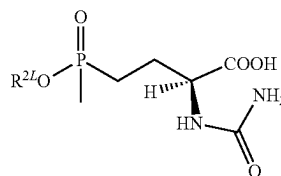
R^{3D} is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (VIII), (IX), more preferably R^{3D} has formula (VIII).

When R^{3D} is a radical of the formula (VIII), the radical R^{3D} may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

When R^{3D} is a radical of the formula (IX), the radical R^{3D} may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

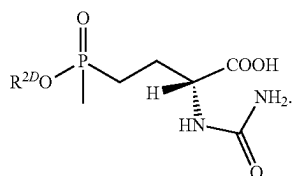
When R^{3D} is a radical of the formula (X), the radical R^{3D} may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

β.2) In a step (ii), the mixture M_{III A} is subjected to a reaction with a hydantoinase E₂, which is more preferably as defined under item α.2), giving a composition M_{II} comprising at least one compound according to formula L-(II) and optionally at least one compound according to formula D-(II), wherein L-(II) and D-(II) have the following formulae:



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-continued



R^{2L} is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (IX), (X), more preferably R^{2L} has formula (IX).

R^{2D} is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (IX), (X), more preferably R^{2D} has formula (IX).

When R^{2L} is a radical of the formula (VIII), the radical R^{2L} may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

When R^{2D} is a radical of the formula (VIII), the radical R^{2D} may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

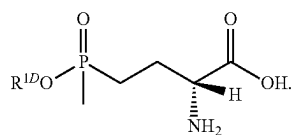
When R^{2L} is a radical of the formula (IX), the radical R^{2L} may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

When R^{2D} is a radical of the formula (IX), the radical R^{2D} may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

When R^{2L} is a radical of the formula (X), the radical R^{2L} may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

When R^{2D} is a radical of the formula (X), the radical R^{2D} may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

β.3) Then, in a further step (iii), M_I is subjected to a reaction, which is catalyzed by an L-carbamoylase E_1 , which is preferably as defined under item α.1), giving a M_I comprising at least one compound according to formula L-(I) and optionally at least one compound according to formula D-(I), wherein D-(I) has the following formula:



wherein R^{1D} is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the above-mentioned formulae (IX), (X), more preferably R^{1D} has formula (X).

When R^{1D} is a radical of the formula (VIII), the radical R^{1D} may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

When R^{1D} is a radical of the formula (IX), the radical R^{1D} may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

When R^{1D} is a radical of the formula (X), the radical R^{1D} may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

The method according to the second aspect of the invention is enantioselective, because M_I comprises either only L-(I) or, in case M_I comprises both stereoisomers L-(I) and

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D-(I), the molar ratio of L-(I) to D-(I) in M_I is greater than the molar ratio of L-(III) to D-(III) in M_{III} .

β.4) Furthermore, it was surprisingly found that the method for enantioselective production according to the second aspect of the invention can be further improved. Namely, in an optional step (i-B) which is carried out after step (i-A) [and which may be carried out before or concomitantly with steps (ii) and (iii), preferably concomitantly with steps (ii) and (iii)], at least a part of the compounds D-(III) comprised by the mixture M_{III} are reacted, preferably enzymatically catalyzed by a hydantoin racemase E_3 , into L-(III), which is the isomer of D-(III), giving a composition M_{III} comprising L-(III) and optionally its isomer D-(III).

In case the reaction according to step (i-B) is enzymatically catalyzed, it is preferably catalyzed by a hydantoin racemase E_3 .

The hydantoin racemase E_3 is preferably as defined under item α.3).

3. BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows the pOM21c plasmid map.

FIG. 2 shows the pOM22c plasmid map.

4. DETAILED DESCRIPTION OF THE INVENTION

4.1 Definitions

Any of the enzymes used according to any aspect of the present invention, may be an isolated enzyme. In particular, the enzymes used according to any aspect of the present invention may be used in an active state and in the presence of all cofactors, substrates, auxiliary and/or activating polypeptides or factors essential for its activity. Such factors may be metal ions such as Mn^{2+} or Co^{2+} .

In particular, an enzyme according to the present application may be a carbamoylase E_1 , a hydantoinase E_2 , or a hydantoin racemase E_3 .

“Enzymatically catalyzed” means that the respective reaction is catalyzed by an enzyme, which may be a carbamoylase E_1 , a hydantoinase E_2 , or a hydantoin racemase E_3 .

The enzyme used according to any aspect of the present invention may be recombinant. The term “recombinant” as used herein, refers to a molecule or is encoded by such a molecule, particularly a polypeptide or nucleic acid that, as such, does not occur naturally but is the result of genetic engineering or refers to a cell that comprises a recombinant molecule. For example, a nucleic acid molecule is recombinant if it comprises a promoter functionally linked to a sequence encoding a catalytically active polypeptide and the promoter has been engineered such that the catalytically active polypeptide is overexpressed relative to the level of the polypeptide in the corresponding wild type cell that comprises the original unaltered nucleic acid molecule.

A “polypeptide” (one or more peptides) is a chain of chemical building blocks called amino acids that are linked together by chemical bonds called peptide bonds. A protein or polypeptide, including an enzyme, may be “native” or “wild-type”, meaning that it occurs in nature or has the amino acid sequence of a native protein, respectively. These terms are sometimes used interchangeably. A polypeptide may or may not be glycosylated.

The term “overexpressed”, as used herein, means that the respective polypeptide encoded or expressed is expressed at a level higher or at higher activity than would normally be

found in the cell under identical conditions in the absence of genetic modifications carried out to increase the expression, for example in the respective wild type cell.

4.2 Methods to Obtain Enzymes

The enzymes that can be used in the method according to the present invention can be synthesized by methods that are known to the skilled person.

One approach, which is a preferred approach according to the invention, is to express the enzyme(s) in micro-organism(s) such as *Escherichia coli* (“*E. coli*”), *Saccharomyces cerevisiae*, *Pichia pastoris*, and others, and to add the whole cells to the reactions as whole cell biocatalysts. Another approach is to express the enzyme(s), lyse the microorganisms, and add the cell lysate. Yet another approach is to purify, or partially purify, the enzyme(s) from a lysate and add pure or partially pure enzyme(s) to the reaction. If multiple enzymes are required for a reaction, the enzymes can be expressed in one or several microorganisms, including expressing all enzymes within a single microor- ganism.

For example, the skilled person can obtain the enzymes according to the invention by expression, in particular, overexpression, [hereinafter, “expression, in particular over- expression” is abbreviated as (over)expression”, and “express, in particular overexpress” is abbreviated as (over) express”] of these enzymes in a cell and subsequent isolation thereof, e.g. as described in DE 100 31 999 A1. Episomal plasmids, for example, are employed for increasing the expression of the respective genes. In such plasmids, the nucleic acid molecule to be (over) expressed or encoding the polypeptide or enzyme to be (over) expressed may be placed under the control of a strong inducible promoter such as the lac promoter, located upstream of the gene. A promoter is a DNA sequence consisting of about 40 to 50 base pairs which constitutes the binding site for an RNA polymerase holoen- zyme and the transcriptional start point (M. Pátek, J. Holátko, T. Busche, J. Kalinowski, J. Nešvera, Microbial Biotechnology 2013, 6, 103-117), whereby the strength of expression of the controlled polynucleotide or gene can be influenced. A “functional linkage” is obtained by the sequen- tial arrangement of a promoter with a gene, which leads to a transcription of the gene.

Suitable strong promoters or methods of producing such promoters for increasing expression are known from the literature (e.g. S. Lisser & H. Margalit, Nucleic Acid Research 1993, 21, 1507-1516; M. Pátek and J. Nesvera in H. Yukawa and M Inui (eds.), *Corynebacterium glutamicum*, Microbiology Monographs 23, Springer Verlag Berlin Heidelberg 2013, 51-88; B. J. Eikmanns, E. Kleinertz, W. Liebl, H. Sahm, Gene 1991, 102, 93-98). For instance, native promoters may be optimized by altering the promoter sequence in the direction of known consensus sequences with respect to increasing the expression of the genes functionally linked to these promoters (M. Pátek, B. J. Eikmanns, J. Pátek, H. Sahm, Microbiology 1996, 142, 1297-1309; M. Pátek, J. Holátko, T. Busche, J. Kalinowski, J. Nešvera, Microbial Biotechnology 2013, 6, 103-117).

Constitutive promoters are also suitable for the (over) expression, in which the gene encoding the enzyme activity is expressed continuously under the control of the promoter such as, for example, the glucose dependent deo promoter. Chemically induced promoters are also suitable, such as tac, lac, rha or trp. The most widespread system for the induction of promoters is the lac operon of *E. coli*. In this case, either lactose or isopropyl β -D-thiogalactopyranoside (IPTG) is

used as inducer. Also, systems using arabinose (e.g. the pBAD system) or rhamnose (e.g. *E. coli* KRX) are common as inducers. A system for physical induction is, for example, the temperature-induced cold shock promoter system based on the *E. coli* cspA promoter from Takara or Lambda PL and also osmotically inducible promoters, for example, osmB (e.g. WO 95/25785 A1).

Suitable plasmids or vectors are in principle all embodi- ments available for this purpose to the person skilled in the art. The state of the art describes standard plasmids that may be used for this purpose, for example the pET system of vectors exemplified by pET-3a or pET-26b (+) (commer- cially available from Novagen). Further plasmids and vec- tors can be taken, for example, pOM21 described in WO 2004/111227 A2, pOM22 described in WO 00/058449 A1 or pOM18 described in WO 2013/072486 A1 or from the brochures of the companies Novagen, Promega, New Eng- land Biolabs, Clontech or Gibco BRL. Further preferred plasmids and vectors can be found in: Glover, D. M. (1985) DNA cloning: a practical approach, Vol. I-III, IRL Press Ltd., Oxford; Rodriguez, R. L. and Denhardt, D. T (eds) (1988) Vectors: a survey of molecular cloning vectors and their uses, 179-204, Butterworth, Stoneham; Goeddel, D. V. (1990) Systems for heterologous gene expression, Methods Enzymol. 185, 3-7; Sambrook, J.; Fritsch, E. F. and Mania- tis, T. (1989), Molecular cloning: a laboratory manual, 2nd ed., Cold Spring Harbor Laboratory Press, New York. Of these plasmids, pOM21 and pOM22 are preferred.

The plasmid vector, which contains the gene to be ampli- fied, is then converted to the desired strain, e.g. by conju- gation or transformation. The method of conjugation is described, for example, by A. Schäfer, J. Kalinowski, A. Pühler, Applied and Environmental Microbiology 1994, 60, 756-759. Methods for transformation are described, for example, in G. Thierbach, A. Schwarzer, A. Pühler, Applied Microbiology and Biotechnology 1988, 29, 356-362, L. K. Dunican & E. Shivnan, Bio/Technology 1989, 7, 1067-1070 and A. Tauch, O. Kirchner, L. Wehmeier, J. Kalinowski, A. Pühler, FEMS Microbiology Letters 1994, 123, 343-347. After homologous recombination by means of a “cross- over” event, the resulting strain contains at least two copies of the gene concerned.

The desired enzyme can be isolated by disrupting cells which contain the desired activity in a manner known to the person skilled in the art, for example with the aid of a ball mill, a French press or of an ultrasonic disintegrator and subsequently separating off cells, cell debris and disruption aids, such as, for example, glass beads, by centrifugation for 10 minutes at 13,000 rpm and 4° C. Using the resulting cell-free crude extract, enzyme assays with subsequent LC- ESI-MS detection of the products can then be carried out. Alternatively, the enzyme can be enriched in the manner known to the person skilled in the art by chromatographic methods (such as nickel-nitrilotriacetic acid affinity chro- matography, streptavidin affinity chromatography, gel filtra- tion chromatography or ion-exchange chromatography) or else purified to homogeneity. Quantification of the enzyme can be performed by methods known to the person skilled in the art, for example by determination of the concentration of the respective polypeptide of the enzyme (e.g. carbamoy- lase, hydantoinase and racemase) in the obtained solution by SDS page and analysis of the respective bands via the software GelQuant® (BiochemLabSolutions).

Moreover, whether or not a nucleic acid or polypeptide is (over) expressed, may be determined by way of quantitative PCR reaction in the case of a nucleic acid molecule, SDS polyacrylamide electrophoreses, Western blotting or com- parative activity assays in the case of a polypeptide. Genetic

modifications may be directed to transcriptional, translational, and/or post-translational modifications that result in a change of enzyme activity and/or selectivity under selected and/or identified culture conditions.

4.3 Variants

In the context of the present invention, the term “variant” with respect to polypeptide sequences refers to a polypeptide sequence with a degree of identity to the reference sequence (“sequence identity”) of at least 60%, preferably at least 70%, more preferably at least 71%, more preferably at least 72%, more preferably at least 73%, more preferably at least 74%, more preferably at least 75%, more preferably at least 76%, more preferably at least 77%, more preferably at least 78%, more preferably at least 79%, more preferably at least 80%, more preferably at least 81%, more preferably at least 82%, more preferably at least 83%, more preferably at least 84%, more preferably at least 85%, more preferably at least 86%, more preferably at least 87%, more preferably at least 88%, more preferably at least 89%, more preferably at least 90%, more preferably at least 91%, more preferably at least 92%, more preferably at least 93%, more preferably at least 94%, more preferably at least 95%, more preferably at least 96%, more preferably at least 97%, more preferably at least 98%, more preferably at least 99%, more preferably at least 99.9%. In still further particular embodiments, the degree of identity is at least 98.0%, more preferably at least 98.2%, more preferably at least 98.4%, more preferably at least 98.6%, more preferably at least 98.8%, more preferably at least 99.0%, more preferably at least 99.1%, more preferably at least 99.2%, more preferably at least 99.3%, more preferably at least 99.4%, more preferably at least 99.5%, more preferably at least 99.6%, more preferably at least 99.7%, more preferably at least 99.8%, or at least more preferably at least 99.9%.

It goes without saying that a “variant” of a certain polypeptide sequence is not identical to the polypeptide sequence.

Such variants may be prepared by introducing deletions, insertions, substitutions, or combinations thereof, in particular in amino acid sequences, as well as fusions comprising such macromolecules or variants thereof.

Modifications of amino acid residues of a given polypeptide sequence which lead to no significant modifications of the properties and function of the given polypeptide are known to those skilled in the art. Thus for example many amino acids can often be exchanged for one another without problems; examples of such suitable amino acid substitutions are: Ala by Ser; Arg by Lys; Asn by Gln or His; Asp by Glu; Cys by Ser; Gln by Asn; Glu by Asp; Gly by Pro; His by Asn or Gln; Ile by Leu or Val; Leu by Met or Val; Lys by Arg or Gln or Glu; Met by Leu or Ile; Phe by Met or Leu or Tyr; Ser by Thr; Thr by Ser; Trp by Tyr; Tyr by Trp or Phe; Val by Ile or Leu. It is also known that modifications, particularly at the N- or C-terminus of a polypeptide in the form of for example amino acid insertions or deletions, often exert no significant influence on the function of the polypeptide.

In line with this, preferable variants according to the invention of any of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24,

SEQ ID NO: 25, SEQ ID NO: 26, respectively, have a polypeptide sequence that comprises the complete polypeptide sequence of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26, respectively, or at least the amino acids of the respective sequence SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26 that are essential for the function, for example the catalytic activity of a protein, or the fold or structure of the protein. The other amino acids may be deleted, substituted or replaced by insertions or essential amino acids are replaced in a conservative manner to the effect that the activity of the enzyme, in particular the L-carbamoylase (SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9), hydantoinase (SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16), hydantoin racemase (SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26), is preserved.

3.4 Sequence Identity

The person skilled in the art is aware that various computer programs are available for the calculation of similarity or identity between two nucleotide or polypeptide sequences.

Preferred methods for determining the sequence identity initially generate the greatest alignment between the sequences to be compared. Computer programs for determining the sequence identity include, but are not limited to, the GCG program package including

GAP [J. Deveroy et al., *Nucleic Acid Research* 1984, 12, page 387, Genetics Computer Group University of Wisconsin, Medicine (WI)], and

BLASTP, BLASTN and FASTA (S. Altschul et al., *Journal of Molecular Biology* 1990, 215, 403-410). The BLAST program can be obtained from the National Center for Biotechnology Information (NCBI) and from other sources (BLAST Handbook, S. Altschul et al., NCBI NLM NIH Bethesda ND 22894; S. Altschul et al., above).

For instance, the percentage identity between two polypeptide sequences can be determined by the algorithm developed by S. B. Needleman & C. D. Wunsch, *J. Mol. Biol.* 1970, 48, 443-453, which has been integrated into the GAP program in the GCG software package, using either a BLOSUM62 matrix or a PAM250 matrix, a gap weight of 16, 14, 12, 10, 8, 6 or 4 and a length weight of 1, 2, 3, 4, 5 or 6. The person skilled in the art will recognize that the use of different parameters will lead to slightly different results, but that the percentage identity between two polypeptide overall will not be significantly different. The BLOSUM62 matrix is typically used applying the default settings (gap weight: 12, length weight: 1).

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In the context of the present invention, a sequence identity of 60% according to the above algorithm means 60% homology. The same applies to higher sequence identities.

Most preferably, the degree of identity between sequences is determined in the context of the present invention by the programme "Needle" using the substitution matrix BLOSUM62, the gap opening penalty of 10, and the gap extension penalty of 0.5. The Needle program implements the global alignment algorithm described in S. B. Needleman & C. D. Wunsch, J. Mol. Biol. 1970, 48, 443-453. The substitution matrix used according to the present invention is BLOSUM62, gap opening penalty is 10, and gap extension penalty is 0.5. The preferred version used in the context of this invention is the one presented by F. Madeira, Y. M. Park, J. Lee, N. Buso, T. Gur, N. Madhusoodanan, P. Basutkar, A. R. N. Tivey, S. C. Potter, R. D. Finn, Nucleic Acids Research 2019, 47, W636-W641, Web Server issue (preferred version accessible online on Mar. 4, 2022 via https://www.ebi.ac.uk/Tools/psa/emboss_needle/).

In a particular embodiment, the percentage of identity of an amino acid sequence of a polypeptide with, or to, a reference polypeptide sequence is determined by

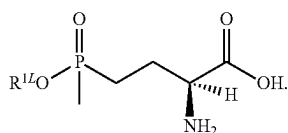
- i) aligning the two amino acid sequences using the Needle program, with the BLOSUM62 substitution matrix, a gap opening penalty of 10, and a gap extension penalty of 0.5;
- ii) counting the number of exact matches in the alignment;
- iii) dividing the number of exact matches by the length of the longest of the two amino acid sequences, and
- iv) converting the result of the division of iii) into percentage.

4.5 First Aspect: Method for Production of a Glufosinate P-Ester

4.5.1 Step (c)

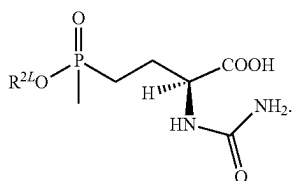
4.5.1.1 Enzymatic Catalysis of Step (c)

The present invention relates in a first aspect to a method for the production of an L-glufosinate P-ester according to formula L-(I):



The method according to the first aspect of the invention comprises a step (c).

In step (c), a compound according to formula L-(II) is reacted to give a compound according to formula L-(I). Formula L-(II) has the following formula:



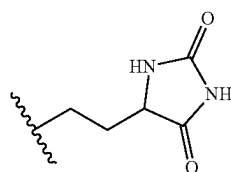
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The reaction according step (c) is enzymatically catalyzed, namely it is catalyzed by a carbamoylase E₁.

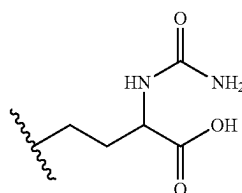
R^{1L} is a radical selected from the formulae (VIII), (IX), (X), preferably selected from the formulae (IX), (X), more preferably R^{1L} has formula (X).

R^{2L} is a radical selected from the formulae (VIII), (IX), (X), preferably selected from the formulae (IX), (X), more preferably R^{2L} has formula (X).

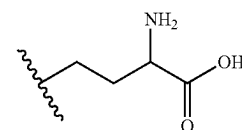
Formulae (VIII), (IX), (X) are as follows:



(VIII)



(IX)



(X)

When R^{1L} is a radical of the formula (VIII), the radical R^{1L} may be selected from the formulae L-(VIII) and D-(VIII), preferably L-(VIII).

When R^{1L} is a radical of the formula (IX), the radical R^{1L} may be selected from the formulae L-(IX) and D-(IX), preferably L-(IX).

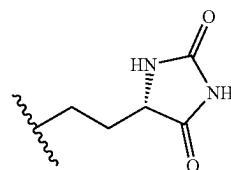
When R^{1L} is a radical of the formula (X), the radical R^{1L} may be selected from the formulae L-(X) and D-(X), preferably L-(X).

When R^{2L} is a radical of the formula (VIII), the radical R^{2L} may be selected from the formulae L-(VIII) and D-(VIII), preferably L-(VIII).

When R^{2L} is a radical of the formula (IX), the radical R^{2L} may be selected from the formulae L-(IX) and D-(IX), preferably L-(IX).

When R^{2L} is a radical of the formula (X), the radical R^{2L} may be selected from the formulae L-(X) and D-(X), preferably L-(X).

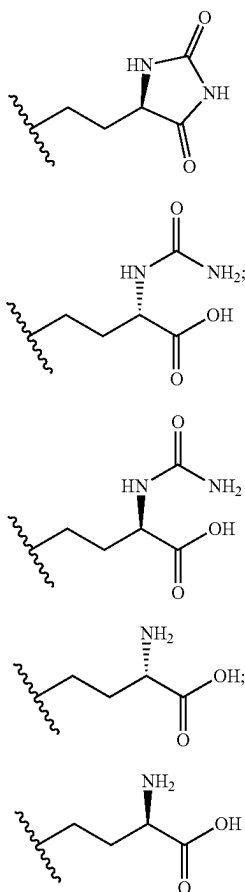
Formulae L-(VIII), D-(VIII), L-(IX), D-(IX), L-(X), and D-(X) are as follows:



L-(VIII)

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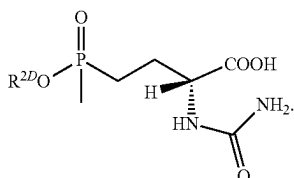
-continued



4.5.1.2 Enantioselective and Enantiospecific Catalysis of Step (c)

Step (c) of the method according to the first aspect of the present invention is preferably L-enantioselective, even more preferably L-enantiospecific.

In such a preferred embodiment, L-(II) is in particular employed in step (c) as a mixture M_{II} comprising at least one compound L-(II) and at least one compound D-(II), wherein D-(II) has the following formula:



R^{2D} is a radical selected from the formulae (VIII), (IX), (X), preferably selected from the formulae (IX), (X), more preferably R^{2D} has formula (X).

When R^{2D} is a radical of the formula (VIII), the radical R^{2D} may be selected from the formulae L-(VIII) and D-(VIII), preferably L-(VIII).

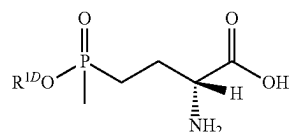
When R^{2D} is a radical of the formula (IX), the radical R^{2D} may be selected from the formulae L-(IX) and D-(IX), preferably L-(IX).

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When R^{2D} is a radical of the formula (X), the radical R^{2D} may be selected from the formulae L-(X) and D-(X), preferably L-(X).

Mixture M_{II} preferably comprises at least one compound L-(II) and at least one compound D-(II), wherein $R^{2L}=R^{2D}$.

For such mixtures M_{II} , it may be observed that the at least one compound D-(II) undergoes a reaction according to step (c)*, i.e. parallel to step (c). Namely, in the reaction according to step (c)*, the at least one compound D-(II) in mixture M_{II} is reacted to give a compound according to formula D-(I):



R^{1D} is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the above-mentioned formulae (IX), (X), more preferably R^{1D} has formula (X).

When R^{1D} is a radical of the formula (VIII), the radical R^{1D} may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

When R^{1D} is a radical of the formula (IX), the radical R^{1D} may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

When R^{1D} is a radical of the formula (X), the radical R^{1D} may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

In case that step (c) is "L-enantioselective", this means that in case a mixture M_{II} is employed in step (c), then there is

either no reaction according to step (c)* of any compound according to formula D-(II) to a compound according to D-(I) in the mixture M_{II} or,

in case there is a reaction according to step (c)*, then there is at least one pair of compounds D-(II)(II)* / L-(II)(II)* in the mixture M_{II} , for which the reaction according to step (c)* of D-(II)(II)* to give D-(I)(II)*, is less than the rate of the reaction according to step (c) of L-(II)(II)* to give L-(I)(II)*.

D-(II)(II)* is a compound according to formula D-(II), and L-(II)(II)* is a compound according to formula L-(II), wherein the radical R^{2L} in L-(II)(II)* and the radical R^{2D} in D-(II)(II)* are identical.

D-(I)(II)* is a compound according to formula D-(I), and L-(I)(II)* is a compound according to formula L-(I), wherein the radical R^{2L} in L-(I)(II)* and the radical R^{2D} in D-(I)(II)* are identical.

Preferably, the radical R^{2L} in L-(II)(II)* and the radical R^{1L} in L-(I)(II)* are identical. Preferably, the radical R^{2D} in D-(II)(II)* and the radical R^{1D} in D-(I)(II)* are identical.

Step (c) is "L-enantiospecific", if the rate of reaction according to step (c)* is essentially zero, i.e. there is no reaction according to step (c)* of any compound according to formula D-(II) to a compound according to D-(I) in the mixture M_{II} .

In a preferred embodiment, the molar ratio of all compounds according to formula L-(II) in the mixture M_{II} to all compounds according to formula D-(II) in the mixture M_{II} is essentially 1:1.

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In other preferred embodiments, the molar ratio of all compounds according to formula L-(II) in the mixture M_H to all compounds according to formula D-(II) in the mixture M_H is in the range of from 3:2 to 1:99, more preferably in the range of from 1.01:1 to 1:99, more preferably in the range of from 1:1 to 1:99, more preferably in the range of from 1:1.01 to 1:99, more preferably in the range of from 1:1.01 to 1:9, more preferably in the range of from 1:1.01 to 1:8, more preferably in the range of from 1:1.01 to 1:3.

Alternatively, D-(II) is comprised in an excess to L-(II) in mixture M_H , meaning that, while at least one compound according to formula L-(II) is present in the mixture M_H , the molar ratio of all compounds according to formula L-(II) in the mixture M_H to all compounds according to formula D-(II) in the mixture M_H is <1:1, preferably <0.9:1, more preferably <0.75:1, more preferably <0.5:1, more preferably <0.2:1, more preferably <0.1:1, more preferably <0.01:1.

Step (c) is in particular L-enantioselective, if it is catalyzed by an L-carbamoylase E_1 , which may be determined by the skilled person as set forth under 4.5.4.3.

In case step (c) is L-enantioselective with respect to a given pair of compounds D-(II)(III)* / L-(II)(III)*, the reaction according to step (c) of L-(II)(III)* to give L-(I)(III)* proceeds preferably at a reaction rate that is at least 2 times greater, preferably at least 10 times greater, more preferably at least 100 times greater, even more preferably at least 103 times greater, even more preferably at least 104 times greater, even more preferably at least 105 times greater than the reaction rate at which step (c)* of D-(II)(III)* to give D-(I)(III)* proceeds.

To quantify for a given pair of compounds D-(II)(III)* / L-(II)(III)* the factor at which the reaction rate of step (c) proceeds compared to the reaction rate of step (c)*, the following test may be carried out:

(1) An equimolar mixture [molar ratio of the two compounds L-(II)(III)* and D-(II)(III)* is 1:1] is subjected to the respective reactions conditions and the development of the two products L-(I)(III)* and D-(I)(III)* is monitored over time (e.g. by LC-MS as set forth under item 5.4).

(2) When n_{L10} =10 mol-% of the initially employed L-(II)(III)* has reacted to the product L-(I)(III)*, the amount of D-(I)(III)* that was formed by reaction from D-(II)(III)* [in mol-% relative to the initially employed D-(II)+] is measured (= n_{D10}).

(3) The ratio of n_{L10}/n_{D10} =10/ n_{D10} gives the factor at which the reaction rate of step (c) proceeds compared to the reaction rate of step (c)* for the given pair of compounds D-(II)(III)* / L-(II)(III)*.

If n_{L10}/n_{D10} >1, the reaction rate of step (c) is greater than the reaction rate of step (c)* for the given pair of compounds D-(II)(III)* / L-(II)(III)*.

If n_{L10}/n_{D10} <1, the reaction rate of step (c) is lesser than the reaction rate of step (c)* for the given pair of compounds D-(II)(III)* / L-(II)(III)*.

If n_{L10}/n_{D10} =1, the reaction rates of steps (c) and (c)* are the same for the given pair of compounds D-(II)(III)* / L-(II)(III)*.

4.5.2 Carbamoylases

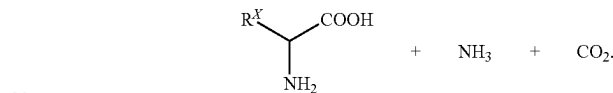
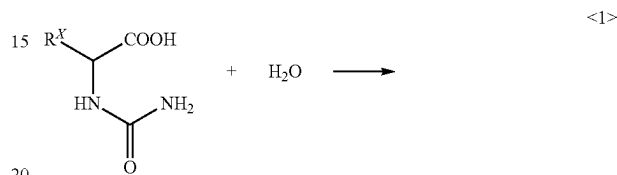
The reaction according to step (c) of the method according to the first aspect of the invention is catalyzed by a carbamoylase E_1 .

Namely, it was surprisingly found that carbamoylases accept compounds of formula L-(II) as substrates and convert them to products L-(I), and hence can be used to

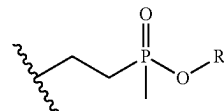
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catalyze the reaction according to step (c). This finding is of high scientific and economic value, as it opens new synthetic routes based on new starting materials for the production of L-glufosinate P-esters and L-glufosinate. Even more surprisingly, it was found that L-glufosinate carbamoylate, i.e. the compound according to L-(II), in which R=H, does not undergo reaction by carbamoylases to give L-glufosinate.

In nature, carbamoylases generally catalyze the following reaction <1>, wherein R^X may be an organic residue, e.g. a side chain of one of the naturally occurring amino acids.

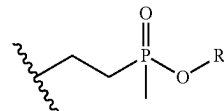


It was now surprisingly found that carbamoylases also accept substrates in which $R^X=R^Y$ =



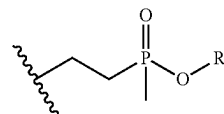
wherein $R=R^{2L}$.

Surprisingly, they do not accept substrates in which $R^X=R^Y$ =



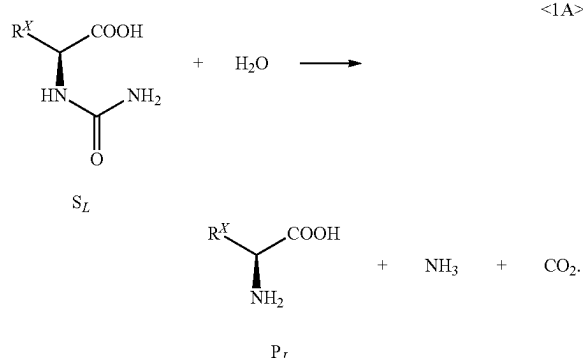
wherein R=H.

In the context of the present invention, a "carbamoylase E_1 " is a carbamoylase that catalyzes the following reaction <1A> of a carbamoyl substrate SL to the respective amino acid product P_L , wherein $R^X=R^Y$ =

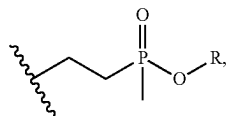


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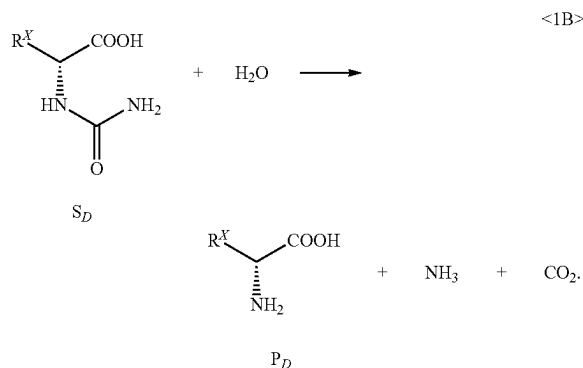
wherein $R=R^{2L}$:



In particular, the carbamoylase E_1 is a "L-carbamoylase", i.e. it has a greater catalytic activity for reaction <1A> than for reaction <1B>, wherein R^X in the substrate Sp in the reaction <1B> and R^X in the substrate SL in the reaction <1A> are identical, and $R^X=R^Y$.



wherein $R=R^{2L}$



As an L-carbamoylase has a higher catalytic activity for reaction <1A> than for reaction <1B>, it is "L-enantioselective". An L-carbamoylase that has no catalytic activity for reaction <1B> and thus only has catalytic activity for reaction <1A> is "L-enantiospecific".

A "D-carbamoylase" is defined as a carbamoylase which is "D-enantioselective", i.e. it has a higher catalytic activity for reaction <1B> than for reaction <1A>. A D-carbamoylase that does not catalyze reaction <1A> and thus only has catalytic activity for reaction <1B> is "D-enantiospecific".

A carbamoylase which has the same catalytic activity for reaction <1B> as for reaction <1A>, is referred to as "non-enantioselective carbamoylase".

For determination whether a carbamoylase may be denoted as "L-carbamoylase", "D-carbamoylase" or "non-enantioselective carbamoylase" in the context of the present invention, the procedure according to Assay B (item 4.5.4) may preferably be used.

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The carbamoylase E_1 , in particular the L-carbamoylase E_1 , that may be used in step (c) of the method according to the first aspect of the invention may originate from *Achromobacter* sp., in particular *Achromobacter xylosoxidans*; *Agrobacterium* sp., in particular *Agrobacterium tumefaciens*; *Arthrobacter* sp., in particular *Arthrobacter crystallopoietes*, *Arthrobacter aurescens*, *Arthrobacter* sp. BT801; *Bacillus* sp., in particular *Bacillus fordii*; *Blastobacter* sp.; *Bradyrhizobium* sp., in particular *Bradyrhizobium japonicum*; *Brevibacillus* sp., in particular *Brevibacillus reuszeri*; *Comamonas* sp.; *Ensifer* sp., in particular *Ensifer adhaerens*; *Flavobacterium* sp. *Geobacillus* sp., in particular *Geobacillus kaustophilus*, *Geobacillus stearothermophilus*; *Microbacterium* sp., in particular *Microbacterium liquefaciens* strain AJ3912; *Paenarthrobacter* sp., in particular *Paenarthrobacter aurescens*; *Pasteurella* sp.; *Pseudomonas* sp.; *Ralstonia* sp., in particular *Ralstonia pickettii*; *Sinorhizobium* sp., in particular *Sinorhizobium meliloti*.

An L-carbamoylase E_1 suitable for the method according to the present invention may be the enzyme HyuC, which originates from *Arthrobacter*. Other enzymes are AmaB, AtcC, Inc, SinmeB_2280.

WO 01/23582 A1 discloses an example of an enzyme having carbamoylase activity according to the invention.

The carbamoylase E_1 that may be used in step (c) of the method according to the present invention may be an L-carbamoylase categorized in the EC class EC 3.5.1.87.

L-carbamoylase enzymes are for example described by J. Ogawa, H. Miyake, S. Shimizu, *Appl. Microbiol. Biotechnol.* 1995 43, 1039-1043 and in WO 01/23582 A1.

A L-carbamoylase E_1 that may preferably be used in step (c) according to the first aspect of the invention may originate from *Arthrobacter* sp., in particular *Arthrobacter crystallopoietes*, *Arthrobacter aurescens*, *Arthrobacter* sp. BT801, *Arthrobacter aurescens* DSM 3747; *Bacillus* sp., in particular *Bacillus fordii*; *Geobacillus* sp., in particular *Geobacillus stearothermophilus*, *Geobacillus kaustophilus*; *Microbacterium* sp., in particular *Microbacterium liquefaciens* strain AJ3912; *Paenarthrobacter* sp., in particular *Paenarthrobacter aurescens*; *Pseudomonas* sp., in particular *Pseudomonas* sp. QR-101; *Sinorhizobium* sp., in particular *Sinorhizobium meliloti*. Even more preferably, the L-carbamoylase E_1 that may preferably be used in step (c) according to the first aspect of the invention may originate from *Arthrobacter* sp., in particular *Arthrobacter crystallopoietes*, *Arthrobacter aurescens*, *Arthrobacter* sp. BT801, *Arthrobacter aurescens* DSM 3747, most preferably from *Arthrobacter aurescens* DSM 3747.

The respective sequences can be derived from databases such as the Braunschweig Enzyme Database (BRENDA, Germany, available under www.brenda-enzymes.org/index.php), the National Center for Biotechnological Information (NCBI, available under www.ncbi.nlm.nih.gov/) or the Kyoto Encyclopedia of Genes and Genomes (KEGG, Japan, available under www.genome.jp/kegg/).

The following table 1 gives preferred examples for polypeptide sequences of L-carbamoylases E_1 that may be preferably used in step (c) of the method according to the first aspect of the invention. The genes encoding the respective L-carbamoylase E_1 and the respective accession code are indicated as far as known.

TABLE 1

L-Carbamoylases (EC 3.5.1.87)			
Strain	Gene name	GenBank/UniProt accession	SEQ ID NO: of the polypeptide
<i>Arthrobacter aureus</i> DSM 3747	hyuC		SEQ ID NO: 1
<i>Geobacillus stearothermophilus</i>	amaB	Q53389	SEQ ID NO: 2
<i>Pseudomonas</i> sp. QR-101	atcC	H9B8T5	SEQ ID NO: 3
<i>Geobacillus kaustophilus</i>	Inc	Q8GQG5	SEQ ID NO: 4
<i>Paenarthrobacter aureus</i>	hyuC	Q9F464	SEQ ID NO: 5
<i>Sinorhizobium meliloti</i>	SinmeB_2280	A0A0E0UEY4	SEQ ID NO: 6
<i>Bacillus fordii</i> strain MH602	not assigned	ABL14248	SEQ ID NO: 7
<i>Arthrobacter</i> sp. BT801	hyuC	AAL55413	SEQ ID NO: 8
<i>Microbacterium liquefaciens</i> strain AJ3912	not assigned		SEQ ID NO: 9

In a preferred embodiment of the method according to the first aspect of the present invention, the reaction according to step (c) is catalyzed by an L-carbamoylase E_1 , wherein the polypeptide sequence of E_1 is selected from the group consisting of SEQ ID NO: 1 and variants thereof, SEQ ID NO: 2 and variants thereof, SEQ ID NO: 3 and variants thereof, SEQ ID NO: 4 and variants thereof, SEQ ID NO: 5 and variants thereof, SEQ ID NO: 6 and variants thereof, SEQ ID NO: 7 and variants thereof, SEQ ID NO: 8 and variants thereof, SEQ ID NO: 9 and variants thereof, preferably SEQ ID NO: 1 and variants thereof.

4.5.3 Assays A_L and A_D for Determining Carbamoylase Activity

The skilled person is aware of carbamoylases, in particular L-carbamoylases, that may be used in step (c) of the method according to the first aspect of the invention.

In particular, Assay A_L , described in the following, may be used to determine carbamoylase and L-carbamoylase activity of a given enzyme E_X and may advantageously be used according to the invention to determine carbamoylase and L-carbamoylase activity in variants of SEQ ID NO: 1, variants of SEQ ID NO: 2, variants of SEQ ID NO: 3, variants of SEQ ID NO: 4, variants of SEQ ID NO: 5, variants of SEQ ID NO: 6, variants of SEQ ID NO: 7, variants of SEQ ID NO: 8, variants of SEQ ID NO: 9.

For comparative reasons, Assay A_D may be used to determine D-carbamoylase activity of a given enzyme E_X .

For the purpose of Assay A_L and Assay A_D , the molar mass of the enzyme E_X to be tested is calculated as the molar mass of the polypeptide sequence of E_X .

Moreover, a substrate according to formula L-(II), wherein R^{2L} =n-butyl, is used in Assay A_L so that there is only one chiral carbon atom in the substrate. The compound according to formula L-(II), wherein R^{2L} =n-butyl, is denoted as "L-(II)*".

A product according to formula L-(I), wherein R^{1L} =n-butyl, is detected in Assay A_L .

The compound according to formula L-(I), wherein R^{1L} =n-butyl, is denoted as "L-(I)(II)*".

Moreover, a substrate according to formula D-(II), wherein R^{2D} =n-butyl, is used in Assay A_D so that there is only one chiral carbon atom in the substrate. The compound according to formula D-(II), wherein R^{2D} =n-butyl, is denoted as "D-(II)(II)*".

A product according to formula D-(I), wherein R^{1D} =n-butyl, is detected in Assay A_D .

The compound according to formula D-(I), wherein R^{1D} =n-butyl, is denoted as "D-(I)(II)*".

4.5.3.1 Assay A_L :

To 0.9 ml of an aqueous reaction solution (phosphate buffer, pH 7.2, 10 mM $MnCl_2$), containing 50 mM of a compound according to formula L-(II)(II)* are added 400 nmol of E_X in 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM $MnCl_2$). The resulting solution is incubated at 25° C., and the pH is held at pH 7.2 by addition of 0.5 M NaOH. After 300 minutes, the reaction is stopped by addition of 2 M HCl to achieve a pH of 2.5, and the molar amount of the respective compound according to formula L-(I)(II)* is determined. L-(I)(II)* may be detected by the LC-MS method as described in the example section (item 5.4) for detection of LGA.

4.5.3.2 Assay A_D :

To 0.9 ml of an aqueous reaction solution (phosphate buffer, pH 7.2, 10 mM $MnCl_2$), containing 50 mM of a compound according to formula D-(II)(II)* are added 400 nmol of E_X in 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM $MnCl_2$). The resulting solution is incubated at 25° C., and the pH is held at pH 7.2 by addition of 0.5 M NaOH. After 300 minutes, the reaction is stopped by addition of 2 M HCl to achieve a pH of 2.5, and the molar amount of the respective compound according to formula D-(I)(II)* is determined. D-(I)(II)* may be detected by the LC-MS method described in the example section (item 5.4) for detection of LGA.

4.5.4 Assay B for Identifying Carbamoylases, L-Carbamoylases, D-Carbamoylases, L- and D-Enantiospecificity

The carbamoylase E_1 according to the invention is preferably an L-carbamoylase, more preferably L-enantiospecific.

Whether a given enzyme E_X may be considered a carbamoylase E_1 , in particular an L-carbamoylase, may be determined in the context of the present invention by the following Assay B.

4.5.4.1 Assay B:

B-1. Firstly, Assay A_L as set forth under item 4.5.3.1 is conducted, and the obtained molar amount of the compound of the formula L-(I)(II)* is determined according to Assay A_L .

B-2. Secondly, Assay A_D as set forth under item 4.5.3.2 is conducted, and the obtained molar amount of the compound of the formula D-(I)(II)* is determined according to Assay A_D .

B-3. Then, step B-1 is repeated, except that instead of the addition of 400 nmol E_x in 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM $MnCl_2$), 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM $MnCl_2$) without E_x is added.

B-4. Then, step B-2 is repeated, except that instead of the addition of 400 nmol E_x in 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM $MnCl_2$), 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM $MnCl_2$) without E_x is added.

4.5.4.2 Carbamoylase Activity

If the molar amount of the compound of the formula L-(I) (II)* that is determined in step B-1, is greater than the molar amount of the compound of the formula L-(I) (II)* that is determined in step B-3, then E_x is deemed to have carbamoylase activity, and hence may be considered a carbamoylase E_1 in the context of the invention.

4.5.4.3 L-Carbamoylase Activity

4.5.4.3.1 L-Carbamoylases

- (i) If the molar amount of the compound of formula L-(I) (II)* that is determined in step B-1, is greater than the molar amount of the compound of formula L-(I) (II)* that is determined in step B-3, and
- (ii) if the molar amount of the compound of formula D-(I) (II)* that is determined in step B-2, is greater than or the same as the molar amount of the compound of formula D-(I) (II)* that is determined in step B-4, and
- (iii) if, in addition, the molar amount of the compound of formula of L-(I) (II)* that is determined in step B-1 is greater than the molar amount of the compound of formula of D-(I) (II)* that is determined in step B-2, then E_x is deemed to have L-carbamoylase activity, and hence may be considered an L-carbamoylase in the context of the invention. In this case, E_x is deemed to be "L-enantioselective" in the context of the invention.

For the sake of clarity, it is pointed out that condition (iii) is automatically fulfilled in those cases in which the molar amount of the compound of formula D-(I) (II)* that is determined in step B-2, is the same as the molar amount of the compound of formula D-(I) (II)* that is determined in step B-4.

- (i) If the molar amount of the compound of formula L-(I) (II)* that is determined in step B-1, is greater than the molar amount of the compound of formula L-(I) (II)* that is determined in step B-3, and
- (ii) if the molar amount of the compound of formula D-(I) (II)* that is determined in step B-2 is the same as the molar amount of the compound of formula D-(I) (II)* that is determined in step B-4, then E_x is deemed to have L-carbamoylase activity, and hence may be considered an L-carbamoylase in the context of the invention. In this case, E_x is not only "L-enantioselective", but also "L-enantiospecific" in the context of the invention.

For L-carbamoylases E_x that are not L-enantiospecific, the L-enantioselectivity may then be quantified by dividing the molar amount of the compound of formula L-(I) (II)* that is determined in step B-1, by the molar amount of the compound of formula D-(I) (II)* that is determined in step B-2, and then multiplying the obtained value by 100, giving the L-enantioselectivity of E_x in %.

4.5.4.3.2 D-Carbamoylases

- (i) If the molar amount of the compound of formula L-(I) (II)* that is determined in step B-1, is greater than or the same as the molar amount of the compound of formula L-(I) (II)* that is determined in step B-3, and
- (ii) if the molar amount of the compound of formula D-(I) (II)* that is determined in step B-2, is greater than the

molar amount of the compound of formula D-(I) (II)* that is determined in step B-4, and

- (iii) if, in addition, the molar amount of the compound of formula D-(I) (II)* that is determined in step B-2 is greater than the molar amount of the compound of formula L-(I) (II)* that is determined in step B-1, then E_x is deemed to have D-carbamoylase activity, and hence may be considered a D-carbamoylase. In this case, E_x is "D-enantioselective" in the context of the invention.

For the sake of clarity it is pointed out that condition (iii) is automatically fulfilled in those cases in which the molar amount of the compound of formula L-(I) (II)* that is determined in step B-1 is the same as the molar amount of the compound of formula L-(I) (II)* that is determined in step B-3.

- (i) If the molar amount of the compound of formula L-(I) (II)* that is determined in step B-1, is the same as the molar amount of the compound of formula L-(I) (II)* that is determined in step B-3, and
- (ii) if the molar amount of the compound of formula D-(I) (II)* that is determined in step B-2, is greater than the molar amount of the compound of formula D-(I) (II)* that is determined in step B-4, then E_x is deemed to have D-carbamoylase activity, and hence may be considered to be a D-carbamoylase. In this case, E_x is not only "D-enantioselective", but also "D-enantiospecific" in the context of the invention.

For D-carbamoylases E_x , that are not D-enantiospecific, the D-enantioselectivity may then be quantified by dividing the molar amount of the compound of formula D-(I) (II)* that is determined in step B-2 by the molar amount of the compound of formula L-(I) (II)* that is determined in step B-1 and then multiplying the obtained value by 100, giving the D-enantioselectivity of E_x in %.

4.5.4.3.3 Non-Enantioselective Carbamoylases

- (i) If the molar amount of the compound of formula L-(I) (II)* that is determined in step B-1, is greater than the molar amount of the compound of formula L-(I) (II)* that is determined in step B-3, and
- (ii) if the molar amount of the compound of formula D-(I) (II)* that is determined in step B-2 is greater than the molar amount of the compound of formula D-(I) (II)* that is determined in step B-4, and
- (iii) if, in addition, the molar amount of the compound of formula D-(I) (II)* that is determined in step B-2 is the same as the molar amount of the compound of formula L-(I) (II)* that is determined in step B-1, then E_x is deemed to be a non-enantioselective carbamoylase in the context of the invention. In this case, E_x is "non-enantioselective".

4.5.5 Assay C for Identifying Preferred Carbamoylase Variants of SEQ ID NOs: 1-9

4.5.5.1 L- and D-Enantioselective Carbamoylase Variants

An enzyme, the polypeptide sequence of which is selected from the group consisting of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9 has carbamoylase, and L-carbamoylase activity.

In a preferred embodiment of the method according to the first aspect of the invention, the polypeptide sequence of the L-carbamoylase E_1 is selected from the group consisting of SEQ ID NO: 1 and variants thereof, SEQ ID NO: 2 and variants thereof, SEQ ID NO: 3 and variants thereof, SEQ

ID NO: 4, and variants thereof, SEQ ID NO: 5 and variants thereof, SEQ ID NO: 6 and variants thereof, SEQ ID NO: 7, and variants thereof, SEQ ID NO: 8 and variants thereof, SEQ ID NO: 9 and variants thereof, more preferably SEQ ID NO: 1 and variants thereof.

The term "variant" is defined under item 4.3.

In the context of the invention, an enzyme E_1 , the polypeptide sequence of which is a variant of one of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9 has carbamoylase activity, preferably L-carbamoylase activity, more preferably is L-enantiospecific.

Whether a given enzyme E_X , the polypeptide sequence of which is a variant of one of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, has carbamoylase activity, L carbamoylase activity and/or is L-enantiospecific may be determined as set forth under items 4.5.4.2 and 4.5.4.3.1, respectively.

The carbamoylase activity of a given L-carbamoylase E_{1P} , the polypeptide sequence of which is a variant of one of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, relative to the carbamoylase activity of an L-carbamoylase E_{1S} , wherein the polypeptide sequence of E_{1S} is selected from one of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, may be quantified in the context of the present invention by the following Assay C:

4.5.5.2 Assay C

C-1 Assay A_L as set forth under item 4.5.3.1 is conducted, wherein E_{1S} is the enzyme to be tested. The obtained molar amount of the compound according to formula L-(I)(II)* is determined according to Assay A_L .

C-2 Step C-1 is repeated, except that, instead of E_{1S} , E_{1P} is used as the enzyme to be tested.

C-3. Then, the molar amount of the compound according to formula L-(I)(II)* that is determined in step C-2, is divided by the molar amount of the compound according to formula L-(I)(II)* that is determined in step C-1, and the obtained ratio is multiplied by 100, giving the carbamoylase activity of L-carbamoylase E_{1P} relative to the carbamoylase activity of the L-carbamoylase E_{1S} , in %.

4.5.6 Preferred Carbamoylase Variants of SEQ ID NOs: 1-9

In the context of the invention, L-carbamoylases E_1 , the polypeptide sequence of which is selected from the group consisting of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, are generally denoted as " E_{1S} ".

L-carbamoylases E_1 , the polypeptide sequence of which is selected from variants of a sequence selected from the group consisting of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, are generally denoted as " E_{1P} ".

In a preferred embodiment of the method according to the first aspect of the present invention, the reaction in step (c) is catalyzed by an L-carbamoylase E_1 , and the polypeptide sequence of the L-carbamoylase E_1 is selected from the group consisting of SEQ ID NO: 1 and variants thereof, SEQ ID NO: 2 and variants thereof, SEQ ID NO: 3 and variants thereof, SEQ ID NO: 4 and variants thereof, SEQ ID NO: 5

and variants thereof, SEQ ID NO: 6 and variants thereof, SEQ ID NO: 7 and variants thereof, SEQ ID NO: 8 and variants thereof, SEQ ID NO: 9 and variants thereof. More preferably, the reaction in step (c) is catalyzed by an L-carbamoylase E_1 , and the polypeptide sequence of the L-carbamoylase E_1 is selected from the group consisting of SEQ ID NO: 1 and variants of SEQ ID NO: 1.

4.5.6.1 Preferred Variants of SEQ ID NO: 1

According to the invention, the polypeptide sequence of the L-carbamoylase E_1 may also be a variant of SEQ ID NO: 1.

The L-carbamoylase E_1 , the polypeptide sequence of which is SEQ ID NO: 1, is denoted as " E_{101S} ". L-carbamoylases E_1 , the polypeptide sequence of which is selected from variants of SEQ ID NO: 1, are generally denoted as " E_{101P} ".

A variant of the polypeptide sequence of SEQ ID NO: 1 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 1.

The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 1 is not identical to SEQ ID NO: 1.

According to the invention, an L-carbamoylase E_{101P} has carbamoylase activity and L-carbamoylase activity, determined as described under items 4.5.4.2 and 4.5.4.3.1.

According to the invention, an L-carbamoylase E_{101P} preferably has carbamoylase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the carbamoylase activity of the L-carbamoylase E_{101S} , wherein the carbamoylase activity of E_{101P} relative to the carbamoylase activity of E_{101S} is determined by Assay C described under item 4.5.5.2.

It is even more preferable according to the invention, that an L-carbamoylase E_{101P} has carbamoylase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the carbamoylase activity of the L-carbamoylase E_{101S} , wherein the carbamoylase activity of E_{101P} relative to the carbamoylase activity of E_{101S} is determined by Assay C described under item 4.5.5.2.

4.5.6.2 Preferred Variants of SEQ ID NO: 2

According to the invention, the polypeptide sequence of the L-carbamoylase E_1 may also be a variant of SEQ ID NO: 2.

The L-carbamoylase E_1 , the polypeptide sequence of which is SEQ ID NO: 2, is denoted as " E_{102S} ". L-carbamoylases E_1 , the polypeptide sequence of which is selected from variants of SEQ ID NO: 2, are generally denoted as " E_{102P} ".

A variant of the polypeptide sequence of SEQ ID NO: 2 is a polypeptide with sequence identity of at least 60%,

preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 2.

The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 2 is not identical to SEQ ID NO: 2.

According to the invention, an L-carbamoylase E_{102V} has carbamoylase activity and L-carbamoylase activity, determined as described under items 4.5.4.2 and 4.5.4.3.1.

According to the invention, an L-carbamoylase E_{102P} preferably has carbamoylase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the carbamoylase activity of the L-carbamoylase E_{102S}, wherein the carbamoylase activity of E_{102P} relative to the carbamoylase activity of E_{102S}, is determined by Assay C described under item 4.5.5.2.

It is even more preferable according to the invention, that an L-carbamoylase E_{102V} has carbamoylase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the carbamoylase activity of the L-carbamoylase E_{102S}, wherein the carbamoylase activity of E_{102V} relative to the carbamoylase activity of E_{102S} is determined by Assay C described under item 4.5.5.2.

4.5.6.3 Preferred Variants of SEO ID NO: 3

According to the invention, the polypeptide sequence of the L-carbamoylase E₁ may also be a variant of SEQ ID NO: 3.

The L-carbamoylase E₁, the polypeptide sequence of which is SEQ ID NO: 3, is denoted as "E₁₀₃₅". L-carbamoylases E₁, the polypeptide sequence of which is selected from variants of SEQ ID NO: 3, are generally denoted as "E_{1035'}".

A variant of the polypeptide sequence of SEQ ID NO: 3 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 3.

The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 3 is not identical to SEQ ID NO: 3.

According to the invention, an L-carbamoylase E_{103V} has carbamoylase activity and L-carbamoylase activity, determined as described under items 4.5.4.2 and 4.5.4.3.1.

According to the invention, an L-carbamoylase E_{103 v} preferably has carbamoylase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least

60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the carbamoylase activity of the L-carbamoylase E_{103S}, wherein the carbamoylase activity of E_{103P}, relative to the carbamoylase activity of E_{103S}, is determined by Assay C described under item 4.5.5.2.

It is even more preferable according to the invention, that an L-carbamoylase E_{103V} has carbamoylase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the carbamoylase activity of the L-carbamoylase E_{103S} , wherein the carbamoylase activity of E_{103V} relative to the carbamoylase activity of E_{103S} is determined by Assay C described under item 4.5.5.2.

4.5.6.4 Preferred Variants of SEO ID NO: 4

According to the invention, the polypeptide sequence of the L-carbamoylase E₁ may also be a variant of SEQ ID NO: 4

The L-carbamoylase E₁, the polypeptide sequence of which is SEQ ID NO: 4, is denoted as “E₁₀₄₅”. L-carbamoylases E₁, the polypeptide sequence of which is selected from variants of SEQ ID NO: 4, are generally denoted as “E_{1045P}”.

A variant of the polypeptide sequence of SEQ ID NO: 4 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 4.

The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 4 is not identical to SEQ ID NO: 4.

According to the invention, an L-carbamoylase E_{104V} has carbamoylase activity and L-carbamoylase activity, determined as described under items 4.5.4.2 and 4.5.4.3.1.

According to the invention, an L-carbamoylase E_{104P} preferably has carbamoylase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the carbamoylase activity of the L-carbamoylase E_{104S}, wherein the carbamoylase activity of E_{104P}, relative to the carbamoylase activity of E_{104S}, is determined by Assay C described under item 4.5.5.2.

It is even more preferable according to the invention, that an L-carbamoylase E_{104V} has carbamoylase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the carbamoylase activity of the L-carbamoylase E_{104S} wherein the carbamoylase

According to the invention, an L-carbamoylase E_{107V} preferably has carbamoylase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the carbamoylase activity of the L-carbamoylase E_{107S}, wherein

the carbamoylase activity of E_{107V} relative to the carbamoylase activity of E_{107S} is determined by Assay C described under item 4.5.5.2.

It is even more preferable according to the invention, that an L-carbamoylase E_{107V} has carbamoylase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the carbamoylase activity of the L-carbamoylase E_{107S} , wherein the carbamoylase activity of E_{107V} relative to the carbamoylase activity of E_{107S} is determined by Assay C described under item 4.5.5.2.

4.5.6.8 Preferred Variants of SEQ ID NO: 8

According to the invention, the polypeptide sequence of the L-carbamoylase E_1 may also be a variant of SEQ ID NO: 8.

The L-carbamoylase E_1 , the polypeptide sequence of which is SEQ ID NO: 8, is denoted as " E_{108S} ". L-carbamoylases E_1 , the polypeptide sequence of which is selected from variants of SEQ ID NO: 8, are generally denoted as " E_{108V} ".

A variant of the polypeptide sequence of SEQ ID NO: 8 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 8.

The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 8 is not identical to SEQ ID NO: 8.

According to the invention, an L-carbamoylase E_{108V} has carbamoylase activity and L-carbamoylase activity, determined as described under items 4.5.4.2 and 4.5.4.3.1.

According to the invention, an L-carbamoylase E_{108V} preferably has carbamoylase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the carbamoylase activity of the L-carbamoylase E_{108S} , wherein the carbamoylase activity of E_{108V} relative to the carbamoylase activity of E_{108S} is determined by Assay C described under item 4.5.5.2.

It is even more preferable according to the invention, that an L-carbamoylase E_{108V} has carbamoylase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the carbamoylase activity of the L-carbamoylase E_{108S} , wherein the carbamoylase activity of E_{108V} relative to the carbamoylase activity of E_{108S} is determined by Assay C described under item 4.5.5.2.

4.5.6.9 Preferred Variants of SEQ ID NO: 9

According to the invention, the polypeptide sequence of the L-carbamoylase E_1 may also be a variant of SEQ ID NO: 9.

The L-carbamoylase E_1 , the polypeptide sequence of which is SEQ ID NO: 9, is denoted as " E_{109S} ". L-carbamoylases E_1 , the polypeptide sequence of which is selected from variants of SEQ ID NO: 9, are generally denoted as " E_{109V} ".

A variant of the polypeptide sequence of SEQ ID NO: 9 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 9.

The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 9 is not identical to SEQ ID NO: 9.

According to the invention, an L-carbamoylase E_{109V} has carbamoylase activity and L-carbamoylase activity, determined as described under items 4.5.4.2 and 4.5.4.3.1.

According to the invention, an L-carbamoylase E_{109V} preferably has carbamoylase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the carbamoylase activity of the L-carbamoylase E_{109S} , wherein the carbamoylase activity of E_{109V} relative to the carbamoylase activity of E_{109S} is determined by Assay C described under item 4.5.5.2.

It is even more preferable according to the invention, that an L-carbamoylase E_{109V} has carbamoylase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the carbamoylase activity of the L-carbamoylase E_{109S} , wherein the carbamoylase activity of E_{109V} relative to the carbamoylase activity of E_{109S} is determined by Assay C described under item 4.5.5.2.

4.5.7 Preferred Method Conditions in Step (c)

The reaction in step (c) of the method according to the first aspect of the present invention may be carried out under conditions known to the skilled person.

The reaction medium is preferably aqueous, more preferably an aqueous buffer.

Exemplary buffers commonly used in biotransformation reactions and advantageously used herein include Tris, phosphate, or any of Good's buffers, such as 2-(N-morpholino)ethanesulfonic acid ("MES"), N-(2-acetamido)iminodiacetic acid ("ADA"), piperazine-N,N'-bis(2-ethanesulfonic acid) ("PIPES"), N-(2-acetamido)-2-aminoethanesulfonic acid ("ACES"), P-hydroxy-4-morpholinepropanesulfonic acid ("MOPSO"), choline chloride, 3-(N-morpholino)propanesulfonic acid ("MOPS"), N,N-Bis(2-hydroxyethyl)-2-aminoethanesulfonic acid ("BES"), 2-[[1,3-dihydroxy-2-(hydroxymethyl)propan-2-yl]amino]ethanesulfonic acid

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("TES"), 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid ("HEPES"), 3-(Bis(2-hydroxyethyl)amino)-2-hydroxypropane-1-sulfonic acid ("DIPSO"), acetamidoglycine, 3-(N-Tris(hydroxymethyl)methylamino)-(2-hydroxypropane) sulfonic acid ("TAPSO"), piperazine-N,N'-bis(2-hydroxypropanesulfonic acid) ("POPSO"), 4-(2-Hydroxyethyl) piperazine-1-(2-hydroxypropanesulfonic acid) ("HEPPSO"), 3-[4-(2-Hydroxyethyl)-1-piperazinyl]propanesulfonic acid ("HEPPS"), tricine, glycineamide, bicine, or 3-[[1,3-dihydroxy-2-(hydroxymethyl) propan-2-yl]amino]propane-1-sulfonic acid ("TAPS").

In some embodiments, ammonium can act as a buffer. One or more organic solvents can also be added to the reaction.

The buffer preferably contains metal salts, more preferably metal salts such as halogenides of metals, preferably halogenides of monovalent or bivalent or trivalent metals, preferably chlorides of monovalent or bivalent metals, preferably CoCl_2 or MnCl_2 , preferably CoCl_2 .

The concentration of these metal salts in the reaction medium is preferably in the range from 1 μM to 1 M, more preferably 1 mM to 100 mM, even more preferably 1 to 10 mM.

Preferably, step (c) of the method according to the first aspect of the invention is carried out in a phosphate buffer.

The pH of the reaction medium in step (c) of the method according to the first aspect of the invention is preferably in the range of from 2 to 10, more preferably in the range of from 5 to 8, more preferably 7.2 to 7.5, most preferably 7.5.

Preferably, step (c) of the method according to the first aspect of the invention is carried out at a temperature in the range of from 20° C. to 70° C., more preferably in the range of from 30° C. to 55° C., most preferably 50° C.

Preferably, the total concentration of all carbamoylases E_1 in the reaction solution in step (c) is in the range of from 1 μM to 10 mM, preferably 10 μM to 1 mM, more preferably 0.1 mM to 0.5 mM, most preferably 0.4 mM.

In alternative preferred embodiments, the total concentration of all carbamoylases E_1 in the reaction solution in step (c) is in the range of from 1 $\mu\text{g/l}$ to 10 g/l, preferably 0.1 mg/l to 5 g/l, more preferably 1 mg/l to 1 g/l, more preferably 5 mg/l to 500 mg/l.

Preferably, the initial concentration of all the compounds according to formula L-(II) in the reaction medium in step (c) is in the range of from 1 μM to 1 M, preferably of from 10 μM to 0.5 M, more preferably of from 0.1 mM to 0.1 M, more preferably of from 1 mM to 10 mM, most preferably 1.25 mM.

If compounds according to formula D-(II) are present in the reaction medium in step (c), the initial concentration of all the compounds according to formula D-(II) in the reaction medium is preferably from 1% to 100% the concentration of all the compounds according to formula L-(II), more preferably 10% to 100% the concentration, even more preferably 50 to 100%, even more preferably 100% the concentration of all the compounds according to formula L-(II).

"Initial concentration of all the compounds according to formula L-(II)/D-(II)" refers to the concentration of the respective compound L-(II) or D-(II) respectively, in the reaction medium when the respective compounds are employed in step (c).

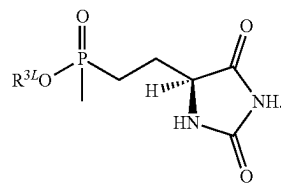
4.5.8 Step (b)

4.5.8.1 Enzymatic Catalysis of Step (b)

In a preferred embodiment of the method according to the first aspect of the invention, the compound according to

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formula L-(II) is obtained by a step (b) in which a compound according to formula L-(III) is reacted to give a compound according to formula L-(II):



L-(III)

Step (b) gives the starting material for step (c).

R^{3L} is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (VIII), (IX), more preferably R^{3L} has formula (VIII).

When R^{3L} is a radical of the formula (VIII), the radical R^{3L} may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

When R^{3L} is a radical of the formula (IX), the radical R^{3L} may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

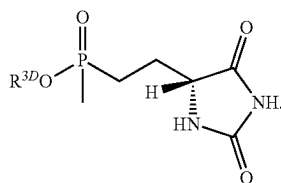
When R^{3L} is a radical of the formula (X), the radical R^{3L} may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

The reaction according step (b) is enzymatically catalyzed, namely it is catalyzed by a hydantoinase E_2 .

4.5.8.2 Enantioselective and Enantiospecific Catalysis of Step (b)

Step (b) of the method according to the first aspect of the present invention is preferably L-enantioselective, even more preferably L-enantiospecific.

In such a preferred embodiment, L-(III) is in particular employed in step (b) as a mixture M_{III} comprising at least one compound L-(III) and at least one compound D-(III), wherein D-(III) has the following formula:



D-(III)

R^{3D} is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (VIII), (IX), more preferably R^{3D} has formula (VIII).

When R^{3D} is a radical of the formula (VIII), the radical R^{3D} may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

When R^{3D} is a radical of the formula (IX), the radical R^{3D} may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

When R^{3D} is a radical of the formula (X), the radical R^{3D} may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

Mixture M_{III} preferably comprises at least one compound L-(III) and at least one compound D-(III), wherein $R^{3L}=R^{3D}$.

For such mixtures M_{III} , it may be observed that the at least one compound D-(II) undergoes a reaction according to step (b)*, i.e. parallel to step (b).

Namely, in the reaction according to step (b)*, the at least one compound D-(III) in mixture M_{II} is reacted to give a compound according to formula D-(II), wherein compounds according to formula D-(II) are explained above (in particular under item 4.5.1.2).

In case that step (b) is "L-enantioselective", this means that in case a mixture M_{II} is employed in step (b), then there is

either no reaction according to step (b)* of any compound according to formula D-(III) to a compound according to D-(II) in the mixture M_{II} or,

in case there is a reaction according to step (b)*, then there is at least one pair of compounds D-(III)(II)* / L-(III)(II)* in the mixture M_{II} , for which the reaction according to step (b)* of D-(III)(II)* to give D-(II)(II)*, is lesser than the rate of the reaction according to step (b) of L-(III)(II)* to give L-(II)(II)*.

D-(III)(II)* is a compound according to formula D-(III), and L-(III)(II)* is a compound according to formula L-(III), wherein the radical R^{3L} in L-(III)(II)* and the radical R^{3D} in D-(III)+ are identical.

D-(II)(II)* is a compound according to formula D-(II), and L-(II)(II)* is a compound according to formula L-(II), wherein the radical R^{2L} in L-(II)(II)* and the radical R^{2D} in D-(II)+ are identical.

Preferably, the radical R^{2L} in L-(II)+ and the radical R^{3L} in L-(III)+ are identical. Preferably, the radical R^{2D} in D-(II)(II)* and the radical R^{3D} in D-(III)+ are identical.

Step (b) is "L-enantiospecific", if the rate of reaction according to step (b)* is essentially zero, i.e. there is no reaction according to step (b)* of any compound according to formula D-(III) to a compound according to D-(II) in the mixture M_{III} .

In a preferred embodiment, the molar ratio of all compounds according to formula L-(III) in the mixture M_{III} to all compounds according to formula D-(III) in the mixture M_{III} is essentially 1:1.

In other preferred embodiments, the molar ratio of all compounds according to formula L-(III) in the mixture M_{III} to all compounds according to formula D-(III) in the mixture M_{III} is in the range of from 3:2 to 1:99, more preferably in the range of from 1.01:1 to 1:99, more preferably in the range of from 1:1 to 1:99, more preferably in the range of from 1:1.01 to 1:99, more preferably in the range of from 1:1.01 to 1:9, more preferably in the range of from 1:1.01 to 1:8, more preferably in the range of from 1:1.01 to 1:3.

Alternatively, D-(III) is comprised in an excess to L-(III) in mixture M_{III} , meaning that, while at least one compound according to formula L-(III) is in the mixture M_{III} , the molar ratio of all compounds according to formula L-(III) is in the mixture M_{III} to all compounds according to formula D-(III) in the mixture M_{III} is <1:1, preferably <0.9:1, more preferably <0.75:1, more preferably <0.5:1, more preferably <0.2:1, more preferably <0.1:1, more preferably <0.01:1.

Step (b) is in particular L-enantioselective, if it is preferably catalyzed by an L-hydantoinase E_2 , which may be determined by the skilled person as set forth under 4.5.10.3.

In case step (b) is L-enantioselective with respect to a given pair of compounds D-(III)(II)* / L-(III)+, the reaction according to step (b) of L-(III)(II)* to give L-(II)(II)* proceeds preferably at a reaction rate that is at least 2 times greater, preferably at least 10 times greater, more preferably at least 100 times greater, even more preferably at least 103 times greater, even more preferably at least 104 times greater, even more preferably at least 105 times greater than the reaction rate at which step (b)* of D-(III)(II)* to give D-(II)(II)* proceeds.

To quantify for a given pair of compounds D-(III)(II)* / L-(III)(II)* the factor at which the reaction rate of step (b) proceeds compared to the reaction rate of step (b)*, the following test may be carried out:

(1) An equimolar mixture [molar ratio of the two compounds L-(III)(II)* and D-(III)(II)* is 1:1] is subjected to the respective reactions conditions and the development of the two products L-(I)(II)* and D-(I)(II)* is monitored over time (e.g. by LC-MS as set forth under item 5.4).

(2) When $n_{LII10}=10$ mol-% of the initially employed L-(III)(II)* has reacted to the product L-(II)(II)*, the molar amount of D-(II)(II)* that was formed by reaction from D-(III)+ [in mol-% relative to the initially employed D-(II)(II)*] is measured ($=n_{DII10}$).

(3) The ratio of $n_{DII10}/n_{LII10}=10/n_{DII10}$ gives the factor by which the reaction rate of step (b) proceeds compared to the reaction rate of step (b)* for the given pair of compounds D-(III)(II)* / L-(III)(II)*.

If $n_{LII10}/n_{DII10}>1$, the reaction rate of step (b) is greater than the reaction rate of step (b)* for the given pair of compounds D-(III)(II)* / L-(III)(II)*.

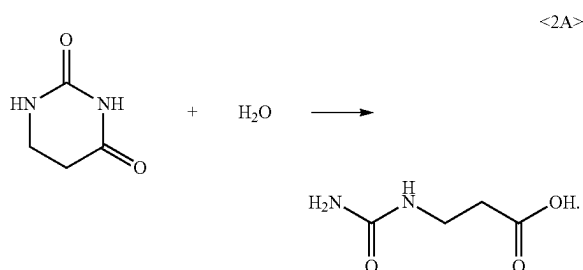
If $n_{LII10}/n_{DII10}<1$, the reaction rate of step (b) is lesser than the reaction rate of step (b)* for the given pair of compounds D-(III)(II)* / L-(III)(II)*.

If $n_{LII10}/n_{DII10}=1$, the reaction rates of steps (b) and (b)* are the same for the given pair of compounds D-(III)(II)* / L-(III)(II)*.

The reaction according to step (b) of the preferred embodiment of the first aspect of the invention is catalyzed by a hydantoinase ("dihydropyrimidinase") E_2 .

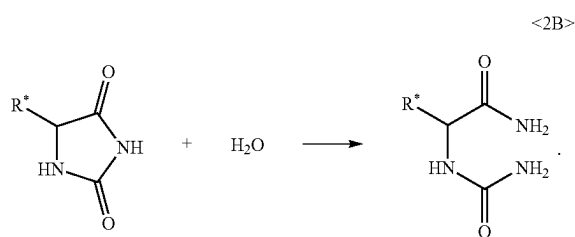
Namely, it was surprisingly found that hydantoinases accept compounds of formula L-(III) as substrates and convert them to products according to formulae L-(II), and hence catalyze the reaction according to step (b). This finding is of high scientific and economic value, as it further broadens the scope of synthetic routes based on new starting materials for the production of L-glufosinate P-esters and L-glufosinate. Even more surprisingly, it is suggested that L-glufosinate hydantoin, i.e. the compound according to formula L-(III), in which $R=H$, does not undergo reaction by hydantoinases to give the respective LGA carbamoylate.

In nature, hydantoinases ("dihydropyrimidinases") generally catalyze the reaction of 5,6-dihydrouracil to produce ureidopropionate (see the following reaction <2A>):



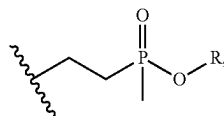
They also catalyze the ring opening of monosubstituted hydantoins to give the respective carbamoyl amino acid according to the following reaction <2B>, wherein R^* may be an organic residue, e.g. a side chain of one of the naturally occurring amino acids.

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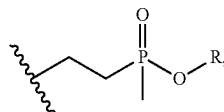
Chapter 1.3 (pages 7 to 11) of the dissertation “Untersuchungen zur Substratspezifität und Enantioselektivität mikrobieller Hydantoinasen/Investigations of substrate specificity and enantioselectivity of microbial hydantoinases” by T. Waniek, University of Stuttgart, 2000 (available under: <https://elib.uni-stuttgart.de/bitstream/11682/1511/1/Dis-s.pdf>) gives an overview over hydantoinases.

It was now surprisingly found that hydantoinases also accept substrates in which $R^*=R^Y$



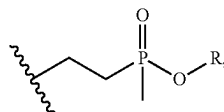
wherein $R=R^{3L}$.

Surprisingly, they supposedly do not accept substrates in which $R^*=R^Y$

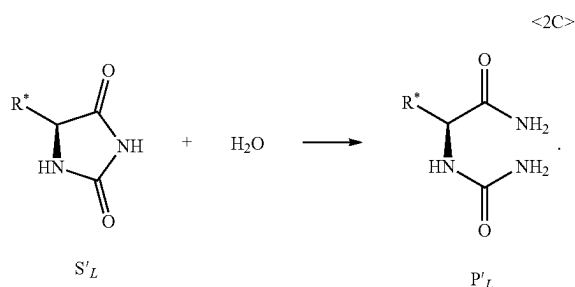


wherein $R=H$.

In the context of the present invention, a “hydantoinase E_2 ” is a hydantoinase that catalyzes the following reaction <2C> of a carbamoyl substrate S'_L to the respective amino acid product P'_L , wherein $R^*=R^Y$

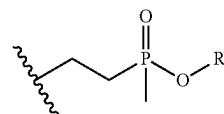


wherein $R=R^{3L}$:

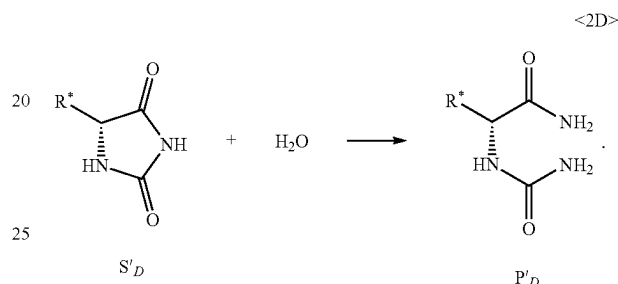


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In particular, the hydantoinase E_2 is a “L-hydantoinase”, i.e. it has a greater catalytic activity for reaction <2C> than for reaction <2D>, wherein R^* in the substrate S'_D in the reaction <2D> and R^* in the substrate S'_L in the reaction <2C> are identical and $R^*=R^Y$



wherein $R=R^{3L}$.



As an L-hydantoinase has a higher catalytic activity for reaction <2C> than for reaction <2D>, it is “L-enantioselective”. An L-hydantoinase that has no catalytic activity for reaction <2D> and thus only has catalytic activity for reaction <2C> is “L-enantiospecific”.

A “D-hydantoinase” is defined as a hydantoinase which is “D-enantioselective”, i.e. it has a higher catalytic activity for reaction <2D> than for reaction <2C>. A D-carbamoylase that does not catalyze reaction <2C> and thus only has catalytic activity for reaction <2D> is “D-enantiospecific”.

A hydantoinase which has the same catalytic activity for reaction <2C> as for reaction <2D>, is referred to as “non enantioselective hydantoinase”.

For determination whether a hydantoinase may be denoted as “L-hydantoinase”, “D-hydantoinase” or “non-enantioselective hydantoinase” in the context of the present invention, the procedure according to Assay E (item 4.5.10) may preferably be used.

The hydantoinase E_2 , in particular the L-hydantoinase E_2 , that may be used in step (b) of the preferred embodiment of the method according to the first aspect of the invention may originate from *Arthrobacter* sp., in particular *Arthrobacter crystallopoietes*, *Arthrobacter aureus*, *Arthrobacter* sp. BT801; *Alcaligenes* sp., in particular *Alcaligenes faecalis* subsp. *faecalis*; *Bacillus* sp., in particular *Bacillus fordii*; *Microbacterium* sp., in particular *Microbacterium liquefaciens* strain AJ3912; *Pseudomonas* sp., in particular *Pseudomonas fluorescens*, *Pseudomonas aeruginosa*.

A hydantoinase E_2 , in particular an L-hydantoinase E_2 suitable for the method according to the present invention may be the enzyme HyuH, which originates from *Arthrobacter*. Another enzyme may be Dht.

Even more preferably, the hydantoinase E_2 , in particular the L-hydantoinase E_2 , that may preferably be used in step (b) according to the first aspect of the invention may originate from *Arthrobacter* sp., in particular *Arthrobacter crystallopoietes*, *Arthrobacter aureus*, *Arthrobacter* sp.

BT801, *Arthrobacter aureescens* DSM 9771, most preferably from *Arthrobacter aureescens* DSM 9771.

A hydantoinase suitable for the method according to the present invention is described e.g. in WO 01/23582 A1 and by J. M. Clemente-Jiménez, S. Martínez-Rodríguez, F. Rodríguez-Vico, F. J. L. Heras-Vázquez, Recent Pat. Biotechnology 2008, 2, 35-46; G. Latacz, E. Pekala, K. Kiec-Kononowicz, Biotechnologia 2006, 2, 189-205.

Further suitable hydantoinases are described by K. Yokozeki, H. Yoshiteru, K. Kubota, Agric. Biol. Chem. 1987, 51, 737-746.

The hydantoinase E₂ that may be used in preferred step (b) of the method according to the present invention may be a hydantoinase categorized in the EC class EC 3.5.2.2.

The following table 2 gives preferred examples for polypeptide sequences of hydantoinases E₂ that may be preferably used in step (b) of the preferred embodiment of the method according to the first aspect of the invention. The genes encoding the respective hydantoinase E₂ and the respective accession code are indicated as far as known.

TABLE 2

Hydantoinases (EC 3.5.2.2)			
Strain	Gene name	GenBank/UniProt accession	SEQ ID NO: of the polypeptide
<i>Arthrobacter aureescens</i> DSM 9771	hyuH		SEQ ID NO: 10
<i>Pseudomonas fluorescens</i>	not assigned	S5MPT0	SEQ ID NO: 11
<i>Pseudomonas aeruginosa</i>	dht	Q91676	SEQ ID NO: 12
<i>Bacillus fordii</i> MH602	not assigned	ABL14245	SEQ ID NO: 13
<i>Arthrobacter</i> sp. BT801	hyuH	AAL55412	SEQ ID NO: 14
<i>Alcaligenes faecalis</i> subsp. <i>faecalis</i>	not assigned		SEQ ID NO: 15
<i>Microbacterium liquefaciens</i> strain AJ3912	not assigned		SEQ ID NO: 16

In a preferred embodiment of the method according to the first aspect of the present invention, the reaction according to step (b) is catalyzed by an hydantoinase E₂, wherein the polypeptide sequence of E₂ is selected from the group consisting of SEQ ID NO: 10 and variants thereof, SEQ ID NO: 11 and variants thereof, SEQ ID NO: 12 and variants thereof, SEQ ID NO: 13 and variants thereof, SEQ ID NO: 14 and variants thereof, SEQ ID NO: 15 and variants thereof, SEQ ID NO: 16 and variants thereof, preferably SEQ ID NO: 10 and variants thereof.

4.5.9 Assays D_L and D_D for Determining Hydantoinase Activity

The skilled person is aware of hydantoinases, in particular L-hydantoinases, that may be used in step (b) of the preferred method according to the first aspect of the invention.

In particular, Assay D_L, described in the following, may be used to determine hydantoinase and L-hydantoinase activity of a given enzyme E_γ and may advantageously be used according to the invention to determine hydantoinase and L-hydantoinase activity in variants of SEQ ID NO: 10, variants of SEQ ID NO: 11, variants of SEQ ID NO: 12, variants of SEQ ID NO: 13, variants of SEQ ID NO: 14, variants of SEQ ID NO: 15, variants of SEQ ID NO: 16.

For comparative reasons, Assay D_D may be used to determine D-hydantoinase activity of a given enzyme E_γ.

For the purpose of Assay D_L and Assay D_D, the molar mass of the enzyme E_γ to be tested is calculated as the molar mass of the polypeptide sequence of E_γ.

Moreover, a substrate according to formula L-(III), wherein R^{3L}=n-butyl, is used in Assay D_L so that there is only one chiral carbon atom in the substrate. The compound according to formula L-(III), wherein R^{3L}=n-butyl, is denoted as "L-(III)+".

A product according to formula L-(II), wherein R^{2L}=n-butyl, is detected in Assay A_L.

The compound according to formula L-(II), wherein R^{2L}=n-butyl, is denoted as "L-(II)+".

Moreover, a substrate according to formula D-(III), wherein R^{3D}=n-butyl, is used in Assay A_D so that there is only one chiral carbon atom in the substrate. The compound according to formula D-(III), wherein R^{3D}=n-butyl, is denoted as "D-(III)+".

A product according to formula D-(II), wherein R^{2D}=n-butyl, is detected in Assay A_D.

The compound according to formula D-(II), wherein R^{2D}=n-butyl, is denoted as "D-(II)+".

4.5.9.1 Assay D_L:

To 0.9 ml of an aqueous reaction solution (phosphate buffer, pH 7.2, 10 mM MnCl₂), containing 50 mM of a

compound according to formula L-(III)(II)* are added 400 nmol of E_γ in 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM MnCl₂). The resulting solution is incubated at 25° C., and the pH is held at pH 7.2 by addition of 1 M NaOH. After 300 minutes, the reaction is stopped by addition of 2 M HCl to achieve a pH of 2.5, and the molar amount of the compound according to formula L-(II)(II)* is determined. L-(II)(II)* may be detected by the LC-MS method as described in the example section (item 5.4) for detection of LGA.

4.5.9.2 Assay D_D:

To 0.9 ml of an aqueous reaction solution (phosphate buffer, pH 7.2, 10 mM MnCl₂), containing 50 mM of a compound according to formula D-(III)(II)* are added 400 nmol of E_γ in 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM MnCl₂). The resulting solution is incubated at 25° C., and the pH is held at pH 7.2 by addition of 1 M NaOH. After 300 minutes, the reaction is stopped by addition of 2 M HCl to achieve a pH of 2.5, and the molar amount of the compound according to formula D-(II)(II)* is determined. D-(II)(II)* may be detected by the LC-MS method as described in the example section (item 5.4) for detection of LGA.

4.5.10 Assay E for Identifying Hydantoinases, L-Hydantoinases, D-Hydantoinases, L- and D-Enantiospecificity

The hydantoinase E₂ according to the invention is preferably an L-hydantoinase, more preferably L-enantiospecific.

Whether a given enzyme E_Y may be considered a hydantoinase E_2 , in particular an L-hydantoinase, may be determined in the context of the present invention by the following Assay E:

4.5.10.1 Assay E:

E-1. Firstly, Assay D_L as set forth under item 4.5.9.1 is conducted, and the obtained molar amount of the compound of the formula L-(II)(III)* is determined according to Assay D_L .

E-2 Secondly, Assay D_D as set forth under item 4.5.9.2 is conducted, and the obtained molar amount of the compound of the formula D-(II)(III)* is determined according to Assay D_D .

E-3. Then, step E-1 is repeated, except that instead of the addition of 400 nmol E_Y in 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM $MnCl_2$), 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM $MnCl_2$) without E_Y is added.

E-4 Then, step E-2 is repeated, except that instead of the addition of 400 nmol E_Y in 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM $MnCl_2$), 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM $MnCl_2$) without E_Y is added.

4.5.10.2 Hydantoinase Activity:

If the molar amount of the compound of the formula L-(II)(III)* that is determined in step E-1, is greater than the molar amount of the compound of formula L-(II)(III)* that is determined in step E-3, then E_Y is deemed to have hydantoinase activity, and hence may be considered a hydantoinase E_2 in the context of the invention.

4.5.10.3 L-Hydantoinase Activity

4.5.10.3.1 L-Hydantoinases

- (i) If the molar amount of the compound of the formula L-(II)(III)* that is determined in step E-1, is greater than the molar amount of the compound of the formula L-(II)(III)* that is determined in step E-3, and
- (ii) if the molar amount of the compound of the formula D-(II)(III)* that is determined in step E-2, is greater than or the same as the molar amount of the compound of the formula D-(II)(III)* that is determined in step E-4, and
- (iii) if, in addition, the molar amount of the compound of the formula L-(II)(III)* that is determined in step E-1 is greater than the molar amount of the compound of the formula D-(II)(III)* that is determined in step E-2, then E_Y is deemed to have L-hydantoinase activity, and hence may be considered an L-hydantoinase in the context of the invention. In this case, E_Y is deemed to be "L-enantioselective" in the context of the invention.

For the sake of clarity, it is pointed out that condition (iii) is automatically fulfilled in those cases in which the molar amount of the compound of formula D-(II)(III)* that is determined in step E-2 is the same as the molar amount of the compound of formula D-(II)(III)* that is determined in step E-4.

- (i) If the molar amount of the compound of formula L-(II)(III)* that is determined in step E-1, is greater than the molar amount of the compound of formula L-(II)(III)* that is determined in step E-3, and
- (ii) if the molar amount of the compound of the formula D-(II)(III)* that is determined in step E-2, is the same as the molar amount of the compound of the formula D-(II)(III)* that is determined in step E-4, then E_Y is deemed to have L-hydantoinase activity, and hence may be considered a L-hydantoinase in the context of the invention. In this case, E_Y is deemed to be not only "L-enantioselective", but also "L-enantiospecific" in the context of the invention.

For L-hydantoinases E_Y that are not L-enantiospecific, the L-enantioselectivity may then be quantified by dividing the molar amount of the compound of formula L-(II)(III)* that is determined in E-1, by the molar amount of the compound of formula D-(II)(III)* that is determined in E-2, and then multiplying the obtained value by 100, giving the L-enantioselectivity of E_Y in %.

4.5.10.3.2 D-Carbamoylases

- (i) If the molar amount of the compound of formula L-(II)(III)* that is determined in step E-1, is greater than or the same as the molar amount of the compound of formula L-(II)(III)* that is determined in step E-3, and
- (ii) if the molar amount of the compound of formula D-(II)(III)* that is determined in step E-2 is greater than the molar amount of the compound of formula D-(II)(III)* that is determined in step E-4, and
- (iii) if, in addition, the molar amount of the compound of formula D-(II)(III)* that is determined in step E-2, is greater than the molar amount of the compound of formula L-(II)(III)* that is determined in step E-1, then E_Y is deemed to have D-hydantoinase activity, and hence may be considered a D-hydantoinase. In this case, E_Y is "D-enantioselective" in the context of the invention.

For the sake of clarity it is pointed out that condition (iii) is automatically fulfilled in those cases in which the molar amount of the compound of formula L-(II)(III)* that is determined in step E-1 is the same as the molar amount of the compound of formula L-(II)(III)* that is determined in step E-3.

- (i) If the molar amount of the compound of formula L-(II)(III)* that is determined in step E-1, is the same as the molar amount of the compound of formula L-(II)(III)* that is determined in step E-3, and
- (ii) if the molar amount of the compound of formula D-(II)(III)* that is determined in step E-2, is greater than the molar amount of the compound of formula D-(II)(III)* that is determined in step E-4, then E_Y is deemed to have D-hydantoinase activity, and hence may be considered a D-hydantoinase in the context of the invention. In this case, E_Y is deemed to be not only "D-enantioselective", but also "D-enantiospecific" in the context of the invention.

For D-hydantoinases E_Y that are not D-enantiospecific, the D-enantioselectivity may then be quantified by dividing the molar amount of the compound of formula D-(II)(III)* that is determined in step E-2 by the molar amount of the compound of formula L-(II)(III)* that is determined in step E-1 and then multiplying the obtained value by 100, giving the D-enantioselectivity of E_Y in %.

4.5.10.3.3 Non-Enantioselective Carbamoylases

- (i) If the molar amount of the compound of formula L-(II)(III)* that is determined in step E-1, is greater than the molar amount of the compound of formula L-(II)(III)* that is determined in step E-3, and
- (ii) if the molar amount of the compound of formula D-(II)(III)* that is determined in step E-2, is greater than the molar amount of the compound of formula D-(II)(III)* that is determined in step E-4, and
- (iii) if, in addition, the molar amount of the compound of formula D-(II)(III)* that is determined in step E-2 is the same as the molar amount of the compound of formula L-(II)(III)* that is determined in step E-1, then E_Y is deemed to be a non-enantioselective hydantoinase in the context of the invention. In this case, E_Y is "non-enantioselective".

4.5.11 Assay F for Identifying Preferred Hydantoinases Variants of SEQ ID NOs: 10-16

4.5.11.1 L- and D-Enantioselective Hydantoinase Variants

An enzyme, the polypeptide sequence of which is selected from the group consisting of SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16 has hydantoinase, in particular L-hydantoinase activity.

In a preferred embodiment of the method according to the first aspect of the invention, the polypeptide sequence of the hydantoinase, in particular the L-hydantoinase E_2 is selected from the group consisting SEQ ID NO: 10 and variants thereof, SEQ ID NO: 11 and variants thereof, SEQ ID NO: 12 and variants thereof, SEQ ID NO: 13 and variants thereof, SEQ ID NO: 14, and variants thereof, SEQ ID NO: 15 and variants thereof, SEQ ID NO: 16 and variants thereof, more preferably SEQ ID NO: 10 and variants thereof.

The term "variant" is defined under item 4.3.

In the context of the invention, an enzyme, the polypeptide sequence of which is a variant of one of SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, has hydantoinase activity, preferably L-hydantoinase activity, more preferably is L-enantiospecific.

Whether a given enzyme E_2 , the polypeptide sequence of which is a variant of one of SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, has hydantoinase activity, L-hydantoinase activity and is L-enantiospecific may be determined as set forth under items 4.5.10.2 and 4.5.10.3.1, respectively.

The hydantoinase activity of a given hydantoinase E_{2P} , the polypeptide sequence of which is a variant of one of SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, relative to the hydantoinase activity of an hydantoinase E_{2S} , wherein the polypeptide sequence of E_{2S} is selected from SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, may be determined in the context of the present invention by the following Assay F:

4.5.11.2 Assay F

F-1 Assay D_L as set forth under item 4.5.9.1 is conducted, wherein E_{2S} is the enzyme to be tested. The obtained molar amount of the compound according to formula L-(II)(III)* is determined according to Assay D_L .

F-2 Step F-1 is repeated, except that, instead of E_{2S} , E_{2P} is used as the enzyme to be tested.

F-3. Then, the molar amount of the compound according to formula L-(II)(III)* that is determined in step F-2, is divided by the molar amount of the compound according to formula L-(II)(III)* that is determined in step F-1, and the obtained ratio is multiplied by 100, giving the hydantoinase activity of hydantoinase E_{2P} relative to the hydantoinase activity of the hydantoinase E_{2S} , in %.

4.5.12 Preferred Hydantoinase Variants of SEQ ID NOs: 10-16

In the context of the invention, hydantoinases E_2 , the polypeptide sequence of which is selected from the group consisting of SEQ ID NO: 10, SEQ ID NO: 11; SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, are generally denoted as " E_{2S} ". Hydantoinases E_2 , the polypeptide sequence of which is selected from

variants of a sequence selected from the group consisting of SEQ ID NO: 10, SEQ ID NO: 11; SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, are generally denoted as " E_{2P} ".

In a preferred embodiment of the method according to the first aspect of the present invention, the reaction in step (b) is catalyzed by a hydantoinase E_2 , preferably an L-hydantoinase E_2 , and the polypeptide sequence of the hydantoinase E_2 , preferably the L-hydantoinase E_2 , is selected from the group consisting of SEQ ID NO: 10 and variants thereof, SEQ ID NO: 11 and variants thereof, SEQ ID NO: 12 and variants thereof, SEQ ID NO: 13 and variants thereof, SEQ ID NO: 14 and variants thereof, SEQ ID NO: 15 and variants thereof, SEQ ID NO: 16 and variants thereof. More preferably, the reaction in step (b) is catalyzed by a hydantoinase E_2 , preferably an L-hydantoinase E_2 , and the polypeptide sequence of the hydantoinase E_2 , preferably the L-hydantoinase E_2 , is selected from the group consisting of SEQ ID NO: 10 and variants of SEQ ID NO: 10.

4.5.12.1 Preferred Variants of SEQ ID NO: 10

According to the invention, the polypeptide sequence of the hydantoinase E_2 , preferably the polypeptide sequence of the L-hydantoinase E_2 , may also be a variant of SEQ ID NO: 10.

The hydantoinase E_2 , the polypeptide sequence of which is SEQ ID NO: 10, is denoted as " E_{210S} ". The hydantoinase E_2 , the polypeptide sequence of which is selected from variants of SEQ ID NO: 10, are generally denoted as " E_{210P} ".

A variant of the polypeptide sequence of SEQ ID NO: 10 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 10.

The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 10 is not identical to SEQ ID NO: 10.

According to the invention, a hydantoinase E_{210P} has hydantoinase activity and preferably L-hydantoinase activity, determined as described under items 4.5.10.2 and 4.5.10.3.1.

According to the invention, a hydantoinase E_{210P} preferably has hydantoinase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoinase activity of the hydantoinase E_{210S} , wherein the hydantoinase activity of E_{210P} relative to the hydantoinase activity of E_{210S} is determined by Assay F described under item 4.5.11.2.

It is even more preferably according to the invention, that a hydantoinase E_{210P} has hydantoinase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoinase activity of the hydantoinase E_{210S} , wherein the hydantoinase activity of

80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoinase activity of the hydantoinase E_{213S} , wherein the hydantoinase activity of E_{213P} , relative to the hydantoinase activity of E_{213S} is determined by Assay F described under item 4.5.11.2.

It is even more preferably according to the invention, that a hydantoinase E_{213P} has hydantoinase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoinase activity of the hydantoinase E_{213S} , wherein the hydantoinase activity of E_{213P} , relative to the hydantoinase activity E_{213S} is determined by Assay F described under item 4.5.11.2.

4.5.12.5 Preferred Variants of SEQ ID NO: 14

According to the invention, the polypeptide sequence of the hydantoinase E_2 , preferably the polypeptide sequence of the L-hydantoinase E_2 , may also be a variant of SEQ ID NO: 14.

The hydantoinase E_2 , the polypeptide sequence of which is SEQ ID NO: 14, is denoted as " E_{214S} ". The hydantoinase E_2 , the polypeptide sequence of which is selected from variants of SEQ ID NO: 14, are generally denoted as " E_{214P} ".

A variant of the polypeptide sequence of SEQ ID NO: 14 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 14.

The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 14 is not identical to SEQ ID NO: 14.

According to the invention, a hydantoinase E_{214P} has hydantoinase activity and preferably L-hydantoinase activity, determined as described under items 4.5.10.2 and 4.5.10.3.1.

According to the invention, a hydantoinase E_{214P} preferably has hydantoinase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoinase activity of the hydantoinase E_{214S} , wherein the hydantoinase activity of E_{214P} , relative to the hydantoinase activity of E_{214S} is determined by Assay F described under item 4.5.11.2.

It is even more preferably according to the invention, that a hydantoinase E_{214P} has hydantoinase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoinase activity of the hydantoinase E_{214S} , wherein the hydantoinase activity of

E_{214P} , relative to the hydantoinase activity E_{214S} is determined by Assay F described under item 4.5.11.2.

4.5.12.6 Preferred Variants of SEQ ID NO: 15

According to the invention, the polypeptide sequence of the hydantoinase E_2 , preferably the polypeptide sequence of the L-hydantoinase E_2 , may also be a variant of SEQ ID NO: 15.

The hydantoinase E_2 , the polypeptide sequence of which is SEQ ID NO: 15, is denoted as " E_{215S} ". The hydantoinase E_2 , the polypeptide sequence of which is selected from variants of SEQ ID NO: 15, are generally denoted as " E_{215P} ".

A variant of the polypeptide sequence of SEQ ID NO: 15 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 15.

The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 15 is not identical to SEQ ID NO: 15.

According to the invention, a hydantoinase E_{215P} has hydantoinase activity and preferably L-hydantoinase activity, determined as described under items 4.5.10.2 and 4.5.10.3.1.

According to the invention, a hydantoinase E_{215P} preferably has hydantoinase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoinase activity of the hydantoinase E_{215S} , wherein the hydantoinase activity of E_{215P} , relative to the hydantoinase activity of E_{215S} is determined by Assay F described under item 4.5.11.2.

It is even more preferably according to the invention, that a hydantoinase E_{215P} has hydantoinase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoinase activity of the hydantoinase E_{215S} , wherein the hydantoinase activity of E_{215P} , relative to the hydantoinase activity E_{215S} is determined by Assay F described under item 4.5.11.2.

4.5.12.7 Preferred Variants of SEQ ID NO: 16

According to the invention, the polypeptide sequence of the hydantoinase E_2 , preferably the polypeptide sequence of the L-hydantoinase E_2 , may also be a variant of SEQ ID NO: 16.

The hydantoinase E_2 , the polypeptide sequence of which is SEQ ID NO: 16, is denoted as " E_{216S} ". The hydantoinase E_2 , the polypeptide sequence of which is selected from variants of SEQ ID NO: 16, are generally denoted as " E_{216P} ".

A variant of the polypeptide sequence of SEQ ID NO: 16 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably

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≥92%, more preferably ≥93%, more preferably ≥94%, more preferably ≥95%, more preferably ≥96%, more preferably ≥97%, more preferably ≥98%, more preferably ≥99%, more preferably ≥99.9% sequence identity to polypeptide sequence SEQ ID NO: 16.

The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 16 is not identical to SEQ ID NO: 16.

According to the invention, a hydantoinase E_{216V} has hydantoinase activity and preferably L-hydantoinase activity, determined as described under items 4.5.10.2 and 4.5.10.3.1.

According to the invention, a hydantoinase E_{216V} preferably has hydantoinase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoinase activity of the hydantoinase E_{216S}, wherein the hydantoinase activity of E_{216V} relative to the hydantoinase activity of E_{216S} is determined by Assay F described under item 4.5.11.2.

It is even more preferably according to the invention, that a hydantoinase E_{216V} has hydantoinase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoinase activity of the hydantoinase E_{216S}, wherein the hydantoinase activity of E_{216V} relative to the hydantoinase activity E_{216S} is determined by Assay F described under item 4.5.11.2.

4.5.13 Preferred Method Conditions in Step (b)

The reaction in step (b) of the method according to the first aspect of the present invention may be carried out under conditions known to the skilled person.

The reaction medium is preferably aqueous, more preferably an aqueous buffer.

Exemplary buffers commonly used in biotransformation reactions and advantageously used herein include Tris, phosphate, or any of Good's buffers, such as 2-(N-morpholino)ethanesulfonic acid ("MES"), N-(2-acetamido)iminodiacetic acid ("ADA"), piperazine-N,N'-bis(2-ethanesulfonic acid) ("PIPES"), N-(2-acetamido)-2-aminoethanesulfonic acid ("ACES"), P-hydroxy-4-morpholinepropanesulfonic acid ("MOPSO"), choline chloride, 3-(N-morpholino)propanesulfonic acid ("MOPS"), N,N-Bis(2-hydroxyethyl)-2-aminoethanesulfonic acid ("BES"), 2-[[1,3-dihydroxy-2-(hydroxymethyl)propan-2-yl]amino]ethanesulfonic acid ("TES"), 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid ("HEPES"), 3-(Bis(2-hydroxyethyl)amino)-2-hydroxypropane-1-sulfonic acid ("DIPSO"), acetamidoglycine, 3-(N-Tris(hydroxymethyl)methylamino(-2-hydroxypropane) sulfonic acid ("TAPSO"), piperazine-N,N'-bis(2-hydroxypropanesulfonic acid) ("POPSO"), 4-(2-Hydroxyethyl) piperazine-1-(2-hydroxypropanesulfonic acid) ("HEPPSO"), 3-[4-(2-Hydroxyethyl)-1-piperazinyl]propanesulfonic acid ("HEPPS"), tricine, glycine, bicine, or 3-[[1,3-dihydroxy-2-(hydroxymethyl)propan-2-yl]amino]propane-1-sulfonic acid ("TAPS").

In some embodiments, ammonium can act as a buffer. One or more organic solvents can also be added to the reaction.

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The buffer preferably contains metal salts, more preferably metal salts such as halogenides of metals, preferably halogenides of monovalent or bivalent or trivalent metals, preferably chlorides of monovalent or bivalent metals, preferably CoCl₂ or MnCl₂, preferably CoCl₂.

The concentration of these metal salts in the reaction medium is preferably in the range from 1 μM to 1 M, more preferably 1 mM to 100 mM, even more preferably 1 to 10 mM.

Preferably, step (b) of the method according to the first aspect of the invention is carried out in a phosphate buffer.

The pH of the reaction medium in step (b) of the method according to the first aspect of the invention is preferably in the range of from 2 to 10, more preferably in the range of from 5 to 8, more preferably 7.2 to 7.5, most preferably 7.5.

Preferably, step (b) of the method according to the first aspect of the invention is carried out at a temperature in the range of from 20° C. to 70° C., more preferably in the range of from 30° C. to 55° C., most preferably 50° C.

Preferably, the total concentration of all hydantoinases E₂ in the reaction solution in step (b) is in the range of from 1 μM to 10 mM, preferably 10 μM to 1 mM, more preferably 0.1 mM to 0.5 mM, most preferably 0.4 mM.

In alternative preferred embodiments, the total concentration of all hydantoinases E₂ in the reaction solution in step (b) is in the range of from 1 μg/l to 10 g/l, preferably 0.1 mg/l to 5 g/l, more preferably 1 mg/l to 1 g/l, more preferably 5 mg/l to 500 mg/l.

Preferably, the initial concentration of all the compounds according to formula L-(III) in the reaction medium in step (b) is in the range of from 1 μM to 1 M, preferably of from 10 μM to 0.5 M, more preferably of from 0.1 mM to 0.1 M, more preferably of from 1 mM to 10 mM, most preferably 1.25 mM.

If compounds according to formula D-(III) are present in the reaction medium in step (b), the initial concentration of all the compounds according to formula D-(III) in the reaction medium is preferably from 1% to 100% the concentration of all the compounds according to formula L-(III), more preferably 10% to 100% the concentration, even more preferably 50 to 100%, even more preferably 100% the concentration of all the compounds according to formula L-(III).

Preferably, step (b) is carried out in the same reaction medium in which step (c) is carried out.

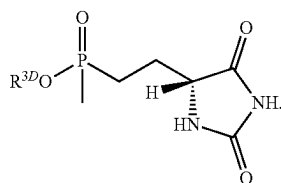
In this case, preferably, the initial concentration of all the compounds according to formula L-(III) in the reaction medium in step (b) is in the range of from 1 μM to 1 M, preferably of from 10 μM to 0.5 M, more preferably of from 0.1 mM to 0.1 M, more preferably of from 1 mM to 10 mM, most preferably 1.25 mM.

"Initial concentration of all the compounds according to formula L-(III)/D-(III)" refers to the concentration of the respective compound L-(III) or D-(III), respectively, in the reaction medium when the respective compounds are employed in step (b).

4.5.14 Step (a)

In a preferred embodiment of the method according to the first aspect of the invention, the compound according to formula L-(III) is obtained by a step (a) in which a compound according to formula D-(III) is reacted to give a compound according to formula L-(III):

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Step (a) gives the starting material for step (b).

R^{3D} is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (VIII), (IX), more preferably R^{3D} has formula (VIII).

When R^{3D} is a radical of the formula (VIII), the radical R^{3D} may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

When R^{3D} is a radical of the formula (IX), the radical R^{3D} may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

When R^{3D} is a radical of the formula (X), the radical R^{3D} may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

The reaction according step (a) may be carried out enzymatically or non-enzymatically, preferably enzymatically. More preferably, the reaction according step (a) is catalyzed by a hydantoin racemase E₃.

In preferred step (a), the compound according to formula D-(III) is employed in step (a) as a mixture M_{III} of at least one compound D-(III) with at least one compound L-(III), wherein R^{3L}=R^{3D}.

In a preferred embodiment, the molar ratio of all compounds according to formula L-(III) in the mixture M_{III} to all compounds according to formula D-(III) in the mixture M_{III} is essentially 1:1.

In other preferred embodiments, the molar ratio of all compounds according to formula L-(III) in the mixture M_{II} to all compounds according to formula D-(III) in the mixture M_{III} is in the range of from 3:2 to 1:99, more preferably in the range of from 1.01:1 to 1:99, more preferably in the range of from 1:1 to 1:99, more preferably in the range of from 1:1.01 to 1:99, more preferably in the range of from 1:1.01 to 1:9, more preferably in the range of from 1:1.01 to 1:8, more preferably in the range of from 1:1.01 to 1:3.

Alternatively, D-(III) is comprised in an excess to L-(III) in mixture M_m, meaning that, while at least one compound according to formula L-(III) in the mixture M_{III} to all compounds according to formula D-(III) in the mixture M_{III} is <1:1, preferably <0.9:1, more preferably <0.75:1, more preferably <0.5:1, more preferably <0.2:1, more preferably <0.1:1, more preferably <0.01:1.

4.5.14.1 Step (a) without Enzymatic Catalysis

Step (a) may be carried out non-enzymatically, i.e. without the use of an enzyme. The reaction of compounds according to step D-(III) to compounds according to L-(III) proceeds in alkaline solution, as known to the skilled person and as described by Slomka et al., M. Bovarnick & H. T. Clarke, Journal of the American Chemical Society 1938, 60, 2426-2430, by R. A. Lazarus, J. Org. Chem. 1990, 55, 4755-4757, and by A. S. Bommarius, M. Kottenhahn, H. Klenk, K. Drauz: "A direct route from hydantoins to D-amino acids employing a resting cell biocatalyst with D-hydantoinase and D-carbamoylase activity" on page 164 and 167 in "Microbial Reagents in Organic Synthesis" Series C: Mathematical and Physical Sciences-Vol. 381, S. Servi (Ed.), 1992, Springer Science+Business Media, B. V., Dordrecht.

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Therefore, if step (a) is carried out non-enzymatically, the conditions that are preferably applied in the reaction medium in which non-enzymatic step (a) is carried out are preferably those that are described for the preferred conditions for step (b) (item 4.5.13), except that the pH is ≥8, preferably in the range of 8 to 12, more preferably 8 to 11, more preferably 8 to 10, even more preferably 8 to 9. As in these embodiments, the preferred conditions in step (a) and (b) with respect to the pH ranges are different, it is preferable that the reaction media in steps (a) and (b) are different.

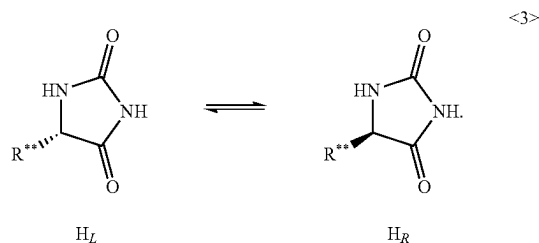
4.5.14.2 Enzymatic Conversion of D-(III) into L-(III)

Step (a) is preferably carried out enzymatically, i.e. the reaction according to step (a) is preferably catalyzed by a hydantoin racemase E₃.

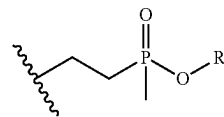
Namely, it was surprisingly found that hydantoin racemases accept compounds of formula D-(III) as substrates and convert them to products according to formulae L-(III), and hence catalyze the reaction according to step (a), while there is no catalysis of the corresponding compound according to formula D-(III), in which R=H.

This finding is of high scientific and economic value, as it further expands the scope of new starting materials for the production of L-glufosinate alkyl esters and L-glufosinate via new synthetic routes. Moreover, this finding also opens new possibilities of enantioselective production of LGA from racemic mixtures of L-(III) and D-(III), as embodied in the method according to the second aspect of the invention.

In nature, hydantoin racemases catalyze the conversion of one of the two hydantoin enantiomers H_L and H_R into the other (see the following reaction <3>):

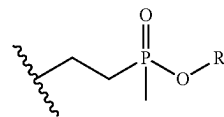


It was now surprisingly found that hydantoin racemases also accept substrates in which R^{**}=R^Y=



wherein R=R^{3D}.

Surprisingly, they do not accept substrates in which R^{**}=R^Y=



wherein R=H.

Suitable hydantoin racemases are described e.g. in WO 01/23582 A1 and by U. Engel, J. Rudat, C. Syldatk in "The

hydantoinase process: recent developments for the production of non-canonical amino acids" in the book "Industrial biocatalysis" by P. Grunwald (Ed.), Pan Stanford Series on Biocatalysis, 2015, pages 817-862, and by F. J. Las Heras-Vazquez, J. M. Clemente-Jimenez, S. Martinez-Rodriguez, F. Rodriguez-Vico in "Hydantoin racemase: the key enzyme for the production of optically pure α -amino acids" in chapter 12 of the book "Modern Biocatalysis: Stereoselective and environmentally friendly reactions" by W. Fessner, T. Anthonsen (Eds), Weinheim: WILEY-VCH Verlag GmbH & Co, 2009, pages 173-193.

A hydantoin racemase E_3 that may be used in optional step (a) of the method according to the first aspect of the invention may originate from *Agrobacterium* sp., in particular *Agrobacterium* strain IP_1-671; *Arthrobacter* sp., in particular *Arthrobacter aurescens*, more in particular *Arthrobacter aurescens* DSM 3745 or *Arthrobacter* sp. BT801; *Flavobacterium* sp., in particular *Flavobacterium* sp. AJ 11199; *Microbacterium* sp., in particular *Microbacterium liquefaciens*, preferably *Microbacterium liquefaciens* AJ 3912; *Pasteurella* sp., in particular *Pasteurella* sp. AJ11221; *Pseudomonas* sp., in particular *Pseudomonas* sp. NS671; *Pyrococcus* sp., in particular *Pyrococcus horikoshii* OT3; *Rhodococcus* sp., in particular *Rhodococcus* R04; *Sinorhizobium* sp., in particular *Sinorhizobium meliloti*, more in particular *Sinorhizobium meliloti* CECT 4114, most preferably from *Arthrobacter aurescens* DSM 3745.

A hydantoin racemase E_3 suitable for the method according to the present invention may be the enzyme HyuR, which originates from *Arthrobacter aurescens* DSM 3745. Another enzyme may be selected from HyuE, Hyu2, HRase, HyuA, PH1054.

The hydantoin racemase E_3 that may preferably be used in preferred step (a) of the method according to the first aspect of the invention may be categorized in the EC class 5.1.99.5.

The following table 3 gives preferred examples for polypeptide sequences of hydantoin racemase E_3 that may be preferably used in step (a) of the method according to the first aspect of the invention. The genes encoding the respective hydantoin racemase E_3 and the respective accession code are indicated as far as known.

TABLE 3

Hydantoin Racemases (EC 5.1.99.5)			
Strain	Gene name	GenBank/UniProt accession	SEQ ID NO: of the polypeptide
<i>Arthrobacter aurescens</i> DSM 3745	hyuR		SEQ ID NO: 17
<i>Pseudomonas</i> sp. NS671	hyuE	Q00924	SEQ ID NO: 18
<i>Rhodococcus</i> R04	hyu2		SEQ ID NO: 19
<i>Microbacterium liquefaciens</i> AJ 3912	"HRase"		SEQ ID NO: 20
<i>Sinorhizobium meliloti</i> CECT 4114	hyuA	Q6TMG4	SEQ ID NO: 21
<i>Flavobacterium</i> sp. AJ11199	"HRase"		SEQ ID NO: 22
<i>Agrobacterium</i> IP_1-671	hyuA		SEQ ID NO: 23
<i>Pasteurella</i> sp. AJ 11221	not assigned		SEQ ID NO: 24
<i>Pyrococcus horikoshii</i> OT3	PH1054	O58781	SEQ ID NO: 25
<i>Arthrobacter</i> sp. BT801	hyuA	AAL55411	SEQ ID NO: 26

In a preferred embodiment of the preferred method according to the second aspect of the present invention, the reaction according to step (a) is catalyzed by a hydantoin racemase E_3 , wherein the polypeptide sequence of E_3 is selected from the group consisting of SEQ ID NO: 17 and variants thereof, SEQ ID NO: 18 and variants thereof, SEQ ID NO: 19 and variants thereof, SEQ ID NO: 20 and variants thereof, SEQ ID NO: 21 and variants thereof, SEQ ID NO: 22 and variants thereof, SEQ ID NO: 23 and variants thereof, SEQ ID NO: 24 and variants thereof, SEQ ID NO: 25 and variants thereof, SEQ ID NO: 26 and variants thereof.

4.5.14.3 Assay G for Determining Hydantoin Racemase Activity

The skilled person is aware of hydantoin racemases, that may be used in preferred step (a) of the method according to the first aspect of the invention.

In particular, Assay G, described in the following, may be used to determine hydantoin racemase activity of a given enzyme E_Z and may advantageously be used according to the invention to determine carbamoylase and L-carbamoylase activity in variants of SEQ ID NO: 17, variants of variants of SEQ ID NO: 18, variants of SEQ ID NO: 19, variants of SEQ ID NO: 20, variants of SEQ ID NO: 21, variants of SEQ ID NO: 22, variants of variants of SEQ ID NO: 23, variants of SEQ ID NO: 24, variants of SEQ ID NO: 25, variants of SEQ ID NO: 26.

For the purpose of Assay G, the molar mass of the enzyme E_Z to be tested is calculated as the molar mass of the polypeptide sequence of E_Z .

Moreover, a substrate according to formula D-(III), wherein $R^{3D}=n$ -butyl, is used in Assay G so that there is only one chiral carbon atom in the substrate. The compound according to formula D-(III), wherein $R^{3D}=n$ -butyl, is denoted as "D-(III)+".

A product according to formula L-(III), wherein $R^{3L}=n$ -butyl, is detected in Assay G.

The compound according to formula L-(III), wherein $R^{3L}=n$ -butyl, is denoted as "L-(III)+".

Assay G:

To 0.9 ml of an aqueous reaction solution (phosphate buffer, pH 7.2, 10 mM Mg_2Cl_2), containing 50 mM of the pure D-enantiomer of an n-butyl ester of hydantoin glutamate of the formula D-(III)(II)* are added 400 nmol of E_Z in 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM $MnCl_2$). The resulting solution is incubated at 25° C., and the pH is held at pH 7.2 by addition of 0.5 M. After 300 minutes, the reaction is stopped by addition of 2 M HCl to achieve a pH of 2.5. The molar amount of the L-enantiomer of formula L-(III)(II)* is measured, at least every 3 minutes (determination by LC-MS, e.g. by the LC-MS method described in the example section, item 5.4, for the detection of LGA).

4.5.14.4 Assay H for Identifying Hydantoin Racemases

Whether a given enzyme E_Z may be considered a hydantoin racemase E_3 , may be determined in the context of the present invention by the following Assay H:

H-1 Firstly, Assay G as set forth under item 4.5.14.3 is conducted, and the obtained molar amount of L-(III)(II)* is determined according to Assay G.

H-2 Then, step H-1 is repeated, except that instead of the addition of 400 nmol E_Z in 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM $MnCl_2$), 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM $MnCl_2$) without E_Z is added.

4.5.14.5 Hydantoin Racemase Activity

If the molar amount of the compound of formula L-(III)(II)* that is determined in step H-1 is greater than the molar amount of the compound of formula L-(III)(II)* that is

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determined in step H-2, then E_Z is deemed to have hydantoin racemase activity, and hence may be considered a hydantoin racemase E_3 in the context of the invention.

4.5.14.6 Assay J for Identifying Preferred Hydantoin Racemase Variants of SEQ ID NO: 17-26

An enzyme, the polypeptide sequence of which is selected from the group consisting of SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26, has hydantoin racemase activity.

In a preferred embodiment of the method according to the first aspect of the invention, the polypeptide sequence of the hydantoin racemase E_3 is selected from the group consisting of SEQ ID NO: 17 and variants thereof, SEQ ID NO: 18 and variants thereof, SEQ ID NO: 19 and variants thereof, SEQ ID NO: 20 and variants thereof, SEQ ID NO: 21 and variants thereof, SEQ ID NO: 22 and variants thereof, SEQ ID NO: 23 and variants thereof, SEQ ID NO: 24 and variants thereof, SEQ ID NO: 25 and variants thereof, SEQ ID NO: 26 and variants thereof.

The term "variant" is defined under item 4.3.

In the context of the invention, an enzyme, the polypeptide sequence of which is a variant of one of SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26, has hydantoin racemase activity.

Whether a given enzyme E_Z , the polypeptide sequence of which is a variant of one SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26, has hydantoin racemase activity may be determined as set forth under items 4.5.14.4 and 4.5.14.5.

The hydantoin racemase activity of a given hydantoin racemase E_{3V} , the polypeptide sequence of which is a variant of one of SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26, relative to the hydantoin racemase activity of a hydantoin racemase E_{3S} , wherein the polypeptide sequence of E_{3S} is selected from SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26, may be quantified in the context of the present invention by the following Assay J.

Assay J:

J-1 Firstly, Assay G as set forth under item 4.5.14.3 is conducted, wherein E_{3S} is the enzyme to be tested. The molar amount of the compound according to formula L-(III) (II)* is determined according to Assay G.

J-2 Step J-1 is repeated, except that, instead of E_{3S} , E_{3V} is used as the enzyme to be tested.

J-3. Then, the molar amount of the compound according to formula L-(III) (II)* is determined in step J-2, is divided by the molar amount of the compound according to formula L-(III), wherein $R=n$ -butyl, is determined in step J-1, and the obtained ratio is multiplied by 100, giving the hydantoin racemase activity of hydantoin racemase E_{3V} , relative to the hydantoin racemase activity of the hydantoin racemase E_{3S} , in %.

4.5.14.7 Preferred Hydantoin Racemase Variants SEQ ID NO: 17-26

In the context of the present invention, hydantoin racemase E_3 , the polypeptide sequence of which is selected from the group consisting of SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID

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NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26, are generally denoted as " E_{3S} ".

Hydantoin racemases E_3 , the polypeptide sequence of which is selected from variants of a sequence selected from the group consisting of f SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26, are generally denoted as " E_{3V} ".

In a preferred embodiment of the method according to the first aspect of the present invention, the reaction in step (a) is catalyzed by a hydantoin racemase E_3 , and the polypeptide sequence of the hydantoin racemase E_3 is selected from the group consisting of SEQ ID NO: 17 and variants thereof, SEQ ID NO: 18 and variants thereof, SEQ ID NO: 19 and variants thereof, SEQ ID NO: 20 and variants thereof, SEQ ID NO: 21 and variants thereof, SEQ ID NO: 22 and variants thereof, SEQ ID NO: 23 and variants thereof, SEQ ID NO: 24 and variants thereof, SEQ ID NO: 25 and variants thereof, SEQ ID NO: 26 and variants thereof. More preferably, the reaction in step (a) is catalyzed by a hydantoin racemase E_3 , and the polypeptide sequence of the hydantoin racemase E_3 is selected from the group consisting of SEQ ID NO: 1 and variants of SEQ ID NO: 1.

4.5.14.7.1 Preferred Variants of SEQ ID NO: 17

According to the invention, the polypeptide sequence of the hydantoin racemase E_3 may also be a variant of SEQ ID NO: 17.

The hydantoin racemase E_3 , the polypeptide sequence of which is SEQ ID NO: 17, is denoted as " E_{317S} ". Hydantoin racemases E_3 , the polypeptide sequence of which is selected from variants of SEQ ID NO: 17, are generally denoted as " E_{317V} ".

A variant of the polypeptide sequence of SEQ ID NO: 17 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 17.

The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 17 is not identical to SEQ ID NO: 17.

According to the invention, a hydantoin racemase E_{317V} has hydantoin racemase activity, determined as described under item 4.5.14.4 and 4.5.14.5.

According to the invention, a hydantoin racemase E_{317V} preferably has hydantoin racemase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoin racemase of the hydantoin racemase E_{317S} , wherein the hydantoin racemase activity of E_{317V} , relative to the hydantoin racemase activity E_{317S} is determined by Assay J described under item 4.5.14.6.

It is even more preferably according to the invention, that a hydantoin racemase E_{317V} has hydantoin racemase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more prefab in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the

range of 90 to 110%, more preferably 100% the hydantoin racemase activity of the hydantoin racemase E_{317S} , wherein the hydantoin racemase activity of E_{317V} , relative to the hydantoin racemase activity of E_{317S} is determined by Assay J described under item 4.5.14.6.

4.5.14.7.2 Preferred Variants of SEQ ID NO: 18

According to the invention, the polypeptide sequence of the hydantoin racemase E_3 may also be a variant of SEQ ID NO: 18.

The hydantoin racemase E_3 , the polypeptide sequence of which is SEQ ID NO: 18, is denoted as " E_{318S} ". Hydantoin racemases E_3 , the polypeptide sequence of which is selected from variants of SEQ ID NO: 18, are generally denoted as " E_{318V} ".

A variant of the polypeptide sequence of SEQ ID NO: 18 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 18.

The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 18 is not identical to SEQ ID NO: 18.

According to the invention, a hydantoin racemase E_{318V} has hydantoin racemase activity, determined as described under item 4.5.14.4 and 4.5.14.5.

According to the invention, a hydantoin racemase E_{318V} preferably has hydantoin racemase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoin racemase of the hydantoin racemase E_{318S} , wherein the hydantoin racemase activity of E_{318V} , relative to the hydantoin racemase activity E_{318S} is determined by Assay J described under item 4.5.14.6.

It is even more preferably according to the invention, that a hydantoin racemase E_{318V} has hydantoin racemase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoin racemase activity of the hydantoin racemase E_{318S} , wherein the hydantoin racemase activity of E_{318V} , relative to the hydantoin racemase activity of E_{318S} is determined by Assay J described under item 4.5.14.6.

4.5.14.7.3 Preferred Variants of SEQ ID NO: 19

According to the invention, the polypeptide sequence of the hydantoin racemase E_3 may also be a variant of SEQ ID NO: 19.

The hydantoin racemase E_3 , the polypeptide sequence of which is SEQ ID NO: 19, is denoted as " E_{319S} ". Hydantoin racemases E_3 , the polypeptide sequence of which is selected from variants of SEQ ID NO: 19, are generally denoted as " E_{319V} ".

A variant of the polypeptide sequence of SEQ ID NO: 19 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more

preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 19.

The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 19 is not identical to SEQ ID NO: 19.

According to the invention, a hydantoin racemase E_{319V} has hydantoin racemase activity, determined as described under item 4.5.14.4 and 4.5.14.5.

According to the invention, a hydantoin racemase E_{319V} preferably has hydantoin racemase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoin racemase of the hydantoin racemase E_{319S} , wherein the hydantoin racemase activity of E_{319V} , relative to the hydantoin racemase activity E_{319S} is determined by Assay J described under item 4.5.14.6.

It is even more preferably according to the invention, that a hydantoin racemase E_{319V} has hydantoin racemase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoin racemase activity of the hydantoin racemase E_{319S} , wherein the hydantoin racemase activity of E_{319V} , relative to the hydantoin racemase activity of E_{319S} is determined by Assay J described under item 4.5.14.6.

4.5.14.7.4 Preferred Variants of SEQ ID NO: 20

According to the invention, the polypeptide sequence of the hydantoin racemase E_3 may also be a variant of SEQ ID NO: 20.

The hydantoin racemase E_3 , the polypeptide sequence of which is SEQ ID NO: 20, is denoted as " E_{320S} ". Hydantoin racemases E_3 , the polypeptide sequence of which is selected from variants of SEQ ID NO: 20, are generally denoted as " E_{320V} ".

A variant of the polypeptide sequence of SEQ ID NO: 20 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 20.

The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 20 is not identical to SEQ ID NO: 20.

According to the invention, a hydantoin racemase E_{320V} has hydantoin racemase activity, determined as described under item 4.5.14.4 and 4.5.14.5.

According to the invention, a hydantoin racemase E_{320V} preferably has hydantoin racemase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more

A variant of the polypeptide sequence of SEQ ID NO: 23 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably

≥97%, more preferably ≥98%, more preferably ≥99%, more preferably ≥99.9% sequence identity to polypeptide sequence SEQ ID NO: 23.

The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 23 is not identical to SEQ ID NO: 23.

According to the invention, a hydantoin racemase E_{323V} has hydantoin racemase activity, determined as described under item 4.5.14.4 and 4.5.14.5.

According to the invention, a hydantoin racemase E_{323V} preferably has hydantoin racemase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoin racemase activity of the hydantoin racemase E_{323S} , wherein the hydantoin racemase activity of E_{323V} relative to the hydantoin racemase activity E_{323S} is determined by Assay J described under item 4.5.14.6.

It is even more preferably according to the invention, that a hydantoin racemase E_{323V} has hydantoin racemase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoin racemase activity of the hydantoin racemase E_{323S} , wherein the hydantoin racemase activity of E_{323V} relative to the hydantoin racemase activity of E_{323S} is determined by Assay J described under item 4.5.14.6.

4.5.14.7.8 Preferred Variants of SEQ ID NO: 24

According to the invention, the polypeptide sequence of the hydantoin racemase E_3 may also be a variant of SEQ ID NO: 24.

The hydantoin racemase E_3 , the polypeptide sequence of which is SEQ ID NO: 24, is denoted as " E_{324S} ". Hydantoin racemases E_3 , the polypeptide sequence of which is selected from variants of SEQ ID NO: 24, are generally denoted as " E_{324V} ".

A variant of the polypeptide sequence of SEQ ID NO: 24 is a polypeptide with sequence identity of at least 60%, preferably ≥65%, more preferably ≥70%, more preferably ≥75%, more preferably ≥80%, more preferably ≥85%, more preferably ≥90%, more preferably ≥91%, more preferably ≥92%, more preferably ≥93%, more preferably ≥94%, more preferably ≥95%, more preferably ≥96%, more preferably ≥97%, more preferably ≥98%, more preferably ≥99%, more preferably ≥99.9% sequence identity to polypeptide sequence SEQ ID NO: 24.

The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 24 is not identical to SEQ ID NO: 24.

According to the invention, a hydantoin racemase E_{324V} has hydantoin racemase activity, determined as described under item 4.5.14.4 and 4.5.14.5.

According to the invention, a hydantoin racemase E_{324V} preferably has hydantoin racemase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoin racemase activity of the hydantoin racemase E_{324S} , wherein the hydantoin racemase activity of E_{324V} relative to

the hydantoin racemase activity E_{324S} is determined by Assay J described under item 4.5.14.6.

It is even more preferably according to the invention, that a hydantoin racemase E_{324V} has hydantoin racemase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoin racemase activity of the hydantoin racemase E_{324S} , wherein the hydantoin racemase activity of E_{324V} relative to the hydantoin racemase activity of E_{324S} is determined by Assay J described under item 4.5.14.6.

4.5.14.7.9 Preferred Variants of SEQ ID NO: 25

According to the invention, the polypeptide sequence of the hydantoin racemase E_3 may also be a variant of SEQ ID NO: 25.

The hydantoin racemase E_3 , the polypeptide sequence of which is SEQ ID NO: 25, is denoted as " E_{325S} ". Hydantoin racemases E_3 , the polypeptide sequence of which is selected from variants of SEQ ID NO: 25, are generally denoted as " E_{325V} ".

A variant of the polypeptide sequence of SEQ ID NO: 25 is a polypeptide with sequence identity of at least 60%, preferably ≥65%, more preferably ≥70%, more preferably ≥75%, more preferably ≥80%, more preferably ≥85%, more preferably ≥90%, more preferably ≥91%, more preferably ≥92%, more preferably ≥93%, more preferably ≥94%, more preferably ≥95%, more preferably ≥96%, more preferably ≥97%, more preferably ≥98%, more preferably ≥99%, more preferably ≥99.9% sequence identity to polypeptide sequence SEQ ID NO: 25.

The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 25 is not identical to SEQ ID NO: 25.

According to the invention, a hydantoin racemase E_{325V} has hydantoin racemase activity, determined as described under item 4.5.14.4 and 4.5.14.5.

According to the invention, a hydantoin racemase E_{325V} preferably has hydantoin racemase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoin racemase activity of the hydantoin racemase E_{325S} , wherein the hydantoin racemase activity of E_{325V} relative to the hydantoin racemase activity E_{325S} is determined by Assay J described under item 4.5.14.6.

It is even more preferably according to the invention, that a hydantoin racemase E_{325V} has hydantoin racemase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoin racemase activity of the hydantoin racemase E_{325S} , wherein the hydantoin racemase activity of E_{325V} relative to the hydantoin racemase activity of E_{325S} is determined by Assay J described under item 4.5.14.6.

4.5.14.7.10 Preferred Variants of SEQ ID NO: 26

According to the invention, the polypeptide sequence of the hydantoin racemase E_3 may also be a variant of SEQ ID NO: 26.

The hydantoin racemase E_3 , the polypeptide sequence of which is SEQ ID NO: 26, is denoted as " E_{326S} ". Hydantoin racemases E_3 , the polypeptide sequence of which is selected from variants of SEQ ID NO: 26, are generally denoted as " E_{326V} ".

A variant of the polypeptide sequence of SEQ ID NO: 26 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 26.

The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 26 is not identical to SEQ ID NO: 26.

According to the invention, a hydantoin racemase E_{326V} has hydantoin racemase activity, determined as described under item 4.5.14.4 and 4.5.14.5.

According to the invention, a hydantoin racemase E_{326V} preferably has hydantoin racemase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoin racemase of the hydantoin racemase E_{326S} , wherein the hydantoin racemase activity of E_{326V} , relative to the hydantoin racemase activity E_{326S} is determined by Assay J described under item 4.5.14.6.

It is even more preferably according to the invention, that a hydantoin racemase E_{326V} has hydantoin racemase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoin racemase activity of the hydantoin racemase E_{326S} , wherein the hydantoin racemase activity of E_{326V} , relative to the hydantoin racemase activity of E_{326S} is determined by Assay J described under item 4.5.14.6.

4.5.15 Preferred Method Conditions in Step (a)

In this preferred embodiment, in which step (a) is catalyzed by a hydantoin racemase E_3 , the reaction in step (a) may be carried out under conditions known to the skilled person.

The reaction medium is preferably aqueous, more preferably an aqueous buffer.

Exemplary buffers commonly used in biotransformation reactions and advantageously used herein include Tris, phosphate, or any of Good's buffers, such as 2-(N-morpholino)ethanesulfonic acid ("MES"), N-(2-acetamido)iminodiacetic acid ("ADA"), piperazine-N,N'-bis(2-ethanesulfonic acid) ("PIPES"), N-(2-acetamido)-2-aminoethanesulfonic acid ("ACES"), P-hydroxy-4-morpholinepropanesulfonic acid ("MOPSO"), choline chloride, 3-(N-morpholino)propanesulfonic acid ("MOPS"), N, N-Bis(2-hydroxyethyl)-

2-aminoethanesulfonic acid ("BES"), 2-[[1,3-dihydroxy-2-(hydroxymethyl)propan-2-yl]amino]ethanesulfonic acid ("TES"), 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid ("HEPES"), 3-(Bis(2-hydroxyethyl)amino)-2-hydroxypropane-1-sulfonic acid ("DIPSO"), acetamidoglycine, 3-(N-Tris(hydroxymethyl)methylamino(2-hydroxypropane) sulfonic acid ("TAPSO"), piperazine-N,N'-bis(2-hydroxypropanesulfonic acid) ("POPSO"), 4-(2-Hydroxyethyl) piperazine-1-(2-hydroxypropanesulfonic acid) ("HEPPSO"), 3-[4-(2-Hydroxyethyl)-1-piperazinyl]propanesulfonic acid ("HEPPS"), tricine, glycylamide, bicine, or 3-[[1,3-dihydroxy-2-(hydroxymethyl)propan-2-yl]amino]propane-1-sulfonic acid ("TAPS").

In some embodiments, ammonium can act as a buffer. One or more organic solvents can also be added to the reaction.

The buffer preferably contains metal salts, more preferably metal salts such as halogenides of monovalent or bivalent metals (e.g. CoCl_2 , MnCl_2).

The concentration of these metal salts in the reaction medium is preferably in the range from 1 μM to 1 M, more preferably 1 mM to 100 mM, even more preferably 1 to 10 mM.

Preferably, step (a) of the method according to the first aspect of the invention is carried out in a phosphate buffer.

The pH of the reaction medium in step (a) of the method according to the first aspect of the invention is preferably in the range of from 2 to 10, more preferably in the range of from 5 to 8, more preferably 7.2 to 7.5, most preferably 7.5.

Preferably, step (a) of the method according to the first aspect of the invention is carried out at a temperature in the range of from 20° C. to 70° C., more preferably in the range of from 30° C. to 55° C., most preferably 50° C.

In this preferred embodiment, in which step (a) is catalyzed by a hydantoin racemase E_3 , the preferred reaction conditions in step (a) are the same as described for steps (b) and (c), confer items 4.5.7 and 4.5.13, respectively. It is even more preferred to carry out step (a) concomitantly with steps (b) and (c).

Preferably, the total concentration of all hydantoin racemases E_3 in the reaction solution in step (a) is in the range of from 1 μM to 10 mM, preferably 10 μM to 1 mM, more preferably 0.1 mM to 0.5 mM, most preferably 0.4 mM.

In alternative preferred embodiments, the total concentration of all hydantoin racemases E_3 in the reaction solution in step (a) is in the range of from 1 $\mu\text{g/l}$ to 10 g/l, preferably 0.1 mg/l to 5 g/l, more preferably 1 mg/l to 1 g/l, more preferably 5 mg/l to 500 mg/l.

Preferably, step (a) is carried out in the same reaction medium in which steps (c) and (b) are carried out. The advantage is that this allows for a one-pot synthesis in which all the steps (a), (b), and (c) are carried out.

Preferably, the initial concentration of all the compounds according to formula D-(III) in the reaction medium in step (a) is in the range of from 1 μM to 1 M, preferably of from 10 μM to 0.5 M, more preferably of from 0.1 mM to 0.1 M, more preferably of from 1 mM to 10 mM, most preferably 1.25 mM.

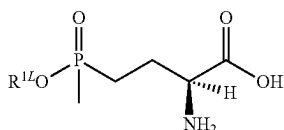
If compounds according to formula L-(III) are present in the reaction medium in step (a), the initial concentration of all the compounds according to formula L-(III) in the reaction medium is preferably from 1 to 100 times the concentration of all the compounds according to formula D-(III), more preferably 1 to 10 times the concentration, even more preferably 1 to 2 times even more preferably the same as the concentration of all the compounds according to formula D-(III).

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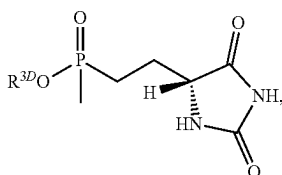
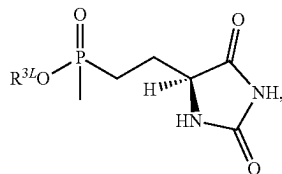
“Initial concentration of all the compounds according to formula L-(III)/D-(III)” refers to the concentration of the respective compound L-(III) or D-(III), respectively, in the reaction medium when the respective compounds are employed in step (a).

4.6 Second Aspect: Method for Production of an L-Glufosinate P-Ester

In a second aspect, the present invention relates to a further method for the production of an L-glufosinate P-ester according to formula L-(I):



from a mixture M_{III} comprising at least one compound L-(III) and at least one compound D-(III):



wherein R^{1L} is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the above-mentioned formulae (IX), (X), more preferably R^{1L} has formula (X).

When R^{1L} is a radical of the formula (VIII), the radical R^{1L} may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

When R^{1L} is a radical of the formula (IX), the radical R^{1L} may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

When R^{1L} is a radical of the formula (X), the radical R^{1L} may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

R^{3L} is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (VIII), (IX), more preferably R^{3L} has formula (VIII).

When R^{3L} is a radical of the formula (VIII), the radical R^{3L} may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

When R^{3L} is a radical of the formula (IX), the radical R^{3L} may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

When R^{3L} is a radical of the formula (X), the radical R^{3L} may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

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R^{3D} is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (VIII), (IX), more preferably R^{3D} has formula (VIII).

When R^{3D} is a radical of the formula (VIII), the radical R^{3D} may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

When R^{3D} is a radical of the formula (IX), the radical R^{3D} may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

When R^{3D} is a radical of the formula (X), the radical R^{3D} may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

In this method according to the second aspect of the invention, a mixture M_{III} comprising at least one compound L-(III) and at least one compound D-(III), wherein preferably $R^{3L}=R^{3D}$, is provided, and subjected to at least two enzymatic reactions (ii) and (iii), giving a composition M_I comprising at least one compound L-(I) and optionally D-(I). Therefore, the reaction according to the preferred second aspect of the invention is enzymatic.

The method for the production of an L-glufosinate P-ester according to formula L-(I), according to the second aspect of the invention is L-enantioselective, in particular L-enantio-specific, because M_I either comprises compounds according to formula L-(I), but not compounds according to formula D-(I) (in which case it is “L-enantiospecific”) or, when M_I comprises compounds according to formulae L-(I) and D-(I), the molar ratio of all compounds according to formula L-(I) in M_I to all compounds according to formula D-(I) in M_I is greater than the molar ratio of all compounds according to formula L-(III) in M_{III} to all compounds of formula D-(III) in M_{III} (in which case it is “L-enantioselective, but not L-enantiospecific”).

In a particular embodiment of the method according to the second aspect of the invention, M_I comprises at least one compound according to formula L-(I) and at least one compound according to formula D-(I) and the molar ratio of all compounds according to formula L-(I) in M_I to all compounds according to formula D-(I) in M_I is greater than the molar ratio of all compounds according to formula L-(III) in M_{III} to all compounds of formula D-(III) in M_{III} (in which case the method according to the second aspect of the invention is “L-enantioselective, but not L-enantiospecific”).

In a preferred embodiment of the method according to the second aspect of the invention, a mixture M_{III} comprising at least one compound L-(III) and at least one compound D-(III), wherein $R^{3L}=R^{3D}$, is provided in step (a).

In the method according to the second aspect of the invention, it is possible to enrich the amount of compounds according to formula L-(I) over compounds according to formula D-(I) in the product composition M_I , which means that the molar ratio of all compounds according to formula L-(I) in M_I to all compounds according to formula D-(I) in M_I is greater than the molar ratio of all compounds according to formula L-(III) in M_{III} to all compounds of formula D-(III) in M_{III} provided in step (i-A).

In particular, the molar amount of compounds of formula L-(I) per compound of formula D-(I) in M_I is greater than the molar amount of compounds of formula L-(III) per compound of formula D-(III) in M_{III} , in particular by a factor of XL, wherein XL is at least 1.1, preferably at least 1.5, more preferably at least 2, more preferably at least 5, even more preferably at least 9, even more preferably at least 99. In another preferred embodiment XL is in the range of 1.1 to 99, preferably in the range of 1.5 to 9, even more preferably in the range of 2 to 5.

4.6.1 Step (i-A)

In a first step of the method according to the second aspect of the invention, a mixture $M_{III A}$ comprising at least one compound L-(III) and at least one compound D-(III) is provided.

In a preferred embodiment, the molar ratio of all compounds according to formula L-(III) in the mixture $M_{III A}$ to all compounds according to formula D-(III) in the mixture $M_{III A}$ is essentially 1:1.

Alternatively, compounds according to formula D-(III) are comprised in an excess to compounds according to formula L-(III) in mixture $M_{III A}$, meaning that, while L-(III) is present in mixture $M_{III A}$, the molar ratio of all compounds according to formula L-(III) to all compounds according to formula D-(III) in mixture $M_{III A}$ is <1:1, preferably <0.9:1, more preferably <0.75:1, more preferably <0.5:1, more preferably <0.2:1, more preferably <0.1:1, more preferably <0.01:1.

In other preferred embodiments, the molar ratio of all compounds according to formula L-(III) in the mixture $M_{III A}$ to all compounds according to formula D-(III) in the mixture $M_{III A}$ is in the range of from 3:2 to 1:99, more preferably in the range of from 1.01:1 to 1:99, more preferably in the range of from 1:1 to 1:99, more preferably in the range of from 1:1.01 to 1:99, more preferably in the range of from 1:1.01 to 1:9, more preferably in the range of from 1:1.01 to 1:8, more preferably in the range of from 1:1.01 to 1:3.

Alternatively, D-(III) is comprised in an excess to L-(III) in mixture $M_{III A}$, meaning that, while at least one compound according to formula L-(III) is in the mixture $M_{III A}$, the molar ratio of all compounds according to formula L-(III) in the mixture $M_{III A}$ to all compounds according to formula D-(III) in the mixture $M_{III A}$ is <1:1, preferably <0.9:1, more preferably <0.75:1, more preferably <0.5:1, more preferably <0.2:1, more preferably <0.1:1, more preferably <0.01:1.

A mixture of these compounds can be obtained by the skilled person, for example by organic synthesis as set forth in DE 31 42 036 A1 and by Slomka et al.

E. Ware, Chem. Rev. 1950, 46, 403-470 and C. Avendaño & J. C. Menendez, "Hydantoin and its derivatives" in Kirk-Othmer Encyclopedia of Chemical Technology 2000 give a general overview over hydantoin chemistry.

Preferably, the mixture $M_{III A}$ is provided in step (i-A) in a reaction medium.

Namely, the reaction medium in step (i-A) is preferably aqueous, more preferably an aqueous buffer.

Exemplary buffers commonly used in biotransformation reactions and advantageously used in step (i-A) include Tris, phosphate, or any of Good's buffers, such as 2-(N-morpholino) ethanesulfonic acid ("MES"), N-(2-acetamido) iminodiacetic acid ("ADA"), piperazine-N,N'-bis(2-ethanesulfonic acid) ("PIPES"), N-(2-acetamido)-2-aminoethanesulfonic acid ("ACES"), P-hydroxy-4-morpholinepropanesulfonic acid ("MOPSO"), choline chloride, 3-(N-morpholino) propanesulfonic acid ("MOPS"), N,N-Bis(2-hydroxyethyl)-2-aminoethanesulfonic acid ("BES"), 2-[[1,3-dihydroxy-2-(hydroxymethyl) propan-2-yl]amino]ethanesulfonic acid ("TES"), 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid ("HEPES"), 3-(Bis(2-hydroxyethyl)amino)-2-hydroxypropane-1-sulfonic acid ("DIPSO"), acetamidoglycine, 3-(N-Tris(hydroxymethyl)methylamino(2-hydroxypropane) sulfonic acid ("TAPSO"), piperazine-N,N'-bis(2-hydroxypropanesulfonic acid) ("POPISO"), 4-(2-Hydroxyethyl) piperazine-1-(2-hydroxypropanesulfonic acid) ("HEPPSO"), 3-[4-(2-Hydroxyethyl)-1-piperazinyl]propanesulfonic acid

("HEPPS"), tricine, glycineamide, bicine, or 3-[[1,3-dihydroxy-2-(hydroxymethyl) propan-2-yl]amino]propane-1-sulfonic acid ("TAPS").

In some embodiments, ammonium can act as a buffer. One or more organic solvents can also be added to the reaction.

The buffer preferably contains metal salts, more preferably metal salts such as halogenides of monovalent or bivalent metals (e.g. CoCl_2 , MnCl_2).

The concentration of these metal salts in the reaction medium in step (i-A) is preferably in the range from 1 μM to 1 M, more preferably 1 mM to 100 mM, even more preferably 1 to 10 mM.

Preferably, step (i-A) of the method according to the second aspect of the invention is carried out in a phosphate buffer.

The pH of the reaction medium in step (i-A) of the method according to the first aspect of the invention is preferably in the range of from 2 to 10, more preferably in the range of from 5 to 8, more preferably 7.2 to 7.5, most preferably 7.5.

More preferably, the concentration of all the compounds according to formula D-(III) in the reaction medium provided in step (i-A) is in the range of from 1 μM to 1 M, preferably of from 10 μM to 0.5 M, more preferably of from 0.1 mM to 0.1 M, more preferably of from 1 mM to 10 mM, most preferably 1.25 mM.

Concerning the compounds according to formula L-(III) that are present in the reaction medium preferably provided in step (i-A), the concentration of all the compounds according to formula L-(III) in the reaction medium is preferably from 1 to a 100 times the concentration of all the compounds according to formula D-(III), more preferably 1 to 10 times the concentration, even more preferably 1 to 2 times even more preferably the same as the concentration of all the compounds according to formula D-(III).

4.6.2 Optional Step (i-B)

In an optional step (i-B), at least a part of the compounds D-(III) comprised by the mixture $M_{III A}$ are reacted into compounds according to the formula L-(III), giving a composition $M_{III B}$ comprising compounds according to formula L-(III) and optionally compounds according to formula D-(III). It goes without saying that the molar ratio of all compounds according to formula L-(III) to all compounds in formula D-(III) in mixture $M_{III B}$ is greater than the molar ratio of all compounds according to formula L-(III) to all compounds in formula D-(III) in mixture $M_{III A}$.

In case step (i-B) is carried out, it is preferably carried out as described for step (a) under items 4.5.14 and 4.5.15.

In particular, the reaction according to step (i-B) is catalyzed by a hydantoin racemase E_3 as described with respect to step (a).

Preferably, the hydantoin racemase E_3 used in step (i-B) is categorized in the EC class 5.1.99.5.

In another embodiment, the polypeptide sequence of the hydantoin racemase E_3 used in step (i-B) is selected from the group consisting of SEQ ID NO: 17 and variants thereof, SEQ ID NO: 18 and variants thereof, SEQ ID NO: 19 and variants thereof, SEQ ID NO: 20 and variants thereof, SEQ ID NO: 21 and variants thereof, SEQ ID NO: 22 and variants thereof, SEQ ID NO: 23 and variants thereof, SEQ ID NO: 24 and variants thereof, SEQ ID NO: 25 and variants thereof, SEQ ID NO: 26 and variants thereof, most preferably selected from SEQ ID NO: 17 and variants thereof.

Preferably, the hydantoin racemase E_3 used in step (i-B) is selected from the group consisting of E_{317S} and E_{317V} (both described under item 4.5.14.7.1), E_{318S} and E_{318V}

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(both described under item 4.5.14.7.2), E_{319S} and E_{319V} (both described under item 4.5.14.7.3), E_{320S} and E_{320V} (both described under item 4.5.14.7.4), E_{321S} and E_{321V} (both described under item 4.5.14.7.5), E_{322S} and E_{322V} (both described under item 4.5.14.7.6), E_{323S} and E_{323V} (both described under item 4.5.14.7.7), E_{324S} and E_{324V} (both described under item 4.5.14.7.8), E_{325S} and E_{325V} (both described under item 4.5.14.7.9), E_{326S} and E_{326V} (both described under item 4.5.14.7.10), more preferably from E_{317S} and E_{317V} , even more preferably from E_{317S} .

The reaction conditions in step (i-B) are preferably as described with respect to step (a) under item 4.5.15.

Namely, the reaction medium in step (i-B) is preferably aqueous, more preferably an aqueous buffer.

Exemplary buffers commonly used in biotransformation reactions and advantageously used in step (i-B) include Tris, phosphate, or any of Good's buffers, such as 2-(N-morpholino) ethanesulfonic acid ("MES"), N-(2-acetamido)imino-diacetic acid ("ADA"), piperazine-N,N'-bis(2-ethanesulfonic acid) ("PIPES"), N-(2-acetamido)-2-aminoethanesulfonic acid ("ACES"), P-hydroxy-4-morpholinepropanesulfonic acid ("MOPSO"), choline chloride, 3-(N-morpholino) propanesulfonic acid ("MOPS"), N,N-Bis(2-hydroxyethyl)-2-aminoethanesulfonic acid ("BES"), 2-[[1,3-dihydroxy-2-(hydroxymethyl) propan-2-yl]amino]ethanesulfonic acid ("TES"), 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid ("HEPES"), 3-(Bis(2-hydroxyethyl)amino)-2-hydroxypropane-1-sulfonic acid ("DIPSO"), acetamidoglycine, 3-(N-Tris(hydroxymethyl)methylamino(-2-hydroxypropane) sulfonic acid ("TAPSO"), piperazine-N,N'-bis(2-hydroxypropane-sulfonic acid) ("POPSO"), 4-(2-Hydroxyethyl) piperazine-1-(2-hydroxypropanesulfonic acid) ("HEPPSO"), 3-[4-(2-Hydroxyethyl)-1-piperazinyl]propanesulfonic acid ("HEPPS"), tricine, glycineamide, bicine, or 3-[[1,3-dihydroxy-2-(hydroxymethyl) propan-2-yl]amino]propane-1-sulfonic acid ("TAPS").

In some embodiments, ammonium can act as a buffer. One or more organic solvents can also be added to the reaction.

The buffer preferably contains metal salts, more preferably metal salts such as halogenides of monovalent or bivalent metals (e.g. CoCl_2 , MnCl_2).

The concentration of these metal salts in the reaction medium in step (i-B) is preferably in the range from 1 μM to 1 M, more preferably 1 mM to 100 mM, even more preferably 1 to 10 mM.

Preferably, step (i-B) of the method according to the second aspect of the invention is carried out in a phosphate buffer.

The pH of the reaction medium in step (i-B) of the method according to the first aspect of the invention is preferably in the range of from 2 to 10, more preferably in the range of from 5 to 8, more preferably 7.2 to 7.5, most preferably 7.5.

Preferably, step (i-B) of the method according to the first aspect of the invention is carried out at a temperature in the range of from 20° C. to 70° C., more preferably in the range of from 30° C. to 55° C., most preferably 50° C.

Preferably, the total concentration of all hydantoin racemases E_3 in the reaction solution in step (i-B) is in the range of from 1 μM to 10 mM, preferably 10 μM to 1 mM, more preferably 0.1 mM to 0.5 mM, most preferably 0.4 mM.

In alternative preferred embodiments, the total concentration of all hydantoin racemases E_3 in the reaction solution in step (i-B) is in the range of from 1 $\mu\text{g/l}$ to 10 g/l, preferably 0.1 mg/l to 5 g/l, more preferably 1 mg/l to 1 g/l, more preferably 5 mg/l to 500 mg/l.

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Preferably, step (i-B) is carried out in the same reaction medium in which steps (ii) and (iii) are carried out. The advantage is that this allows for a one-pot synthesis in which all the steps (i-B), (ii), and (iii) are carried out.

At the end of optional step (i-B), a composition M_{IIIb} comprising compounds according to formula L-(III) and optionally compounds according to formula D-(III) is obtained, preferably a composition M_{IIIb} comprising compounds according to formula L-(III) and compounds according to formula D-(III) is obtained.

4.6.3 Step (ii)

In step (ii) of the method according to the second aspect of the invention, the mixture M_{IIIa} or, in case step (i-B) is carried out, composition M_{IIIb} , is subjected to step (b) as described above under item 4.5.8-4.5.13.

In other words: in those embodiments, in which step (i-B) is not carried out, M_{IIIa} is subjected to step (b) as described above under item 4.5.8-4.5.13.

In case step (i-B) is carried out, M_{IIIb} obtained after step (i-B) is subjected to step (b) as described above under item 4.5.8-4.5.13.

The reaction according to step (ii) is catalyzed by a hydantoinase E_2 as described with respect to step (b), preferably by an L-hydantoinase E_2 as described with respect to step (b).

More preferably, the hydantoinase E_2 , even more preferably the L-hydantoinase E_2 used in step (ii) is categorized in the EC class 3.5.2.2.

In another preferred embodiment, the polypeptide sequence of the hydantoinase E_2 , even more preferably the L-hydantoinase E_2 used in step (ii) is selected from the group consisting of SEQ ID NO: 10 and variants thereof, SEQ ID NO: 11 and variants thereof, SEQ ID NO: 12 and variants thereof, SEQ ID NO: 13 and variants thereof, SEQ ID NO: 14 and variants thereof, SEQ ID NO: 15 and variants thereof, SEQ ID NO: 16 and variants thereof, even more preferably from SEQ ID NO: 10 and variants thereof.

Preferably, the hydantoinase E_2 , even more preferably the L-hydantoinase E_2 used in step (ii) is selected from the group consisting of E_{210S} and E_{210V} (both described under item 4.5.12.1), E_{211S} and E_{211V} (both described under item 4.5.12.2), E_{212S} and E_{212V} (both described under item 4.5.12.3), E_{213S} and E_{213V} (both described under item 4.5.12.4), E_{214S} and E_{214V} (both described under item 4.5.12.5), E_{215S} and E_{215V} (both described under item 4.5.12.6), E_{216S} and E_{216V} (both described under item 4.5.12.7), more preferably from E_{210S} and E_{210V} , even more preferably from E_{210S} .

The reaction conditions in step (ii) are preferably as described with respect to step (b) under item 4.5.13.

Namely, the reaction medium in step (ii) is preferably aqueous, more preferably an aqueous buffer.

Exemplary buffers commonly used in biotransformation reactions and advantageously used in step (ii) include Tris, phosphate, or any of Good's buffers, such as 2-(N-morpholino) ethanesulfonic acid ("MES"), N-(2-acetamido)imino-diacetic acid ("ADA"), piperazine-N,N'-bis(2-ethanesulfonic acid) ("PIPES"), N-(2-acetamido)-2-aminoethanesulfonic acid ("ACES"), P-hydroxy-4-morpholinepropanesulfonic acid ("MOPSO"), choline chloride, 3-(N-morpholino) propanesulfonic acid ("MOPS"), N,N-Bis(2-hydroxyethyl)-2-aminoethanesulfonic acid ("BES"), 2-[[1,3-dihydroxy-2-(hydroxymethyl) propan-2-yl]amino]ethanesulfonic acid ("TES"), 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid ("HEPES"),

3-(Bis(2-hydroxyethyl)amino)-2-hydroxypropane-1-sulfonic acid ("DIPSO"), acetamidoglycine, 3-(N-Tris(hydroxymethyl)methylamino(-2-hydroxypropane) sulfonic acid ("TAPSO"), piperazine-N, N-bis(2-hydroxypropane-sulfonic acid) ("POPSO"), 4-(2-Hydroxyethyl) piperazine-1-(2-hydroxypropanesulfonic acid) ("HEPPSO"), 3-[4-(2-Hydroxyethyl)-1-piperazinyl]propanesulfonic acid ("HEPPS"), tricine, glycineamide, bicine, or 3-[[1,3-dihydroxy-2-(hydroxymethyl) propan-2-yl]amino]propane-1-sulfonic acid ("TAPS").

In some embodiments, ammonium can act as a buffer. One or more organic solvents can also be added to the reaction.

The buffer preferably contains metal salts, more preferably metal salts such as halogenides of monovalent or bivalent metals (e.g. CoCl_2 , MnCl_2).

The concentration of these metal salts in the reaction medium is preferably in the range from 1 μM to 1 M, more preferably 1 mM to 100 mM, even more preferably 1 to 10 mM.

Preferably, step (ii) of the method according to the second aspect of the invention is carried out in a phosphate buffer.

The pH of the reaction medium in step (ii) of the method according to the first aspect of the invention is preferably in the range of from 2 to 10, more preferably in the range of from 5 to 8, more preferably 7.2 to 7.5, most preferably 7.5.

Preferably, step (ii) of the method according to the first aspect of the invention is carried out at a temperature in the range of from 20° C. to 70° C., more preferably in the range of from 30° C. to 55° C., most preferably 50° C.

Preferably, the total concentration of all hydantoinases E_2 in the reaction solution in step (ii) is in the range of from 1 μM to 10 mM, preferably 10 μM to 1 mM, more preferably 0.1 mM to 0.5 mM, most preferably 0.4 mM.

In alternative preferred embodiments, the total concentration of all hydantoinases E_2 in the reaction solution in step (ii) is in the range of from 1 $\mu\text{g/l}$ to 10 g/l, preferably 0.1 mg/l to 5 g/l, more preferably 1 mg/l to 1 g/l, more preferably 5 mg/l to 500 mg/l.

Preferably, step (ii) is carried out in the same reaction medium in which step (iii) is carried out.

At the end of step (ii), a composition M_H comprising at least one compound according to formula L-(II) and optionally at least one compound according to formula D-(II), wherein L-(II) and D-(II) have the above-mentioned formulae, is obtained.

4.6.4 Step (iii)

In step (iii) of the method according to the second aspect of the invention, M_H obtained in step (ii) is subjected to step (c) as described above under items 4.5.1 to 4.5.7 wherein the reaction according to step (iii) is catalyzed by an L-carbamoylase E_1 .

This brings about that compounds according to formula L-(I) are enriched over compounds according to formulas D-(I) in M_H , i.e., compared to the molar ratio of compounds according to formula L-(I) to compounds according to formulas D-(I) in M_{HA} .

More preferably, the L-carbamoylase E_1 used in step (iii) is categorized in the EC class 3.5.1.87. In another preferred embodiment, the polypeptide sequence of the L-carbamoylase E_1 used in step (iii) is selected from the group consisting of SEQ ID NO: 1 and variants thereof, SEQ ID NO: 2 and variants thereof, SEQ ID NO: 3 and variants thereof, SEQ ID NO: 4 and variants thereof, SEQ ID NO: 5 and variants thereof, SEQ ID NO: 6 and variants thereof, SEQ ID NO: 7

and variants thereof, SEQ ID NO: 8 and variants thereof, SEQ ID NO: 9 and variants thereof.

Preferably, the L-carbamoylase E_1 used in step (iii) is selected from the group consisting of E_{101S} and E_{101V} (both described under item 4.5.6.1), E_{102S} and E_{102V} (both described under item 4.5.6.2), E_{103S} and E_{103V} (both described under item 4.5.6.3), E_{104S} and E_{104V} (both described under item 4.5.6.4), E_{105S} and E_{105V} (both described under item 4.5.6.5), E_{106S} and E_{106V} (both described under item 4.5.6.6), E_{107S} and E_{107V} (both described under item 4.5.6.7), E_{108S} and E_{108V} (both described under item 4.5.6.8), E_{109S} and E_{109V} (both described under item 4.5.6.9), more preferably from E_{101S} and E_{101V} , even more preferably from E_{101S} .

The reaction conditions in step (iii) are preferably as described with respect to step (c) under item 4.5.7.

Namely, the reaction medium in step (iii) is preferably aqueous, more preferably an aqueous buffer.

Exemplary buffers commonly used in biotransformation reactions and advantageously used in step (iii) include Tris, phosphate, or any of Good's buffers, such as 2-(N-morpholino) ethanesulfonic acid ("MES"), N-(2-acetamido)iminodiacetic acid ("ADA"), piperazine-N,N'-bis(2-ethanesulfonic acid) ("PIPES"), N-(2-acetamido)-2-aminoethanesulfonic acid ("ACES"), P-hydroxy-4-morpholinepropanesulfonic acid ("MOPSO"), choline chloride, 3-(N-morpholino) propanesulfonic acid ("MOPS"), N,N-Bis(2-hydroxyethyl)-2-aminoethanesulfonic acid ("BES"), 2-[[1,3-dihydroxy-2-(hydroxymethyl) propan-2-yl]amino]ethanesulfonic acid ("TES"), 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid ("HEPES"), 3-(Bis(2-hydroxyethyl)amino)-2-hydroxypropane-1-sulfonic acid ("DIPSO"), acetamidoglycine, 3-(N-Tris(hydroxymethyl)methylamino(-2-hydroxypropane) sulfonic acid ("TAPSO"), piperazine-N,N'-bis(2-hydroxypropane-sulfonic acid) ("POPSO"), 4-(2-Hydroxyethyl) piperazine-1-(2-hydroxypropanesulfonic acid) ("HEPPSO"), 3-[4-(2-Hydroxyethyl)-1-piperazinyl]propanesulfonic acid ("HEPPS"), tricine, glycineamide, bicine, or 3-[[1,3-dihydroxy-2-(hydroxymethyl) propan-2-yl]amino]propane-1-sulfonic acid ("TAPS").

In some embodiments, ammonium can act as a buffer. One or more organic solvents can also be added to the reaction.

The buffer preferably contains metal salts, more preferably metal salts such as halogenides of monovalent or bivalent metals (e.g. CoCl_2 , MnCl_2).

The concentration of these metal salts in the reaction medium is preferably in the range from 1 μM to 1 M, more preferably 1 mM to 100 mM, even more preferably 1 to 10 mM.

Preferably, step (iii) of the method according to the second aspect of the invention is carried out in a phosphate buffer.

The pH of the reaction medium in step (iii) of the method according to the first aspect of the invention is preferably in the range of from 2 to 10, more preferably in the range of from 5 to 8, more preferably 7.2 to 7.5, most preferably 7.5.

Preferably, step (iii) of the method according to the first aspect of the invention is carried out at a temperature in the range of from 20° C. to 70° C., more preferably in the range of from 30° C. to 55° C., most preferably 50° C.

Preferably, the total concentration of all carbamoylases E_1 in the reaction solution in step (iii) is in the range of from 1 μM to 10 mM, preferably 10 μM to 1 mM, more preferably 0.1 mM to 0.5 mM, most preferably 0.4 mM.

In alternative preferred embodiments, the total concentration of all carbamoylases E_1 in the reaction solution in

step (iii) is in the range of from 1 µg/l to 10 g/l, preferably 0.1 mg/l to 5 g/l, more preferably 1 mg/l to 1 g/l, more preferably 5 mg/l to 500 mg/l.

At the end of step (i), a mixture M_I comprising compounds according to formula L-(I) and optionally compounds according to formula D-(I) is obtained, preferably a mixture M_I comprising compounds according to formula L-(I) and compounds according to formula D-(I) is obtained.

4.7 Saponification

The L-glufosinate P ester according to formula L-(I) produced with the method according to the first or second aspect can then be saponified to produce LGA, for example in an acidic aqueous medium, preferably at pH<7, even more preferably at an pH between <6, more preferably at an pH<3, even more preferably at an pH of <1. These saponification conditions are known to the skilled person and described e.g. by H. J. Zeiss, J. Org. Chem. 1991, 56, 1783-1788.

5. EXAMPLES

5.1 Example 1 Identification of Suitable Enzymes and Construction of Plasmids

Genes of different origins encoding a hydantoinase (dihydropyrimidinase, EC 3.5.2.2), L-carbamoylase (N-carbamoyl-L-amino-acid hydrolase, EC 3.5.1.87) and hydantoin racemase (EC 5.1.99.5) were tested for their ability to react with different hydantoin substrates according to formula L-(III) and D-(III) to form the respective enantioselective L-glufosinate derivative according to formula L-(I).

5.1.1 Examined Enzymes

Details of the strains and genes of the respective enzymes that were used in the examples are summarized in table 4.

TABLE 4

Plasmid	Enzyme	Donor Strain	Gene name	SEQ ID NO: of the poly-nucleotides	SEQ ID NO: of the polypeptides
pOM22c	L-carbamoylase	<i>Arthrobacter</i> sp. DSM 3747	hyuC	SEQ ID NO: 27	SEQ ID NO: 1
pOM22c	hydantoinase	<i>Arthrobacter</i> sp. DSM 9771	hyuH	SEQ ID NO: 28	SEQ ID NO: 10
pOM21c	hydantoin racemase	<i>Arthrobacter</i> sp. DSM 3745	hyuR	SEQ ID NO: 29	SEQ ID NO: 17

5.1.2 Cloning of the Enzymes

Cloning of the hydantoin racemase and generation of the plasmid pOM21c (FIG. 1) was carried out as described in example 1 of WO 2004/111227 A2. In particular, a polynucleotide was used which comprised the respective gene (SEQ ID NO: 29) and additional sequences for NdeI and PstI restriction sites.

Cloning of hydantoinase and L-carbamoylase into the rhamnose expression vector pJOE4036 was carried out in a plasmid derivative of the rhamnose expression vector pJOE4036.

Polynucleotides comprising the genes of the respective enzymes (SEQ ID NOs: 27, 28) were synthesized by GeneArt (ThermoFisher Scientific (Waltham, USA)). The polynucleotides carried additional sequences for EcoRI and HindIII restriction sites. Both enzymes were cloned into pJOE4036 using those restriction sites resulting in the plasmid pOM22c, under the control of a rhamnose promoter (FIG. 2).

5.2 Example 2: Production of Strains Positive for Hydantoinase, L-Carbamoylase and Hydantoin Racemase

Chemically competent *E. coli* ET5 cells (as described in WO 2004/042047 A1) were transformed with 10 ng of the plasmid pOM22c generated according to Example 1.

The generated strain which was positive for hydantoinase and carbamoylase was rendered chemically competent and transformed with 10 ng of the plasmid pOM21c.

An *E. coli* ET5 strain transformed with pOM21c or pOM22c was incubated under shaking (250 U/min) at 30° C. for 18 hours in LB medium containing ampicillin (100 µg/l), chloramphenicol (50 µg/l), and 2 g/l rhamnose.

The biomass was separated by centrifugation, resuspended in 50 mM phosphate buffer (pH 7.2) and applied in biotransformation tests in the following examples. The concentration of the biomass in the solution was 40-50 g/l. The solution was used as catalyst ("catalyst 1") in the following.

The concentration of the respective polypeptide carbamoylase, hydantoinase and racemase in the obtained solution may be determined by SDS page and analysis of the respective bands via the software GelQuant® (BiochemLabSolutions).

5.3 Inventive Example 11 and Comparative Examples C1 to C4: Production of P-Alkyl Phosphinothricin Ester Using the Strains Produced in Examples 1 and 2

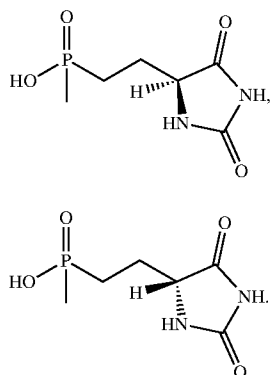
In the following examples, different hydantoin substrates were tested to determine whether the respective polypeptide

catalyzed the reaction of the respective substrate enantioselectively to give the corresponding L amino acid.

5.3.1 Comparative Example C1

In comparative example C1, a racemic mixture Mc1 of 1.25 mmol L-(IV)_{c1} and 1.25 mmol D-(IV)_{c1}, wherein L-(IV)_{c1} and D-(IV)_{c1} are hydantoins with the following formulae, was used as substrate:

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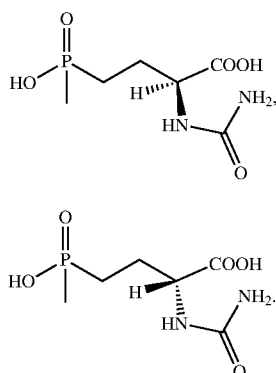
The mixture Mc1 was dissolved in a stirring reactor with 25 ml water. 2.4 g of the catalyst 1 was added, followed by the addition of 50 μ l CoCl_2 solution. The suspension was set to pH 7.5 with 0.5 M NaOH. The total volume was replenished with water to 50 ml, so that the concentration of each enantiomer L-(IV)_{c1} and D-(IV)_{c1} was 0.025 mol/l. The pH was held between 7.0 and 7.5 by HCl-titration or NaOH-titration. The temperature was maintained at 37° C. by a thermostat during the reaction.

The reaction was stopped after 120 hours by addition of 2 N HCl until pH 2.5 was reached. The biomass was separated by centrifugation or filtration.

The enzyme reaction was monitored by ninhydrin test to determine the formation of amino acids. No formation of amino acids was detected by the ninhydrine test.

5.3.2 Comparative Example C2

In comparative example C2, a racemic mixture Mc2 of 1.25 mmol L-(V)_{c2} and 1.25 mmol D-(V)_{c2}, wherein L-(V)_{c2} and D-(V)_{c2} are carbamoylates with the following formulae, was used as substrate:



The mixture Mc2 was dissolved in a stirring reactor with 25 ml water. 2.4 g of the catalyst 1 was added, followed by the addition of 50 μ l CoCl_2 solution. The suspension was set to pH 7.5 with 0.5 M NaOH. The total volume was replenished with water to 50 ml, so that the concentration of each enantiomer L-(V)_{c2} and D-(V)_{c2} was 0.025 mol/l. The pH was held between 7.0 and 7.5 by HCl-titration or NaOH-titration. The temperature was maintained at 37° C. by a thermostat during the reaction.

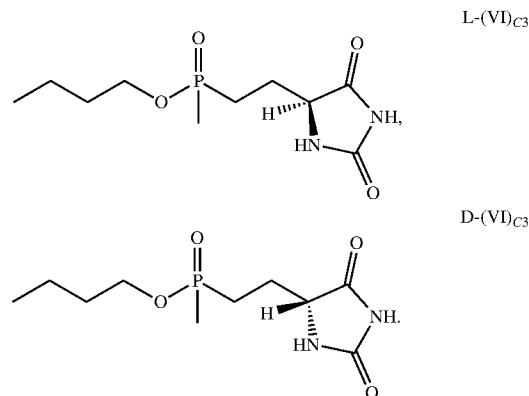
78

The reaction was stopped after 120 hours by addition of 2 N HCl until pH 2.5 was reached. The biomass was separated by centrifugation or filtration.

The enzyme reaction was monitored by ninhydrin test to determine the formation of amino acids. No formation of amino acids was detected by the ninhydrine test.

5.3.3 Example C3 (not According to the Invention)

In example C3, a racemic mixture Mc3 of 1.25 mmol L-(VI)_{c3} and 1.25 mmol D-(VI)_{c3}, wherein L-(VI)_{c3} and D-(VI)_{c3} were hydantoin with the following formulae, was used as substrate:



The mixture M_{II} was dissolved in a stirring reactor with 25 ml water. 2.4 g of the catalyst 1 was added, followed by the addition of 50 μ l CoCl_2 solution. The suspension was set to pH 7.5 with 0.5 M NaOH. The total volume was replenished with water to 50 ml, so that the concentration of each enantiomer L-(VI)_{c3} and D-(VI)_{c3} was 0.025 mol/l. The pH was held between 7.0 and 7.5 by HCl-titration or NaOH-titration. The temperature was maintained at 37° C. by a thermostat during the reaction.

The reaction was stopped after 120 hours by addition of 2 N HCl until pH 2.5 was reached. The biomass was separated by centrifugation or filtration.

The enzyme reaction was monitored by ninhydrin test to determine the formation of amino acids. Formation of amino acids was detected by the ninhydrine test.

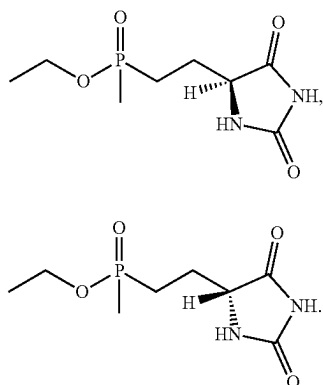
In a final step the reaction mixture was saponified at 100° C. for 10 hours by adding 6 M HCl to obtain L-glufosinate and, if present, D-glufosinate. The final reaction mixture was analysed by LC-MS with a CR-I column as described under item 5.4 to determine the enantiomeric excess ("ee") of either D- or L-glufosinate. An ee of LGA of 79% over the D-enantiomer was detected. The ee of the L-enantiomer ("ee_L") is determined by the following formula in the context of the invention, wherein m_L and m_D are the detected molar masses of L- and D-glufosinate, respectively:

$$ee_L = \frac{(m_L - m_D) \times 100\%}{(m_L + m_D)}.$$

5.3.4 Example C4 (not According to the Invention)

In example C4, a racemic mixture Mc4 of 1.25 mmol L-(VII)_{c4} and 1.25 mmol D-(VII)_{c4}, wherein L-(VII)_{c4} and D-(VII)_{c4} were hydantoin with the following formulae, was used as substrate:

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The mixture Mc4 was dissolved in a stirring reactor with 25 ml water. 2.4 g of the catalyst 1 was added, followed by the addition of 50 μ l CoCl_2 solution. The suspension was set to pH 7.5 with 0.5 M NaOH. The total volume was replenished with water to 50 ml, so that the concentration of each enantiomer L-(VII)_{c4} and D-(VII)_{c4} was 0.025 mol/l. The pH was held between 7.0 and 7.5 by HCl-titration or NaOH-titration. The temperature was maintained at 37° C. by a thermostat during the reaction.

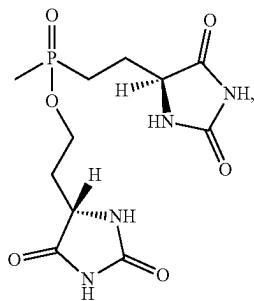
The reaction was stopped after 120 hours by addition of 2 N HCl until pH 2.5 was reached. The biomass was separated by centrifugation or filtration.

The enzyme reaction was monitored by ninhydrin test to determine the formation of amino acids. Formation of amino acids was detected by the ninhydrine test.

In a final step the reaction mixture was saponified at 100° C. for 10 hours by adding 6 M HCl to obtain L-glufosinate and, if present, D-glufosinate. The final reaction mixture was analysed by LC-MS with a CR-I column as described under item 5.4 to determine the enantiomeric excess ("ee") of either D- or L-glufosinate. An ee of LGA of 77% over the D-enantiomer was detected.

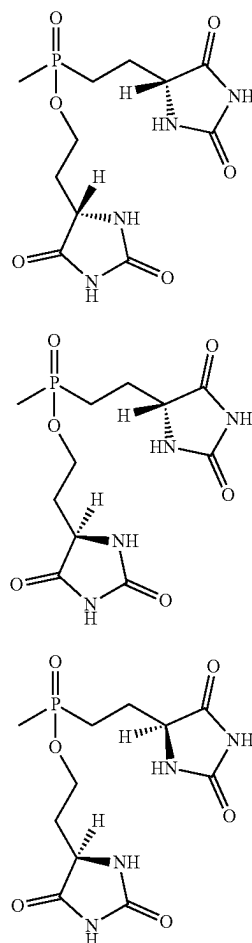
5.3.5 Inventive Example 11

In example 11, a mixture Mm of 0.625 mmol L-(VIII)₁₁, 0.625 mmol L-(VIII)₁₁, 0.625 mmol D-(VIII)₁₁, and 0.625 mmol D-(VIII)₁₁, wherein L-(VIII)₁₁, L-(VIII)₁₁, D-(VIII)₁₁, and D-(VIII)₁₁ were hydantoin with the following formulae, is used as substrate:



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-continued



These compounds may be obtained from mixtures of L-homoserine/D-homoserine by a synthesis method as set forth in CN 111662325 A.

The mixture Mm is dissolved in a stirring reactor with 25 ml water. 2.4 g of the catalyst 1 is added, followed by the addition of 50 μ l CoCl_2 solution. The suspension is set to pH 7.5 with 0.5 M NaOH. The total volume is replenished with water to 50 ml, so that the concentration of each isomer L-(VIII)₁₁, L-(VIII)₁₁, D-(VIII)₁₁, and D-(VIII)₁₁ is 0.0125 mol/l. The pH is held between 7.0 and 7.5 by HCl-titration or NaOH-titration. The temperature is maintained at 37° C. by a thermostat during the reaction.

The reaction is stopped after 120 hours by addition of 2 N HCl until pH 2.5 is reached. The biomass is separated by centrifugation or filtration.

The enzyme reaction is monitored by ninhydrin test to determine the formation of amino acids. Formation of amino acids may be detected by the ninhydrine test.

In a final step the reaction mixture is saponified at 100° C. for 10 hours by adding 6 M HCl to obtain L-glufosinate and, if present, D-glufosinate. The final reaction mixture is analysed by LC-MS with a CR-I column as described under item 5.4 to determine the enantiomeric excess ("ee") of either D- or L-glufosinate. An ee of LGA over the D-enantiomer may be detected (not quantified). The ee of the L-enantiomer ("ee_L") is determined by the following formula in the context of the invention, wherein m_L and m_D are the detected molar masses of L- and D-glufosinate, respectively:

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$$ee_L = \frac{(m_L - m_D) \times 100\%}{(m_L + m_D)}.$$

5.3.5 Results

The results of the examples C1, C2, C3, C4 are summarized in table 5.

TABLE 5

Example	substrate	Ninhydrine test	ee _L
C1	L-(IV) _{C1} , D-(IV) _{C1}	Negative	—
C2	L-(V) _{C2} , D-(V) _{C2}	Negative	—
C3	L-(VI) _{C3} , D-(VI) _{C3}	Positive	79%
C4	L-(VII) _{C4} , D-(VII) _{C4}	Positive	77%

5.3.6 Conclusion

The results as summarized in table 5 show that certain hydantoin esters such as the alkylated hydantoin esters can be used as substrates for the enzymatic enantioselective synthesis of L-glufosinate (C3, C4). In contrast, substrates in which the phosphinic acid group is not protected are not accepted by the catalytic system (C1, C2). In this regard, C2 suggests that at least one reason for this is that the L-carbamoylase enzyme does not accept the respective substrate in which the phosphinic acid function is not protected by an ester group.

Analogous results are obtained, when compound mixtures according to the scope of the invention, i.e. of compounds according to L-(III) and D-(III), wherein R^{3L} and R^{3D} are preferably identical, such as in inventive example 11, are used.

Moreover, in contrast to the process for enantioselective production according to the prior art (CN 111662325 A), the ee excess could be maintained throughout the process, because saponification is carried out not at the hydantoin stage, but at the stage of the amino acid ester.

5.4 Analytical Methods

L-glufosinate and D-glufosinate were detected by LC-MS ("Liquid Chromatography-Mass Spectrometry") with a chi-

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ral column [Daicel CROWNPAK CR-I(-)] as follows. For hydantoin, a Daicel Chiralpak IA-U column may also be used.

- This detection method may also be used for detection and quantification of the
- the LGA P-(n-butyl) ester according to formula L-(I)(II)* according to Assay A_L (item 4.5.3.1);
 - the D-glufosinate P-(n-butyl) ester according to formula D-(I)(II)* according to Assay A_D (item 4.5.3.2);
 - the carbamoyl LGA P-(n-butyl) ester according to formula L-(II)(II)* according to Assay D_L (item 4.5.9.1);
 - the carbamoyl D-glufosinate P-(n-butyl) ester according to formula D-(II)(II)* according to Assay D_D (item 4.5.9.2);
 - the L-enantiomer of a n-butyl ester of hydantoin glufosinate of the formula L-(III)(II)* according to Assay G (item 4.5.14.3).
 - the D-enantiomer of a n-butyl ester of hydantoin glufosinate of the formula D-(III)(II)* according to Assay G (item 4.5.14.3).

5.4.1 Acquisition Method Details

CR-I(-), 3.0 mm I.D.×150 mm, 5 μm, Crownpak, Part. No. 54784
MS QQQ Mass Spectrometer 6420, Agilent
ESI+
MS2 SIM, Glufosinat m/z 182.06 Unit Positive
Centroid

5.4.2 Source Parameter

Parameter Value (+); Gas Temp (° C.): 350; Gas Flow (l/min): 12; Nebulizer (psi): 25; Capillary (V): 3000; Sampler Module: G1329B; Injection Volume 1.00 μL; 5° C.

5.4.3 Binary Pump Module

G1312B
Flow: 0.200 mL/min;
Channel A: H₂O with trifluoroacetic acid (TFA) pH1.15;
Channel B: ACN (acetonitrile);
isocratic 80.0% A/20.0% B; stop time 7.00 min
Column Comp. Module: G1316A
Temperature 5.0° C.

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 29

<210> SEQ ID NO 1
<211> LENGTH: 414
<212> TYPE: PRT
<213> ORGANISM: *Arthrobacter* sp.
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: 1..414
<223> OTHER INFORMATION: *Arthrobacter* sp. DSM 3747 (strain)
<220> FEATURE:
<221> NAME/KEY: gene
<222> LOCATION: 1..414
<223> OTHER INFORMATION: HyuC

<400> SEQUENCE: 1

Met Thr Leu Gln Lys Ala Gln Ala Glu Arg Ile Glu Lys Glu Ile Trp
1 5 10 15

Glu Leu Ser Arg Phe Ser Ala Glu Gly Pro Gly Val Thr Arg Leu Thr

-continued

20					25					30					
Tyr	Thr	Pro	Glu	His	Ala	Ala	Ala	Arg	Glu	Thr	Leu	Ile	Ala	Ala	Met
	35						40					45			
Glu	Ala	Ala	Ala	Leu	Ser	Val	Arg	Glu	Asp	Ala	Leu	Gly	Asn	Ile	Ile
	50					55					60				
Gly	Arg	Arg	Glu	Gly	Thr	Asp	Pro	Gln	Leu	Pro	Ala	Ile	Ala	Val	Gly
	65					70					75				80
Ser	His	Phe	Asp	Ser	Val	Arg	Asn	Gly	Gly	Met	Phe	Asp	Gly	Thr	Ala
				85					90					95	
Gly	Val	Val	Cys	Ala	Leu	Glu	Ala	Ala	Arg	Val	Met	Leu	Glu	Ser	Gly
			100						105					110	
Tyr	Val	Asn	Arg	His	Pro	Phe	Glu	Phe	Ile	Ala	Ile	Val	Glu	Glu	Glu
		115					120					125			
Gly	Ala	Arg	Phe	Ser	Ser	Gly	Met	Leu	Gly	Gly	Arg	Ala	Ile	Ala	Gly
	130					135					140				
Leu	Val	Ala	Asp	Arg	Glu	Leu	Asp	Ser	Leu	Val	Asp	Glu	Asp	Gly	Val
	145					150					155				160
Ser	Val	Arg	Gln	Ala	Ala	Thr	Ala	Phe	Gly	Leu	Lys	Pro	Gly	Glu	Leu
			165						170					175	
Gln	Ala	Ala	Ala	Arg	Ser	Ala	Ala	Asp	Leu	Arg	Ala	Phe	Ile	Glu	Leu
			180						185					190	
His	Ile	Glu	Gln	Gly	Pro	Ile	Leu	Glu	Gln	Glu	Gln	Ile	Glu	Ile	Gly
		195					200					205			
Val	Val	Thr	Ser	Ile	Val	Gly	Val	Arg	Ala	Leu	Arg	Val	Ala	Val	Lys
		210				215					220				
Gly	Arg	Ser	Ala	His	Ala	Gly	Thr	Thr	Pro	Met	His	Leu	Arg	Gln	Asp
	225					230					235				240
Ala	Leu	Val	Pro	Ala	Ala	Leu	Met	Val	Arg	Glu	Val	Asn	Arg	Phe	Val
			245						250					255	
Asn	Glu	Ile	Ala	Asp	Gly	Thr	Val	Ala	Thr	Val	Gly	His	Leu	Thr	Val
		260							265					270	
Ala	Pro	Gly	Gly	Gly	Asn	Gln	Val	Pro	Gly	Glu	Val	Glu	Phe	Thr	Leu
		275					280						285		
Asp	Leu	Arg	Ser	Pro	His	Glu	Glu	Ser	Leu	Arg	Val	Leu	Ile	Asn	Arg
	290					295					300				
Ile	Ser	Val	Met	Val	Gly	Glu	Val	Ala	Ser	Gln	Ala	Gly	Val	Ala	Ala
	305					310					315				320
Asp	Val	Asp	Glu	Phe	Phe	Asn	Leu	Ser	Pro	Val	Gln	Leu	Ala	Pro	Thr
			325						330					335	
Met	Val	Asp	Ala	Val	Arg	Glu	Ala	Ala	Ser	Ala	Leu	Gln	Phe	Thr	His
			340						345					350	
Arg	Asp	Ile	Ser	Ser	Gly	Ala	Gly	His	Asp	Ser	Met	Phe	Ile	Ala	Gln
		355					360						365		
Val	Thr	Asp	Val	Gly	Met	Val	Phe	Val	Pro	Ser	Arg	Ala	Gly	Arg	Ser
		370					375				380				
His	Val	Pro	Glu	Glu	Trp	Thr	Asp	Phe	Asp	Asp	Leu	Arg	Lys	Gly	Thr
		385					390				395				400
Glu	Val	Val	Leu	Arg	Val	Met	Lys	Ala	Leu	Asp	Arg	Gly	Ser		
			405						410						

<210> SEQ ID NO 2

<211> LENGTH: 409

<212> TYPE: PRT

<213> ORGANISM: Geobacillus stearothermophilus

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<220> FEATURE:
<221> NAME/KEY: gene
<222> LOCATION: 1..409
<223> OTHER INFORMATION: AmaB

<400> SEQUENCE: 2

Met Ile Gln Gly Glu Arg Leu Trp Gln Arg Leu Met Glu Leu Gly Glu
 1             5             10             15
Val Gly Lys Gln Pro Ser Gly Gly Val Thr Arg Leu Ser Phe Thr Ala
 20             25             30
Glu Glu Arg Arg Ala Lys Asp Leu Val Ala Ser Tyr Met Arg Glu Ala
 35             40             45
Gly Leu Phe Val Tyr Glu Asp Ala Ala Gly Asn Leu Ile Gly Arg Lys
 50             55             60
Glu Gly Thr Asn Pro Asp Ala Thr Val Val Leu Val Gly Ser His Leu
 65             70             75             80
Asp Ser Val Tyr Asn Gly Gly Cys Phe Asp Gly Pro Leu Gly Val Leu
 85             90             95
Ala Gly Val Glu Val Val Gln Thr Met Asn Glu His Gly Val Val Thr
 100            105            110
His His Pro Ile Glu Val Val Ala Phe Thr Asp Glu Glu Gly Ala Arg
 115            120            125
Phe Arg Phe Gly Met Ile Gly Ser Arg Ala Met Ala Gly Thr Leu Pro
 130            135            140
Pro Glu Ala Leu Glu Cys Arg Asp Ala Glu Gly Ile Ser Leu Ala Glu
 145            150            155            160
Ala Met Lys Gln Ala Gly Leu Asp Pro Asp Arg Leu Pro Gln Ala Ala
 165            170            175
Arg Lys Pro Gly Thr Val Lys Ala Tyr Val Glu Leu His Ile Glu Gln
 180            185            190
Gly Arg Val Leu Glu Glu Thr Gly Leu Pro Val Gly Ile Val Thr Gly
 195            200            205
Ile Ala Gly Leu Ile Trp Val Lys Phe Thr Ile Glu Gly Lys Ala Glu
 210            215            220
His Ala Gly Ala Thr Pro Met Ser Leu Arg Arg Asp Pro Met Ala Ala
 225            230            235            240
Ala Ala Gln Ile Ile Ile Val Ile Glu Glu Glu Ala Arg Arg Thr Gly
 245            250            255
Thr Thr Val Gly Thr Val Gly Gln Leu His Val Tyr Pro Gly Gly Ile
 260            265            270
Asn Val Ile Pro Glu Arg Val Glu Phe Val Leu Asp Leu Arg Asp Leu
 275            280            285
Lys Ala Glu Val Arg Asp Gln Val Trp Lys Ala Ile Ala Val Arg Ala
 290            295            300
Glu Thr Ile Ala Lys Glu Arg Asn Val Arg Val Thr Thr Glu Arg Leu
 305            310            315            320
Gln Glu Met Pro Pro Val Leu Cys Ser Asp Glu Val Lys Arg Ala Ala
 325            330            335
Glu Ala Ala Cys Gln Lys Leu Gly Tyr Pro Ser Phe Trp Leu Pro Ser
 340            345            350
Gly Ala Ala His Asp Ser Val Gln Leu Ala Pro Ile Cys Pro Ile Gly
 355            360            365
Met Ile Phe Val Arg Ser Gln Asp Gly Val Ser His Ser Pro Ala Glu
 370            375            380

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Trp Ser Thr Lys Glu Asp Cys Ala Ala Gly Ala Glu Val Leu Tyr His
385 390 395 400

Thr Val Trp Gln Leu Ala Gln Gly Glu
405

<210> SEQ ID NO 3
 <211> LENGTH: 420
 <212> TYPE: PRT
 <213> ORGANISM: Pseudomonas sp.
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: 1..420
 <223> OTHER INFORMATION: Pseudomonas sp. QR-101 (strain)
 <220> FEATURE:
 <221> NAME/KEY: gene
 <222> LOCATION: 1..420
 <223> OTHER INFORMATION: AtcC

<400> SEQUENCE: 3

Met Ser Gly Val Asn Ser Met Asn Arg Thr Pro Gly Ile Gln Pro Glu
1 5 10 15

Arg Leu Trp Gly Asp Leu Met Ser Thr Ala Lys Phe Gly Ala Leu Gly
20 25 30

Glu Thr Gly Met Arg Arg Leu Ala Leu Ser Asn Glu Asp Arg Gln Val
35 40 45

Arg Asp Trp Phe Val Ala Gln Cys Arg Glu Leu Gly Cys Thr Ile Asp
50 55 60

Ile Asp Gln Ile Gly Asn Ile Phe Ala Thr Tyr Pro Gly Leu Asp Ala
65 70 75 80

Ser Leu Ala Pro Ile Ala Met Gly Ser His Leu Asp Thr Gln Pro Ala
85 90 95

Gly Gly Arg Phe Asp Gly Ile Leu Gly Val Leu Ala Gly Ile Glu Val
100 105 110

Leu Arg Ser Leu His Asp Ala Ser Val Arg Pro Ala His Pro Ile Thr
115 120 125

Val Ile Val Trp Thr Asn Glu Glu Gly Ser Arg Phe Ala Pro Ala Met
130 135 140

Met Gly Ser Gly Val Tyr Cys Gly Ala His Gln Trp Glu Thr Val Ala
145 150 155 160

Ala Thr Cys Asp Lys Ser Gly Val Ser Val Ala Gln Ala Leu Asp Ala
165 170 175

Ile Gly Tyr Ala Gly Glu His Glu Pro Gly Phe Met Gln Phe Ser Ala
180 185 190

Tyr Leu Glu Leu His Ile Glu Gln Gly Pro Val Leu Glu Ala Glu Asn
195 200 205

Ile Glu Ile Gly Val Val Asp Ala Val Gln Gly Val Cys Trp Leu Asp
210 215 220

Ile Lys Val Pro Gly Val Ser Ala His Ala Gly Gly Arg Pro Met Thr
225 230 235 240

Met Arg Asp Asp Ala Leu Val Ala Ala Ser Lys Ile Val Leu Ala Val
245 250 255

Glu Ala Val Ala Ser Ala His Leu Pro Gly Val Gly Thr Ile Gly Tyr
260 265 270

Ile Ser Ala Gly Pro Asn Ser Arg Asn Val Ile Pro Gly Thr Val Ala
275 280 285

Leu Glu Val Asp Leu Arg His Pro Asp Asp Ala Glu Leu Ser Ala Leu
290 295 300

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Glu Asn Glu Val Thr Ala Glu Ile Lys Arg Ile Cys Pro Ala Ala Glu
 305 310 315 320
 Val Gln Arg Val Trp Arg Lys Pro Pro Val Thr Phe Asp Ser Arg Ile
 325 330 335
 Val Asp Ala Ile Ala Ala Arg Thr Glu Ala Leu Gly Tyr Ser Ala Arg
 340 345 350
 Arg Met Val Ser Gly Ala Gly His Asp Ala Ala His Val Ala Gly Ile
 355 360 365
 Ala Pro Ser Ala Met Ile Phe Ile Pro Ser Tyr Leu Gly Leu Ser His
 370 375 380
 Asn Val Arg Glu Tyr Ser Ser Pro Glu Gln Cys Ala Gln Gly Ala Thr
 385 390 395 400
 Val Leu Leu Gly Ala Val Leu Asp Leu Asp Asn Leu Leu Leu Ala Asp
 405 410 415
 Ser Arg Thr Gly
 420

<210> SEQ ID NO 4
 <211> LENGTH: 409
 <212> TYPE: PRT
 <213> ORGANISM: Geobacillus kaustophilus
 <220> FEATURE:
 <221> NAME/KEY: gene
 <222> LOCATION: 1..409
 <223> OTHER INFORMATION: Lnc

<400> SEQUENCE: 4

Met Ile Gln Gly Glu Arg Leu Trp Gln Arg Leu Met Glu Leu Gly Glu
 1 5 10 15
 Val Gly Lys Lys Pro Ser Gly Gly Val Thr Arg Leu Ser Phe Thr Ala
 20 25 30
 Glu Glu Arg Arg Ala Lys Asp Leu Val Ala Ser Tyr Met Arg Glu Ala
 35 40 45
 Gly Leu Phe Val Tyr Glu Asp Thr Ala Gly Asn Leu Ile Gly Arg Lys
 50 55 60
 Glu Gly Ala Asn Pro Asp Ala Pro Val Val Leu Val Gly Ser His Leu
 65 70 75 80
 Asp Ser Val Tyr Asn Gly Gly Cys Phe Asp Gly Pro Leu Gly Val Leu
 85 90 95
 Ala Gly Val Glu Val Val Gln Thr Met Asn Glu His Gly Val Val Thr
 100 105 110
 His His Pro Ile Glu Val Val Ala Phe Thr Asp Glu Glu Gly Ala Arg
 115 120 125
 Phe Arg Phe Gly Met Ile Gly Ser Arg Ala Met Ala Gly Thr Leu Leu
 130 135 140
 Pro Glu Ala Leu Glu Cys Arg Asp Ala Asn Gly Ile Ser Ile Ala Glu
 145 150 155 160
 Ala Met Arg Gln Thr Gly Leu Asp Pro Asp Arg Leu Pro Gln Ala Ala
 165 170 175
 Arg Lys Pro Gly Thr Val Lys Ala Tyr Val Glu Leu His Ile Glu Gln
 180 185 190
 Gly Arg Val Leu Glu Glu Ala Gly Leu Pro Val Gly Ile Val Thr Gly
 195 200 205
 Ile Ala Gly Leu Ile Trp Val Lys Phe Ile Ile Glu Gly Lys Ala Glu
 210 215 220
 His Ala Gly Ala Thr Pro Met Ser Leu Arg Arg Asp Pro Met Ala Ala

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225	230	235	240
Ala Ala Gln Ile Ile Thr Val Ile Glu Glu Glu Ala Arg Arg Thr Gly			
	245	250	255
Thr Thr Val Gly Thr Val Gly Gln Leu His Val Tyr Pro Gly Gly Ile			
	260	265	270
Asn Val Ile Pro Glu Arg Val Glu Phe Val Leu Asp Leu Arg Asp Leu			
	275	280	285
Lys Ala Glu Val Arg Asp Gln Val Trp Asn Asp Ile Ala Ser Arg Ala			
	290	295	300
Glu Thr Ile Ala Lys Glu Arg Asn Val Arg Leu Thr Thr Glu Arg Leu			
	305	310	315
Gln Glu Met Ala Pro Val Leu Cys Ser Glu Val Val Lys Gln Ala Ala			
	325	330	335
Glu Arg Ala Cys Lys Gln Leu Gly Tyr Pro Pro Phe Trp Leu Pro Ser			
	340	345	350
Gly Ala Ala His Asp Gly Val Gln Leu Ala Pro Ile Cys Pro Ile Gly			
	355	360	365
Met Ile Phe Val Arg Ser Gln Asp Gly Val Ser His Ser Pro Ala Glu			
	370	375	380
Trp Ser Thr Lys Glu Asp Cys Ala Val Gly Ala Glu Val Leu Tyr His			
	385	390	395
Thr Val Trp Gln Leu Ala Gln Gly Glu			
	405		

<210> SEQ ID NO 5
 <211> LENGTH: 412
 <212> TYPE: PRT
 <213> ORGANISM: Paenarthrobacter aurescens
 <220> FEATURE:
 <221> NAME/KEY: gene
 <222> LOCATION: 1..412
 <223> OTHER INFORMATION: HyuC

<400> SEQUENCE: 5

Met Thr Leu Gln Lys Ala Gln Ala Glu Arg Ile Glu Lys Glu Ile Arg			
1	5	10	15
Glu Leu Ser Arg Phe Ser Ala Glu Gly Pro Gly Val Thr Arg Leu Thr			
	20	25	30
Tyr Thr Pro Glu His Ala Ala Ala Arg Glu Thr Leu Ile Ala Ala Met			
	35	40	45
Lys Ala Ala Ala Leu Ser Val Arg Glu Asp Ala Leu Gly Asn Ile Ile			
	50	55	60
Gly Arg Arg Glu Gly Thr Asp Pro Glu Leu Pro Ala Ile Ala Val Gly			
	65	70	75
Ser His Phe Asp Ser Val Arg Asn Gly Gly Met Phe Asp Gly Thr Ala			
	85	90	95
Gly Val Val Cys Ala Leu Glu Ala Ala Arg Val Met Leu Glu Asn Gly			
	100	105	110
Tyr Val Asn Arg His Pro Phe Glu Phe Ile Ala Ile Val Glu Glu Glu			
	115	120	125
Gly Ala Arg Phe Ser Ser Gly Met Leu Gly Gly Arg Ala Ile Ala Gly			
	130	135	140
Leu Val Ala Asp Arg Glu Leu Asp Ser Leu Val Asp Glu Asp Gly Val			
	145	150	155
Ser Val Arg Gln Ala Ala Thr Ala Phe Gly Leu Lys Pro Gly Glu Leu			
	165	170	175

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Gln Ala Ala Ala Arg Ser Ala Ala Asp Leu Arg Ala Phe Ile Glu Leu
 180 185 190
 His Ile Glu Gln Gly Pro Ile Leu Glu Gln Glu Gln Ile Glu Ile Gly
 195 200 205
 Val Val Thr Ser Ile Val Gly Val Arg Ala Leu Arg Val Ala Val Lys
 210 215 220
 Gly Arg Ser Asp His Ala Gly Thr Thr Pro Met His Leu Arg Gln Asp
 225 230 235 240
 Ala Leu Val Pro Ala Ala Leu Met Val Arg Glu Val Asn Arg Phe Val
 245 250 255
 Asn Glu Ile Ala Asp Gly Thr Val Ala Thr Val Gly His Leu Thr Val
 260 265 270
 Ala Pro Gly Gly Gly Asn Gln Val Pro Gly Glu Val Asp Phe Thr Leu
 275 280 285
 Asp Leu Arg Ser Pro His Glu Glu Ser Leu Arg Val Leu Ile Asp Arg
 290 295 300
 Ile Ser Val Met Val Gly Glu Val Ala Ser Gln Ala Gly Val Ala Ala
 305 310 315 320
 Asp Val Asp Glu Phe Phe Asn Leu Ser Pro Val Gln Leu Ala Pro Thr
 325 330 335
 Met Val Asp Ala Val Arg Glu Ala Ala Ser Ala Leu Gln Phe Thr His
 340 345 350
 Arg Asp Ile Ser Ser Gly Ala Gly His Asp Ser Met Phe Ile Ala Gln
 355 360 365
 Val Thr Asp Val Gly Met Val Phe Val Pro Ser Arg Ala Gly Arg Ser
 370 375 380
 His Val Pro Glu Glu Trp Thr Asp Phe Asp Asp Leu Arg Lys Gly Thr
 385 390 395 400
 Glu Val Val Leu Arg Val Met Lys Ala Leu Asp Arg
 405 410

<210> SEQ ID NO 6
 <211> LENGTH: 416
 <212> TYPE: PRT
 <213> ORGANISM: Sinorhizobium meliloti
 <220> FEATURE:
 <221> NAME/KEY: gene
 <222> LOCATION: 1..416
 <223> OTHER INFORMATION: SinmeB_2280

<400> SEQUENCE: 6

Met Ala Ala Pro Gly Glu Asn Arg Arg Val Asn Ala Asp Arg Leu Trp
 1 5 10 15
 Asp Ser Leu Met Glu Met Ala Lys Ile Gly Pro Gly Val Ala Gly Gly
 20 25 30
 Asn Asn Arg Gln Thr Leu Thr Asp Ala Asp Gly Glu Gly Arg Arg Leu
 35 40 45
 Phe Gln Ser Trp Cys Glu Glu Ala Gly Leu Ser Met Gly Val Asp Lys
 50 55 60
 Met Gly Thr Met Phe Leu Thr Arg Pro Gly Thr Asp Pro Asp Ala Leu
 65 70 75 80
 Pro Val His Ile Gly Ser His Leu Asp Thr Gln Pro Thr Gly Gly Lys
 85 90 95
 Phe Asp Gly Val Leu Gly Val Leu Ser Gly Leu Glu Ala Val Arg Thr
 100 105 110

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Met	Asn	Asp	Leu	Gly	Ile	Lys	Thr	Lys	His	Pro	Ile	Val	Val	Thr	Asn	
	115						120					125				
Trp	Thr	Asn	Glu	Glu	Gly	Ala	Arg	Phe	Ala	Pro	Ala	Met	Leu	Ala	Ser	
	130					135					140					
Gly	Val	Phe	Ala	Gly	Val	His	Thr	Leu	Glu	Tyr	Ala	Tyr	Ala	Arg	Lys	
145					150					155					160	
Asp	Pro	Glu	Gly	Lys	Ser	Phe	Gly	Asp	Glu	Leu	Lys	Arg	Ile	Gly	Trp	
			165						170					175		
Leu	Gly	Asp	Glu	Glu	Val	Gly	Ala	Arg	Lys	Met	His	Ala	Tyr	Phe	Glu	
		180						185					190			
Tyr	His	Ile	Glu	Gln	Gly	Pro	Ile	Leu	Glu	Ala	Glu	Asn	Lys	Gln	Ile	
	195					200						205				
Gly	Val	Val	Thr	His	Cys	Gln	Gly	Leu	Trp	Trp	Leu	Glu	Phe	Thr	Leu	
	210					215					220					
Thr	Gly	Arg	Glu	Ala	His	Thr	Gly	Ser	Thr	Pro	Met	Asp	Met	Arg	Val	
225					230					235					240	
Asn	Ala	Gly	Leu	Ala	Met	Ala	Arg	Ile	Leu	Glu	Met	Val	Gln	Thr	Val	
			245					250						255		
Ala	Met	Glu	Asn	Gln	Pro	Gly	Ala	Val	Gly	Gly	Val	Gly	Gln	Met	Phe	
		260						265					270			
Phe	Ser	Pro	Asn	Ser	Arg	Asn	Val	Leu	Pro	Gly	Lys	Val	Val	Phe	Thr	
	275						280					285				
Val	Asp	Ile	Arg	Ser	Pro	Asp	Gln	Ala	Lys	Leu	Asp	Gly	Met	Arg	Ala	
	290					295					300					
Arg	Ile	Glu	Ala	Glu	Ala	Pro	Lys	Ile	Cys	Glu	Arg	Leu	Gly	Val	Gly	
305					310					315					320	
Cys	Ser	Ile	Glu	Ala	Val	Gly	His	Phe	Asp	Pro	Val	Thr	Phe	Asp	Pro	
			325						330					335		
Lys	Leu	Val	Glu	Thr	Val	Arg	Gly	Ala	Ala	Glu	Lys	Leu	Gly	Tyr	Ser	
		340						345					350			
His	Met	Asn	Leu	Val	Ser	Gly	Ala	Gly	His	Asp	Ala	Cys	Trp	Ala	Ala	
	355						360					365				
Lys	Val	Ala	Pro	Thr	Thr	Met	Ile	Met	Cys	Pro	Cys	Val	Gly	Gly	Leu	
	370					375					380					
Ser	His	Asn	Glu	Ala	Glu	Asp	Ile	Ser	Arg	Glu	Trp	Ala	Ala	Ala	Gly	
385					390					395					400	
Ala	Asp	Val	Leu	Phe	His	Ala	Val	Leu	Glu	Thr	Ala	Glu	Ile	Val	Glu	
			405						410					415		

<210> SEQ ID NO 7
 <211> LENGTH: 413
 <212> TYPE: PRT
 <213> ORGANISM: Bacillus fordii
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: 1..413
 <223> OTHER INFORMATION: Bacillus fordii MH602 (strain)

<400> SEQUENCE: 7

Met	Glu	Lys	Gln	Lys	Val	Leu	Ile	Asn	Gly	Glu	Arg	Leu	Lys	Asp	Thr	
1				5					10					15		
Ile	Glu	Glu	Phe	Ala	Asp	Phe	Gly	Arg	Thr	Glu	Lys	Asn	Gly	Val	Thr	
			20				25						30			
Arg	Leu	Ala	Leu	Ser	Asp	Val	Asp	Val	Lys	Ala	Arg	Arg	His	Phe	Gln	
	35					40					45					
Ser	Leu	Cys	Glu	Gln	Leu	Gly	Met	Ser	Val	Val	Trp	Asp	Asp	Met	Gly	

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50	55	60
Asn Met Tyr Ala Lys Leu Pro Gly Ile Asp Asn Asp Gln Pro Pro Val		
65	70	75 80
Val Ile Gly Ser His Leu Asp Ser Val Lys Lys Gly Gly Arg Phe Asp		
	85	90 95
Gly Thr Leu Gly Val Leu Thr Gly Leu Glu Val Val Arg Thr Met Val		
	100	105 110
Glu Asn Gly Ile Lys Pro Glu Ile Pro Ile Ile Val Ala Asn Ile Thr		
	115	120 125
Asn Glu Glu Gly Ala Arg Phe Glu Pro Ser Leu Met Ala Ser Gly Val		
	130	135 140
Leu Ser Gly Arg Phe Asp Lys Ala Ala Met Leu Lys Ser Thr Asp Val		
	145	150 155 160
Asp Gly Ile Thr Phe Ala Glu Ala Leu Lys Lys Ser Gly Tyr Glu Gly		
	165	170 175
Lys Lys Glu Asn Arg Leu Lys Glu Ala Ala Ala Phe Leu Glu Leu His		
	180	185 190
Ile Glu Gln Gly Pro Val Leu Glu Ser Glu Asp Ile Gln Ile Gly Ile		
	195	200 205
Val Glu Cys Val Val Gly Met Val Cys Phe Glu Ile Glu Val Thr Gly		
	210	215 220
Glu Ser Asp His Ala Gly Thr Thr Pro Met Ser Met Arg Lys Asp Ala		
	225	230 235 240
Leu Phe Ala Ala Asn Gln Leu Ile Ser Glu Ile Arg Gln Lys Met Asn		
	245	250 255
Arg Leu Asp Asp Gln Leu Val Tyr Thr Val Gly Arg Met Thr Val Ser		
	260	265 270
Pro Asn Ile His Thr Val Ile Pro Asn Lys Val Val Phe Thr Ile Gly		
	275	280 285
Ala Arg His Gln Asp Gly Lys Ile Ile Arg Gln Val Glu Glu Ile Ile		
	290	295 300
Gln Gly Leu Pro Asn Ser Ser Gly Lys Glu Lys Cys Asn Val Thr Thr		
	305	310 315 320
Thr Lys Leu Trp Asp Arg His Thr Val Trp Phe Asn Glu Glu Ile Val		
	325	330 335
Asn Thr Leu Glu Lys Ser Ala Arg Ser Leu Gly Tyr Ser Phe Lys Arg		
	340	345 350
Met Val Ser Gly Ala Gly His Asp Ala Gln Phe Ile Ala Thr Tyr Ile		
	355	360 365
Pro Thr Ala Met Val Phe Val Pro Ser Ile Asn Gly Lys Ser His Asp		
	370	375 380
Glu Asp Glu Leu Thr Thr Trp Glu Asp Cys Glu Asn Gly Val Asn Val		
	385	390 395 400
Ile Leu Gln Thr Val Leu Asp Leu Thr Thr Asp Lys Leu		
	405	410

<210> SEQ ID NO 8

<211> LENGTH: 412

<212> TYPE: PRT

<213> ORGANISM: Arthrobacter sp.

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: 1..412

<223> OTHER INFORMATION: Arthrobacter sp. BT801 (strain)

<220> FEATURE:

<221> NAME/KEY: gene

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<222> LOCATION: 1..412

<223> OTHER INFORMATION: HyuC

<400> SEQUENCE: 8

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Met Thr Ser Leu Arg Ala Gln Ala Asp Arg Ile Glu Lys Asp Ile Arg
 1          5          10          15

Asp Leu Ala Arg Phe Ser Ala Gly Gly Pro Gly Val Thr Arg Leu Ser
      20          25          30

Tyr Thr Pro Glu His Ala Ala Ala Arg Asp Leu Ile Ile Ala Ala Met
      35          40          45

Gln Gln Ala Gly Leu Asp Val Arg Glu Asp Gly Leu Gly Asn Ile Thr
      50          55          60

Gly Arg Arg Glu Gly Ser Asp Pro Asp Leu Pro Ala Ile Ala Val Gly
      65          70          75          80

Ser His Phe Asp Ser Val Arg Asn Gly Gly Met Phe Asp Gly Thr Ala
      85          90          95

Gly Val Val Cys Ala Leu Glu Ala Ala Arg Val Val Asn Glu Ser Glu
      100          105          110

Tyr Val Asn Arg His Pro Phe Glu Phe Ile Ala Ile Val Glu Glu Glu
      115          120          125

Gly Ser Arg Phe Asn Ser Gly Met Leu Gly Gly Arg Ala Leu Ala Gly
      130          135          140

Leu Val Thr Asp Glu Asp Leu Asp Lys Leu Lys Asp Glu Asp Gly Ile
      145          150          155          160

Ser Val Arg Asp Ala Ala Thr Glu Phe Gly Leu Gln Pro Gly Asn Leu
      165          170          175

Gln Thr Ser Val Arg Ser Arg Leu Asp Leu Arg Ala Phe Ile Glu Pro
      180          185          190

His Ile Glu Gln Gly Pro Val Leu Glu Gln Glu Gly Ile Glu Ile Gly
      195          200          205

Val Val Thr Ser Ile Val Gly Ile Arg Thr Leu Arg Val Ala Val Leu
      210          215          220

Gly Arg Ser Asp His Ala Gly Thr Thr Pro Met His Leu Arg Gln Asp
      225          230          235          240

Ala Leu Ile Pro Ser Ala Leu Met Val Arg Glu Val Asn Arg Leu Val
      245          250          255

Asn Glu Leu Gly Asp Glu Thr Val Ala Thr Val Gly His Leu Thr Val
      260          265          270

Ala Pro Gly Gly Ile Asn Gln Val Pro Gly Glu Val Asn Phe Thr Leu
      275          280          285

Asp Leu Arg Ser Pro His Glu Glu Ser Leu Lys Gln Ile Val Lys Gln
      290          295          300

Ile Thr Val Met Ile Gln Glu Val Ala Ser Gln Ala Glu Val Thr Ala
      305          310          315          320

Asn Val Glu Glu Phe Phe Ser Leu Asp Pro Val Pro Leu Ala Pro Ser
      325          330          335

Val Val Asp Ala Val Arg Gln Ala Ala Ser Asn Leu Arg Phe Ser His
      340          345          350

Arg Asp Met Pro Ser Gly Ala Gly His Asp Ser Met Phe Ile Ser Gln
      355          360          365

Val Thr Asp Val Gly Met Val Phe Val Pro Ser Arg Asp Gly Arg Ser
      370          375          380

His Val Pro Glu Glu Trp Ser Asp Phe Asp Asp Leu Arg Lys Gly Thr
      385          390          395          400

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Asp Val Val Leu Gly Val Met Thr Ala Leu Asp Lys
 405 410

<210> SEQ ID NO 9
 <211> LENGTH: 412
 <212> TYPE: PRT
 <213> ORGANISM: Microbacterium liquefaciens
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: 1..412
 <223> OTHER INFORMATION: Microbacterium liquefaciens AJ 3912 (strain)

<400> SEQUENCE: 9

Met Thr Leu Gln Gln Ala Arg Ala Asp Arg Ile Glu Glu Glu Leu Trp
 1 5 10 15
 Thr Leu Ser Arg Phe Ser Val Glu Gly Pro Gly Val Thr Arg Leu Thr
 20 25 30
 Tyr Thr Pro Glu His Ala Ala Ala Arg Glu Val Ile Val Ala Ala Met
 35 40 45
 Gln Arg Thr Gly Leu Ser Val His Glu Asp Ala Leu Gly Asn Ile Ile
 50 55 60
 Gly Arg Arg Glu Gly Ser Asp Pro Ala Leu Pro Ala Ile Ala Phe Gly
 65 70 75 80
 Ser His Phe Asp Ser Val Arg Asn Gly Gly Met Phe Asp Gly Thr Ala
 85 90 95
 Gly Val Val Cys Ala Leu Glu Ala Ala Arg Val Leu Gln Glu Ser Gly
 100 105 110
 Tyr Val Asn Arg His Pro Leu Glu Val Ile Ala Ile Val Glu Glu Glu
 115 120 125
 Gly Thr Arg Phe Ser Ser Gly Met Leu Gly Gly Arg Ala Ile Ala Gly
 130 135 140
 Leu Val Ser Asp Ala Asp Leu Asp Thr Leu Val Asp Glu Asp Gly Val
 145 150 155 160
 Thr Val Arg Glu Ala Ala Thr Ala Phe Gly Leu Glu Pro Gly Glu Leu
 165 170 175
 Arg Thr Ala Ala Arg Thr Arg Asp Asp Leu Arg Ala Phe Ile Glu Leu
 180 185 190
 His Ile Glu Gln Gly Pro Ile Leu Glu Gln Glu Lys Val Glu Ile Gly
 195 200 205
 Val Val Thr Gly Ile Val Gly Val Arg Ala Phe Arg Ile Thr Val Glu
 210 215 220
 Gly Arg Ser Asp His Ala Gly Thr Thr Pro Met His Leu Arg Gln Asp
 225 230 235 240
 Ala Leu Val Pro Ala Ala Leu Met Val Arg Glu Ile Asn Arg Phe Val
 245 250 255
 Asn Glu Ile Ala Asp Gly Thr Val Ala Thr Val Gly His Leu Thr Val
 260 265 270
 Thr Pro Gly Gly Leu Asn Gln Val Pro Gly Gly Val Glu Phe Thr Leu
 275 280 285
 Asp Leu Arg Ser Pro His Glu Glu Ser Ile Arg Leu Leu Val Asp Arg
 290 295 300
 Ile Glu Ala Met Val Ala Glu Val Ala Ala Ala Gly Val Glu Ala
 305 310 315 320
 Ala Val Asn Gly Phe Phe Ala Leu Ser Pro Val Gly Leu Ser Pro Val
 325 330 335

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Val	Val	Asp	Arg	Val	Arg	Asp	Ala	Ala	Ser	Glu	Leu	Gly	Phe	Thr	His
			340					345						350	
Arg	Asp	Ile	Thr	Ser	Gly	Ala	Gly	His	Asp	Ser	Met	Phe	Ile	Ala	Gln
		355					360					365			
Ile	Thr	Asp	Val	Gly	Met	Val	Phe	Val	Pro	Ser	Arg	Ala	Gly	Arg	Ser
	370					375					380				
His	Val	Pro	Glu	Glu	Trp	Ser	Asp	Phe	Asp	Asp	Leu	Arg	Lys	Gly	Thr
385					390					395					400
Asp	Val	Val	Leu	His	Val	Val	Thr	Ala	Leu	Asp	Arg				
			405						410						

<210> SEQ ID NO 10
 <211> LENGTH: 420
 <212> TYPE: PRT
 <213> ORGANISM: Arthrobacter sp.
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: 1..420
 <223> OTHER INFORMATION: Arthrobacter sp. DSM 9771 (strain)
 <220> FEATURE:
 <221> NAME/KEY: gene
 <222> LOCATION: 1..420
 <223> OTHER INFORMATION: HyuH

<400> SEQUENCE: 10

Met	Phe	Asp	Val	Ile	Val	Lys	Asn	Cys	Arg	Met	Val	Ser	Ser	Asp	Gly
1				5					10					15	
Ile	Thr	Glu	Ala	Asp	Ile	Leu	Val	Lys	Asp	Gly	Lys	Val	Ala	Ala	Ile
		20						25					30		
Ser	Ser	Asp	Thr	Ser	Asp	Val	Glu	Ala	Ser	Arg	Thr	Ile	Asp	Ala	Gly
		35				40					45				
Gly	Lys	Phe	Val	Met	Pro	Gly	Val	Val	Asp	Glu	His	Val	His	Ile	Ile
	50					55					60				
Asp	Met	Asp	Leu	Lys	Asn	Arg	Tyr	Gly	Arg	Phe	Glu	Leu	Asp	Ser	Glu
65					70					75					80
Ser	Ala	Ala	Val	Gly	Gly	Ile	Thr	Thr	Ile	Phe	Glu	Met	Pro	Phe	Thr
			85					90						95	
Phe	Pro	Pro	Thr	Thr	Thr	Leu	Asp	Ala	Phe	Leu	Glu	Lys	Lys	Lys	Gln
			100					105						110	
Ala	Gly	Gln	Arg	Leu	Lys	Val	Asp	Phe	Ala	Leu	Tyr	Gly	Gly	Gly	Val
		115					120					125			
Pro	Gly	Asn	Leu	Pro	Glu	Ile	Arg	Lys	Met	His	Asp	Ala	Gly	Ala	Val
	130					135					140				
Gly	Phe	Lys	Ser	Met	Met	Ala	Ala	Ser	Val	Pro	Gly	Met	Phe	Asp	Ala
145					150					155					160
Val	Ser	Asp	Gly	Glu	Leu	Phe	Glu	Ile	Phe	Gln	Glu	Ile	Ala	Ala	Cys
			165						170						175
Gly	Ser	Val	Ala	Val	Val	His	Ala	Glu	Asn	Glu	Thr	Ile	Ile	Gln	Ala
			180					185						190	
Leu	Gln	Lys	Gln	Ile	Lys	Ala	Ala	Gly	Arg	Lys	Asp	Met	Ala	Ala	Tyr
		195					200					205			
Glu	Ala	Ser	Gln	Pro	Val	Phe	Gln	Glu	Asn	Glu	Ala	Ile	Gln	Arg	Ala
	210					215					220				
Leu	Leu	Leu	Gln	Lys	Glu	Ala	Gly	Cys	Arg	Leu	Ile	Val	Leu	His	Val
225					230					235					240
Ser	Asn	Pro	Asp	Gly	Val	Glu	Leu	Ile	His	Arg	Ala	Gln	Ser	Glu	Gly
				245					250						255

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Gln Asp Val His Cys Glu Ser Gly Pro Gln Tyr Leu Asn Ile Thr Thr
260 265 270

Asp Asp Ala Glu Arg Ile Gly Pro Tyr Met Lys Val Ala Pro Pro Val
275 280 285

Arg Ser Ala Glu Met Asn Val Arg Leu Trp Glu Gln Leu Glu Asn Gly
290 295 300

Leu Ile Asp Thr Leu Gly Ser Asp His Gly Gly His Pro Val Glu Asp
305 310 315 320

Lys Glu Pro Gly Trp Lys Asp Val Trp Lys Ala Gly Asn Gly Ala Leu
325 330 335

Gly Leu Glu Thr Ser Leu Pro Met Met Leu Thr Asn Gly Val Asn Lys
340 345 350

Gly Arg Leu Ser Leu Glu Arg Leu Val Glu Val Met Cys Glu Lys Pro
355 360 365

Ala Lys Leu Phe Gly Ile Tyr Pro Gln Lys Gly Thr Leu Gln Val Gly
370 375 380

Ser Asp Ala Asp Leu Leu Ile Leu Asp Leu Asp Ile Asp Thr Lys Val
385 390 395 400

Asp Ala Ser Gln Phe Arg Ser Leu His Lys Tyr Ser Pro Phe Asp Gly
405 410 415

Met Pro Val Thr
420

<210> SEQ ID NO 11
 <211> LENGTH: 479
 <212> TYPE: PRT
 <213> ORGANISM: Pseudomonas fluorescens

<400> SEQUENCE: 11

Met Ser Leu Leu Ile Arg Gly Ala Thr Val Ile Thr Tyr Asp Glu Arg
1 5 10 15

Tyr Arg Ala Asp Val Leu Cys Ala Gly Gly Leu Ile Arg Ala Ile Gly
20 25 30

Thr Gly Leu Asp Val Pro Ala Gly Thr Glu Val Leu Asp Gly Ser Gly
35 40 45

Gln Tyr Leu Met Pro Gly Gly Ile Asp Pro His Thr His Met Gln Leu
50 55 60

Pro Phe Met Gly Thr Val Ala Ser Glu Asp Phe Phe Ser Gly Thr Ala
65 70 75 80

Ala Gly Leu Ala Gly Gly Thr Thr Ser Ile Ile Asp Phe Val Ile Pro
85 90 95

Asn Pro Gln Gln Ser Leu Met Glu Ala Phe His Gln Trp Arg Gly Trp
100 105 110

Ala Glu Lys Ser Ala Ser Asp Tyr Gly Phe His Val Ala Ile Thr Trp
115 120 125

Trp Ser Glu Gln Val Arg Glu Glu Met Ala Glu Leu Val Ser His His
130 135 140

Gly Ile Asn Ser Phe Lys His Phe Met Ala Tyr Lys Asn Ala Ile Met
145 150 155 160

Ala Ala Asp Asp Thr Leu Val Ala Ser Phe Glu Arg Cys Leu Glu Leu
165 170 175

Gly Ala Val Pro Thr Val His Ala Glu Asn Gly Glu Leu Val Tyr His
180 185 190

Leu Gln Arg Lys Leu Met Ala Gln Gly Ile Thr Gly Pro Glu Ala His
195 200 205

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Pro Leu Ser Arg Pro Ser Gln Val Glu Gly Glu Ala Ala Ser Arg Ala
 210                      215                      220

Ile Arg Ile Ala Glu Thr Ile Gly Thr Pro Leu Tyr Leu Val His Val
225                      230                      235                      240

Ser Thr Lys Glu Ala Leu Asp Glu Ile Thr Tyr Ala Arg Gly Lys Gly
                      245                      250                      255

Gln Pro Val Tyr Gly Glu Val Leu Ala Gly His Leu Leu Leu Asp Asp
                260                      265                      270

Ser Val Tyr Gln His Pro Asp Trp His Thr Ala Ala Gly Tyr Val Met
                275                      280                      285

Ser Pro Pro Phe Arg Pro Arg Gly His Gln Glu Ala Leu Trp His Gly
                290                      295                      300

Leu Gln Ser Gly Asn Leu His Thr Thr Ala Thr Asp His Cys Cys Phe
305                      310                      315                      320

Cys Ala Glu Gln Lys Ala Ala Gly Arg Asp Asp Phe Ser Lys Ile Pro
                325                      330                      335

Asn Gly Thr Ala Gly Ile Glu Asp Arg Met Ala Leu Leu Trp His Glu
                340                      345                      350

Gly Val Asn Thr Gly Arg Leu Ser Met Gln Glu Phe Val Ala Leu Thr
                355                      360                      365

Ser Thr Asn Thr Ala Lys Ile Phe Asn Leu Tyr Pro Arg Lys Gly Ala
                370                      375                      380

Ile Arg Val Gly Ala Asp Ala Asp Leu Val Leu Trp Asp Pro Glu Gly
385                      390                      395                      400

Thr Arg Thr Ile Ser Ala Lys Thr His His Gln Lys Val Asp Phe Asn
                405                      410                      415

Ile Phe Glu Gly Lys Thr Val Arg Gly Val Pro Ser His Thr Ile Ser
                420                      425                      430

Gln Gly Lys Leu Val Trp Ala Asp Gly Asp Leu Arg Ala Glu Arg Gly
                435                      440                      445

Ala Gly Arg Tyr Val Glu Arg Pro Ala Tyr Pro Ser Val Phe Glu Gln
                450                      455                      460

Leu Ser Lys Arg Ala Glu His Ser Arg Pro Thr Ala Val Lys Arg
465                      470                      475

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<210> SEQ ID NO 12
<211> LENGTH: 479
<212> TYPE: PRT
<213> ORGANISM: Pseudomonas aeruginosa
<220> FEATURE:
<221> NAME/KEY: gene
<222> LOCATION: 1..479
<223> OTHER INFORMATION: Dht

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<400> SEQUENCE: 12

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Met Ser Leu Leu Ile Arg Gly Ala Thr Val Val Thr His Glu Glu Ser
 1                      5                      10                      15

Tyr Arg Ala Asp Val Leu Cys Ala Asn Gly Leu Ile Gln Ala Ile Gly
                20                      25                      30

Glu Asn Leu Glu Thr Pro Ser Gly Cys Asp Val Leu Asp Gly Gly Gly
                35                      40                      45

Gln Tyr Leu Met Pro Gly Gly Ile Asp Pro His Thr His Met Gln Leu
                50                      55                      60

Pro Phe Met Gly Thr Val Ala Ser Glu Asp Phe Phe Ser Gly Thr Ala
65                      70                      75                      80

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Ala	Gly	Leu	Ala	Gly	Gly	Thr	Thr	Ser	Ile	Ile	Asp	Phe	Val	Ile	Pro	
				85					90					95		
Asn	Pro	Arg	Gln	Ser	Leu	Leu	Glu	Ala	Phe	His	Thr	Trp	Arg	Gly	Trp	
				100					105					110		
Ala	Gln	Lys	Ser	Ala	Ala	Asp	Tyr	Gly	Phe	His	Val	Ala	Ile	Thr	Trp	
				115					120					125		
Trp	Ser	Asp	Glu	Val	Ala	Arg	Glu	Met	Gly	Glu	Leu	Val	Ala	Gln	His	
				130					135					140		
Gly	Val	Asn	Ser	Phe	Lys	His	Phe	Met	Ala	Tyr	Lys	Asn	Ala	Ile	Met	
				145					150					155		
Ala	Ala	Asp	Asp	Thr	Leu	Val	Ala	Ser	Phe	Glu	Arg	Cys	Leu	Glu	Leu	
				165					170					175		
Gly	Ala	Val	Pro	Thr	Val	His	Ala	Glu	Asn	Gly	Glu	Leu	Val	Phe	His	
				180					185					190		
Leu	Gln	Gln	Lys	Leu	Leu	Ala	Gln	Gly	Leu	Thr	Gly	Pro	Glu	Ala	His	
				195					200					205		
Pro	Leu	Ser	Arg	Pro	Pro	Gln	Val	Glu	Gly	Glu	Ala	Ala	Ser	Arg	Ala	
				210					215					220		
Ile	Arg	Ile	Ala	Glu	Thr	Leu	Gly	Thr	Pro	Leu	Tyr	Leu	Val	His	Ile	
				225					230					235		
Ser	Ser	Arg	Glu	Ala	Leu	Asp	Glu	Ile	Ala	Tyr	Ala	Arg	Ala	Lys	Gly	
				245					250					255		
Gln	Pro	Val	Tyr	Gly	Glu	Val	Leu	Ala	Gly	His	Leu	Leu	Leu	Asp	Asp	
				260					265					270		
Ser	Val	Tyr	Arg	His	Pro	Asp	Trp	Ala	Thr	Ala	Ala	Gly	Tyr	Val	Met	
				275					280					285		
Ser	Pro	Pro	Phe	Arg	Pro	Val	Glu	His	Gln	Glu	Ala	Leu	Trp	Arg	Gly	
				290					295					300		
Leu	Gln	Ser	Gly	Asn	Leu	His	Thr	Thr	Ala	Thr	Asp	His	Cys	Cys	Phe	
				305					310					315		
Cys	Ala	Glu	Gln	Lys	Ala	Met	Gly	Arg	Asp	Asp	Phe	Ser	Lys	Ile	Pro	
				325					330					335		
Asn	Gly	Thr	Ala	Gly	Ile	Glu	Asp	Arg	Met	Ala	Leu	Leu	Trp	Asp	Ala	
				340					345					350		
Gly	Val	Asn	Ser	Gly	Arg	Leu	Ser	Met	His	Glu	Phe	Val	Ala	Leu	Thr	
				355					360					365		
Ser	Thr	Asn	Thr	Ala	Lys	Ile	Phe	Asn	Leu	Phe	Pro	Arg	Lys	Gly	Ala	
				370					375					380		
Ile	Arg	Val	Gly	Ala	Asp	Ala	Asp	Leu	Val	Leu	Trp	Asp	Pro	Gln	Gly	
				385					390					395		
Ser	Arg	Thr	Leu	Ser	Ala	Ala	Thr	His	His	Gln	Arg	Val	Asp	Phe	Asn	
				405					410					415		
Ile	Phe	Glu	Gly	Arg	Thr	Val	Arg	Gly	Ile	Pro	Ser	His	Thr	Ile	Ser	
				420					425					430		
Gln	Gly	Lys	Leu	Leu	Trp	Ala	Ala	Gly	Asp	Leu	Arg	Ala	Glu	Pro	Gly	
				435					440					445		
Ala	Gly	Arg	Tyr	Val	Glu	Arg	Pro	Ala	Tyr	Pro	Ser	Val	Tyr	Glu	Val	
				450					455					460		
Leu	Gly	Arg	Arg	Ala	Glu	Arg	Gln	Arg	Pro	Val	Ala	Val	Glu	Arg		
				465					470					475		

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<210> SEQ ID NO 13
<211> LENGTH: 475
<212> TYPE: PRT
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<213> ORGANISM: Bacillus fordii
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: 1..475
<223> OTHER INFORMATION: Bacillus fordii MH602 (strain)

<400> SEQUENCE: 13

Met Lys Lys Ile Ile Lys Asp Gly Thr Val Val Thr Ser Thr Asp Val
1      5      10      15

Tyr Thr Ala Asp Val Leu Ile Glu Asp Glu Lys Ile Gln Ala Ile Gly
      20      25      30

Gln Asn Ile Ser Asp Ile Tyr Ala Glu Val Ile Asp Ala Ser Gly Tyr
      35      40      45

Tyr Val Met Pro Gly Gly Ile Asp Pro His Thr His Met Asp Met Pro
50      55      60

Phe Gly Gly Thr Val Thr Ala Asp Asp Phe Glu Thr Gly Ser Ile Ala
65      70      75      80

Ala Ala Cys Gly Gly Thr Thr Thr Ile Ile Asp Phe Cys Leu Thr Gly
      85      90      95

Lys Gly Lys Ser Leu Lys Ser Ala Leu Glu Lys Trp His Ala Lys Ser
100     105     110

Ala Gly Lys Ala Val Ile Asp Tyr Gly Phe His Leu Gln Ile Ala Glu
115     120     125

Ala Asn Glu Glu Val Phe Met Glu Met Pro Gln Ile Ile Glu Glu Glu
130     135     140

Gly Val Thr Ser Phe Lys Ile Phe Met Ala Tyr Lys Asp Val Leu Gln
145     150     155     160

Ala Asp Asp Glu Thr Leu Phe Gln Thr Leu Val Thr Ala Arg Glu His
165     170     175

Gly Gly Leu Val Met Val His Ala Glu Asn Gly Asp Val Ile Asn Tyr
180     185     190

Leu Thr Glu Lys Ala Leu Lys Glu Gly Asn Thr Glu Pro Ile Tyr His
195     200     205

Ala Leu Thr Arg Pro Pro Glu Leu Glu Gly Glu Ala Thr Gly Arg Ala
210     215     220

Ala Arg Leu Thr Ala Leu Ala Asp Ser Gln Leu Tyr Val Val His Val
225     230     235     240

Thr Cys Lys Glu Ala Val Glu Gln Ile Ala Glu Ala Arg Arg Arg Gly
245     250     255

Tyr Arg Val Phe Gly Glu Thr Cys Pro Gln Tyr Leu Val Leu Asp Gln
260     265     270

Thr Tyr Leu Glu Arg Pro Asp Phe Glu Gly Ala Lys Tyr Val Trp Ser
275     280     285

Pro Pro Leu Arg Glu Arg Ser Asn Gln Glu Val Leu Trp Asn Ala Leu
290     295     300

Lys Ser Gly Asp Leu Gln Ala Ile Gly Ser Asp His Cys Ser Phe Asn
305     310     315     320

Phe Arg Gly Gln Lys Glu Leu Gly Lys Asn Asp Phe Ser Lys Ile Pro
325     330     335

Asn Gly Gly Pro Phe Val Glu Asp Arg Phe Ser Val Leu Phe Ser Glu
340     345     350

Gly Val Lys Lys Gly Arg Ile Ser Ile His Asp Phe Val Asn Ile Ile
355     360     365

Ser Thr Gln Ala Ala Lys Leu Phe Gly Leu Phe Pro Arg Lys Gly Thr
370     375     380

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Ile Ala Pro Gly Ser Asp Ala Asp Ile Val Ile Phe Asp Pro Asn Val
385 390 395 400

Glu Arg Ile Ile Ser Val Glu Thr His His Met Asn Val Asp Tyr Ser
405 410 415

Ala Leu Glu Gly Leu Lys Ile Ile Gly Glu Pro Ile Thr Val Leu Ser
420 425 430

Arg Gly Asn Tyr Val Val Lys Asp Lys Glu Phe Val Gly Lys Pro Gly
435 440 445

Lys Gly Lys Phe Leu Lys Cys Asn Lys Phe Asn His Asp Leu Tyr Lys
450 455 460

Glu Thr Gly Gln Leu Glu Gly Leu Leu Lys Gly
465 470 475

<210> SEQ ID NO 14

<211> LENGTH: 458

<212> TYPE: PRT

<213> ORGANISM: *Arthrobacter* sp.

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: 1..458

<223> OTHER INFORMATION: *Arthrobacter* sp. BT801 (strain)

<220> FEATURE:

<221> NAME/KEY: gene

<222> LOCATION: 1..458

<223> OTHER INFORMATION: HyuH

<400> SEQUENCE: 14

Met Phe Asp Val Ile Val Lys Asn Cys Arg Ile Val Ser Ser Glu Gly
1 5 10 15

Ile Thr Glu Ala Asp Ile Leu Val Lys Glu Gly Lys Val Ala Ala Ile
20 25 30

Ser Ser Glu Gly Leu Asn Asp Glu Val Gln Glu Thr Ile Asp Ala Gly
35 40 45

Gly Lys Phe Val Leu Pro Gly Val Val Asp Glu His Val His Ile Ile
50 55 60

Asp Met Asp Leu Lys Asp Gln Asn Gly Arg Phe Glu Phe Asp Ser Glu
65 70 75 80

Ser Ala Ala Val Gly Gly Ile Thr Thr Ile Ile Glu Met Pro Ile Thr
85 90 95

Ser Pro Pro Thr Thr Thr Leu Glu Ala Phe Leu Ala Lys Lys Lys Glu
100 105 110

Ala Gly Asp Arg Leu Lys Val Asp Phe Ala Leu Tyr Gly Gly Gly Val
115 120 125

Pro Gly Asn Leu Ser Glu Ile Arg Lys Met His Glu Ala Gly Ala Val
130 135 140

Gly Phe Lys Ser Met Met Ala Ala Ser Val Pro Gly Met Phe Glu Ala
145 150 155 160

Val Thr Asp Gly Glu Leu Phe Glu Ile Phe Gln Glu Ile Ala Thr Cys
165 170 175

Gly Ser Val Ile Val Val His Ala Glu Asn Glu Thr Ile Ile Gln Ser
180 185 190

Leu Gln Arg Gln Leu Lys Glu Ala Gly Arg Thr Asp Met Ala Ala Tyr
195 200 205

Glu Ala Ser Gln Pro Ala Phe Gln Glu Asn Glu Ala Ile Gln Arg Ala
210 215 220

Leu Ala Leu Gln Lys Glu Thr Gly Cys Arg Leu Ile Val Leu His Val
225 230 235 240

Ser	Asn	Pro	Asp	Gly	Val	Gln	Leu	Ile	His	Gln	Ala	Gln	Ser	Ser	Gly
				245					250					255	
Gln	Asp	Val	His	Cys	Glu	Ser	Gly	Pro	Gln	Tyr	Leu	Asn	Ile	Thr	Ala
			260					265					270		
Glu	Asp	Ala	Ser	Arg	Ile	Gly	Pro	Tyr	Met	Lys	Ile	Ala	Pro	Pro	Val
			275				280					285			
Arg	Ser	Ala	Glu	Met	Asn	Ala	Arg	Leu	Trp	Glu	Gln	Leu	Glu	Lys	Gly
			290			295					300				
Leu	Ile	Asp	Thr	Leu	Gly	Ser	Asp	His	Gly	Gly	His	Pro	Lys	Glu	Asn
305					310					315					320
Lys	Glu	Arg	Gly	Trp	Asp	Asn	Val	Trp	Asp	Ala	Ser	Asn	Gly	Ala	Leu
				325					330					335	
Gly	Leu	Glu	Thr	Ser	Leu	Pro	Met	Met	Leu	Thr	Asn	Gly	Val	Asn	Arg
			340					345					350		
Gly	Arg	Leu	Ser	Leu	Glu	Arg	Leu	Val	Glu	Val	Met	Cys	Glu	Asn	Pro
			355				360					365			
Ala	Lys	Leu	Phe	Gly	Ile	Tyr	Pro	Gln	Lys	Gly	Thr	Leu	Gln	Val	Gly
						375					380				
Ser	Asp	Ala	Asp	Leu	Val	Ile	Val	Asp	Leu	Glu	Val	Asp	Asp	Lys	Val
385					390					395					400
Asp	Ala	Ser	Glu	Phe	Arg	Ser	Leu	Leu	Lys	His	Ser	Pro	Phe	Asp	Gly
				405					410					415	
Met	Pro	Val	Thr	Gly	Lys	Pro	Val	Leu	Thr	Met	Val	Arg	Gly	Thr	Val
			420					425					430		
Val	Ala	Glu	Asn	Gly	Lys	Ile	Leu	Val	Glu	Pro	Gly	Phe	Gly	Gln	Phe
			435				440					445			
Val	Thr	Cys	Arg	Glu	Ser	Gln	Asp	Pro	Lys						
			450			455									

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<210> SEQ ID NO 15
<211> LENGTH: 455
<212> TYPE: PRT
<213> ORGANISM: Alcaligenes faecalis
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: 1..455
<223> OTHER INFORMATION: Alcaligenes faecalis subsp. faecalis
(subspecies)
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<400> SEQUENCE: 15

Met	Ser	Asp	Phe	Asp	Leu	Val	Val	Arg	Gly	Asn	Ile	Val	Glu	Arg	Asp
1			5						10					15	
Arg	Ile	Ile	Thr	Asp	Gly	Trp	Val	Ala	Val	Arg	Glu	Gly	Arg	Val	Ala
			20					25					30		
Ala	Arg	Gly	Ile	Gly	Ala	Ala	Pro	Ser	Ala	Arg	Glu	His	Val	Asp	Ala
		35					40					45			
Arg	Gly	Met	Trp	Val	Met	Pro	Gly	Val	Leu	Asp	Gly	Gln	Val	His	Ser
	50					55					60				
Gly	Ser	Gln	Ala	Asn	Gln	Glu	Gly	Leu	Gly	Trp	Ala	Ser	Arg	Ala	Ala
65					70					75					80
Ala	Ala	Gly	Gly	Val	Thr	Thr	Met	Val	Glu	Met	Pro	His	Asp	Asp	Pro
			85						90					95	
Glu	Pro	Val	Ala	Ser	Arg	Ala	Gln	Leu	Glu	Asp	Lys	Ile	Ala	Arg	Ile
			100					105					110		
Glu	Ser	Asp	Cys	His	Val	Asp	Val	Ala	Cys	Tyr	Gly	Thr	Leu	Asn	Asp
		115					120				125				

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Gln His Gly Leu Glu Ala Ala Ala Gly Leu Ile Glu Gly Gly Val Cys
 130 135 140
 Gly Phe Lys Phe Ser Thr Phe Glu Ala Ser Pro Asn Arg Ser Pro Arg
 145 150 155 160
 Val Glu Glu Asp Ile Leu Met Glu Ala Phe Arg Leu Ile Ala Pro Ser
 165 170 175
 Gly Leu Ala Cys Gly Val His Asn Gln Asp Gln Glu Leu Thr Arg Lys
 180 185 190
 Asn Ile Lys Arg Met Val Glu Ala Gly Asp Thr Gly Trp Asp Ala Phe
 195 200 205
 Leu Arg Ala His Thr Pro Leu Ile Glu Asn Leu Ala Thr Ser Ala Ile
 210 215 220
 Tyr Glu Ile Gly Ala Gln Thr Gly Ala Arg Ala His Ala Val His Val
 225 230 235 240
 Ser Leu Ser Arg Gly Phe Glu Ile Cys Asn Met Tyr Arg Arg Ala Gly
 245 250 255
 Tyr Lys Ala Ser Ile Glu Thr Cys Val Gln Tyr Leu Met Leu Asn His
 260 265 270
 Glu Glu His Thr Arg Arg Phe Gly Ala Lys Thr Lys His Tyr Pro Pro
 275 280 285
 Ile Arg Pro Lys Ala Glu Met Asp Leu Leu Trp Thr His Ile Ala Asn
 290 295 300
 Asp Glu Cys Thr Phe Val Ser Ser Asp His Val Ser Trp Gly Leu Glu
 305 310 315 320
 Arg Lys Gln Phe Glu Asn Val Phe Gln Asn Ser Ser Gly Gly Pro Gly
 325 330 335
 Leu Glu Thr Leu Leu Pro Ala Phe Trp Thr Gly Cys Ala Glu His Gly
 340 345 350
 Ile Ser Pro Thr Met Val Val Lys Gln Leu Cys Trp Gly Pro Ala Gln
 355 360 365
 His Phe Leu Leu Glu His Arg Lys Gly Ser Leu Asn Val Gly Ala Asp
 370 375 380
 Ala Asp Ile Val Ile Val Lys Pro Asp Thr Tyr Arg Phe Asp Pro Ser
 385 390 395 400
 Thr Ser Leu Ser Ala Val Thr Trp Ser Ser Phe Glu Asp Arg Glu Leu
 405 410 415
 Gln Val Arg Val Glu Ala Thr Tyr Val Arg Gly Gln Leu Ala Trp Asp
 420 425 430
 Gly Lys Thr Ile Arg Asn Ala Ala Gly Asp Gly Gln Phe Ile Arg Pro
 435 440 445
 His Lys Ala Gly Glu Leu Ser
 450 455

<210> SEQ ID NO 16
 <211> LENGTH: 459
 <212> TYPE: PRT
 <213> ORGANISM: Microbacterium liquefaciens
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: 1..459
 <223> OTHER INFORMATION: Microbacterium liquefaciens AJ 3912 (strain)
 <400> SEQUENCE: 16

Met Phe Asp Val Ile Val Lys Asn Cys Arg Val Val Ser Ser Gln Gly
 1 5 10 15

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Ile	Ile	Glu	Ala	Asp	Ile	Leu	Val	Lys	Asp	Gly	Arg	Ile	Ala	Ala	Ile
		20						25					30		
Ser	Glu	Glu	Pro	Leu	Glu	Ala	Glu	Ala	Ala	Arg	Thr	Ile	Asp	Ala	Ala
	35						40					45			
Gly	Arg	Phe	Val	Met	Pro	Gly	Val	Val	Asp	Glu	His	Val	His	Ile	Ile
	50					55					60				
Asp	Met	Asp	Leu	Lys	Glu	Val	Tyr	Gly	Arg	Phe	Glu	Leu	Asp	Ser	Glu
65					70					75					80
Ser	Ala	Ala	Val	Gly	Gly	Val	Thr	Thr	Ile	Ile	Glu	Met	Pro	Ile	Thr
			85						90					95	
Phe	Pro	Pro	Thr	Thr	Thr	Leu	Glu	Ala	Phe	Leu	Glu	Lys	Lys	Lys	Gln
			100					105					110		
Gly	Glu	Gln	Arg	Leu	Lys	Val	Asp	Phe	Ala	Leu	Tyr	Gly	Gly	Gly	Val
	115						120					125			
Pro	Gly	Asn	Leu	Ser	Glu	Ile	Arg	Lys	Met	His	Asp	Ala	Gly	Ala	Val
	130					135					140				
Gly	Phe	Lys	Ser	Met	Met	Ala	Ala	Ser	Val	Pro	Gly	Met	Phe	Glu	Ala
145					150					155					160
Val	Asp	Asp	Gly	Gln	Leu	Phe	Glu	Ile	Phe	Gln	Glu	Ile	Ala	Ala	Cys
			165						170					175	
Gly	Ser	Val	Ile	Val	Val	His	Ala	Glu	Asn	Glu	Met	Leu	Ile	Gln	Thr
		180						185					190		
Leu	Gln	Lys	Gln	Leu	Lys	Ala	Ala	Gly	Arg	Lys	Asp	Leu	Ala	Ala	Tyr
	195						200					205			
Glu	Ala	Ser	Gln	Pro	Val	Phe	Gln	Glu	Asn	Glu	Ala	Ile	Gln	Arg	Ala
	210					215					220				
Leu	Leu	Leu	Gln	Lys	Glu	Ala	Gly	Cys	Arg	Leu	Ile	Val	Val	His	Val
225					230					235					240
Ser	Asn	Pro	Gly	Gly	Val	Glu	Leu	Ile	His	Lys	Ala	Gln	Ser	Glu	Gly
			245						250					255	
Gln	Asp	Val	His	Cys	Glu	Ser	Gly	Pro	Gln	Tyr	Leu	Asn	Leu	Thr	Met
		260						265					270		
Asp	Asp	Ala	Glu	Lys	Val	Gly	Pro	Tyr	Met	Lys	Ile	Ala	Pro	Pro	Val
		275					280					285			
Arg	Ser	Ala	Glu	Leu	Asn	Ala	Val	Leu	Trp	Glu	Gln	Leu	Glu	Lys	Gly
	290					295					300				
Tyr	Ile	Asp	Thr	Leu	Gly	Ser	Asp	His	Gly	Gly	His	Pro	Val	Glu	Asn
305					310					315					320
Lys	Glu	Gly	Gly	Trp	Asp	Asp	Ile	Trp	Thr	Ala	Ser	Asn	Gly	Ala	Leu
		325							330					335	
Gly	Leu	Glu	Thr	Ser	Leu	Pro	Met	Met	Leu	Thr	Asn	Gly	Val	Asn	Lys
		340						345					350		
Gly	Arg	Val	Ser	Leu	Glu	Arg	Leu	Val	Glu	Val	Met	Cys	Glu	Asn	Pro
		355					360					365			
Ala	Lys	Leu	Phe	Gly	Ile	Tyr	Pro	Gln	Lys	Gly	Thr	Leu	Gln	Val	Gly
	370					375					380				
Ser	Asp	Ala	Asp	Leu	Leu	Ile	Leu	Asp	Leu	Glu	Ile	Glu	Asp	Arg	Lys
385					390					395					400
Val	Asp	Ala	Ser	Gln	Phe	Arg	Ser	Leu	Leu	His	Tyr	Ser	Pro	Phe	Asp
			405						410					415	
Gly	Arg	Pro	Val	Thr	Gly	Ala	Pro	Val	Leu	Thr	Met	Ile	Arg	Gly	Thr
		420						425					430		
Val	Val	Ala	Gln	Asp	Gly	Glu	Ile	Leu	Val	Asp	Gln	Gly	Phe	Gly	Gln

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435	440	445
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Phe Val Thr Arg Arg Asp Ser Glu Val Ser Ser
 450 455

<210> SEQ ID NO 17
 <211> LENGTH: 236
 <212> TYPE: PRT
 <213> ORGANISM: *Arthrobacter aurescens*
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: 1..236
 <223> OTHER INFORMATION: *Arthrobacter aurescens* DSM 3745 (strain)
 <220> FEATURE:
 <221> NAME/KEY: gene
 <222> LOCATION: 1..236
 <223> OTHER INFORMATION: HyuR

<400> SEQUENCE: 17

Met Arg Ile Leu Val Ile Asn Pro Asn Ser Ser Ser Ala Leu Thr Glu
 1 5 10 15

Ser Val Ala Asp Ala Ala Gln Gln Val Val Ala Thr Gly Thr Ile Ile
 20 25 30

Ser Ala Ile Asn Pro Ser Arg Gly Pro Ala Val Ile Glu Gly Ser Phe
 35 40 45

Asp Glu Ala Leu Ala Thr Phe His Leu Ile Glu Glu Val Glu Arg Ala
 50 55 60

Glu Arg Glu Asn Pro Pro Asp Ala Tyr Val Ile Ala Cys Phe Gly Asp
 65 70 75 80

Pro Gly Leu Asp Ala Val Lys Glu Leu Thr Asp Arg Pro Val Val Gly
 85 90 95

Val Ala Glu Ala Ala Ile His Met Ser Ser Phe Val Ala Ala Thr Phe
 100 105 110

Ser Ile Val Ser Ile Leu Pro Arg Val Arg Lys His Leu His Glu Leu
 115 120 125

Val Arg Gln Ala Gly Ala Thr Asn Arg Leu Ala Ser Ile Lys Leu Pro
 130 135 140

Asn Leu Gly Val Met Ala Phe His Glu Asp Glu His Ala Ala Leu Glu
 145 150 155 160

Thr Leu Lys Gln Ala Ala Lys Glu Ala Val Gln Glu Asp Gly Ala Glu
 165 170 175

Ser Ile Val Leu Gly Cys Ala Gly Met Val Gly Phe Ala Arg Gln Leu
 180 185 190

Ser Asp Glu Leu Gly Val Pro Val Ile Asp Pro Val Glu Ala Ala Cys
 195 200 205

Arg Val Ala Glu Ser Leu Val Ala Leu Gly Tyr Gln Thr Ser Lys Ala
 210 215 220

Asn Ser Tyr Gln Lys Pro Thr Glu Lys Gln Tyr Leu
 225 230 235

<210> SEQ ID NO 18
 <211> LENGTH: 249
 <212> TYPE: PRT
 <213> ORGANISM: *Pseudomonas* sp.
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: 1..249
 <223> OTHER INFORMATION: *Pseudomonas* sp. NS671 (strain)
 <220> FEATURE:
 <221> NAME/KEY: gene
 <222> LOCATION: 1..249
 <223> OTHER INFORMATION: HyuE

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<400> SEQUENCE: 18

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Met Lys Ile Lys Val Ile Asn Pro Asn Thr Thr Leu Ala Met Thr Lys
 1             5             10             15

Gly Ile Glu His Ala Ala Lys Ser Ala Ala Arg Ser Asp Thr Gln Ile
          20             25             30

Val Ala Val Ser Pro Lys Met Gly Pro Ala Ser Ile Glu Ser Tyr Tyr
      35             40             45

Asp Glu Tyr Leu Ser Ile Pro Gly Val Ile Glu Glu Ile Lys Lys Gly
 50             55             60

Glu Glu Glu Gly Val Asp Ala Phe Val Ile Ala Cys Trp Gly Asp Pro
65             70             75             80

Gly Leu His Ala Ala Arg Glu Val Thr Asp Lys Pro Val Val Gly Ile
          85             90             95

Ala Glu Ser Ser Val Tyr Leu Ala Ser Met Leu Ala Ala Arg Phe Ser
      100             105             110

Val Val Thr Val Leu Pro Arg Ile Lys Thr Met Leu Glu Asp Leu Val
      115             120             125

Asp Ser Tyr Gly Met Gln Lys Arg Val Leu Asn Ile Arg Thr Thr Pro
130             135             140

Met Gly Val Leu Asp Phe Glu Arg Asp Pro Glu Ala Gly Ile Glu Met
145             150             155             160

Leu Arg Gln Glu Gly Lys Arg Ala Val Glu Glu Asp Asn Ala Glu Ala
      165             170             175

Ile Leu Leu Gly Cys Ala Gly Met Ala Glu Phe Ala Asp Ser Leu Glu
      180             185             190

Lys Glu Leu Gly Val Pro Val Ile Asp Gly Val Val Ala Gly Val Lys
      195             200             205

Phe Ala Glu Thr Ile Val Asp Leu Gly Lys Lys Thr Ser Lys Leu Lys
      210             215             220

Thr Tyr Lys Tyr Pro Glu Lys Lys Glu Tyr Val Gly Ala Leu Glu Asn
225             230             235             240

Phe Gly Arg Asn Gln Thr Thr Thr Lys
      245

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<210> SEQ ID NO 19

<211> LENGTH: 244

<212> TYPE: PRT

<213> ORGANISM: Rhodococcus

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: 1..244

<223> OTHER INFORMATION: Rhodococcus R04 (strain)

<220> FEATURE:

<221> NAME/KEY: gene

<222> LOCATION: 1..244

<223> OTHER INFORMATION: Hyu2

<400> SEQUENCE: 19

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Met Phe Ile Lys Val Val Asn Pro Asn Thr Thr Trp Ser Met Thr Ala
 1             5             10             15

Thr Ile Glu Ala Cys Ala Arg Ala Val Ala Gly Pro Gly Thr Arg Val
      20             25             30

Glu Ala Val Ser Pro Thr Met Gly Pro Pro Ser Ile Glu Ser His Tyr
      35             40             45

Asp Asp Ala Leu Ala Val Pro Gly Ile Leu Thr Glu Ile Glu Lys Gly
 50             55             60

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Glu Arg Asp Gly Val Asp Gly Tyr Val Ile Ala Cys Phe Gly Asp Pro
 65 70 75 80
 Gly Leu Asp Ala Ala Arg Glu Leu Ala Gly Gly Pro Val Val Gly Ile
 85 90 95
 Ala Glu Ala Ala Met His Thr Ala Ala Val Leu Gly Arg Gly Phe Ser
 100 105 110
 Val Val Thr Thr Leu Ala Arg Thr Thr Gly Arg Ala Trp Asp Leu Ala
 115 120 125
 His Arg Tyr Gly Met Arg Asp Ala Cys Arg Gly Val His Ala Cys Asp
 130 135 140
 Leu Pro Val Leu Ala Leu Asp Ser Glu Pro Asp Ala Arg Lys Ile Val
 145 150 155 160
 Thr Glu Ala Cys Leu Asp Ala Leu Tyr Glu Asp Gly Ser Asp Ala Ile
 165 170 175
 Val Leu Gly Cys Ala Gly Met Ala Asp Leu Cys Ala His Ile Ser Ala
 180 185 190
 Glu Ile Gly Val Pro Val Val Asp Gly Val Ala Ala Ala Thr Leu Thr
 195 200 205
 Val Gln Ser Leu Val Thr Met Gly Leu Ala Thr Gly Lys Arg Gly Glu
 210 215 220
 Phe Ala Ala Pro Pro Pro Lys Arg Tyr Ala Gly Leu Leu Asp Gly Phe
 225 230 235 240
 Thr Thr Gly Gly

<210> SEQ ID NO 20
 <211> LENGTH: 236
 <212> TYPE: PRT
 <213> ORGANISM: Microbacterium liquefaciens
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: 1..236
 <223> OTHER INFORMATION: Microbacterium liquefaciens AJ 3912 (strain)
 <220> FEATURE:
 <221> NAME/KEY: gene
 <222> LOCATION: 1..236
 <223> OTHER INFORMATION: HRase
 <400> SEQUENCE: 20

Met Arg Ile His Val Ile Asn Pro Asn Ser Ser Val Asp Leu Thr Asp
 1 5 10 15
 Ala Val Ala Glu Ala Ala Arg Ser Val Val Ser Pro Gly Thr Thr Ile
 20 25 30
 Thr Ala Val Asn Pro Ser Lys Gly Pro Thr Val Ile Glu Gly Ser Tyr
 35 40 45
 Asp Glu Val Leu Ala Thr Tyr His Leu Val Glu Glu Val Arg Arg Ala
 50 55 60
 Glu Arg Glu Asp Arg Pro Asp Ala Tyr Val Ile Ala Cys Phe Gly Asp
 65 70 75 80
 Pro Gly Leu Asp Ala Val Arg Glu Leu Thr Asp Arg Pro Val Val Gly
 85 90 95
 Ile Ala Glu Ala Ala Ile Gln Met Thr Ser Phe Val Ala Ala Ser Phe
 100 105 110
 Ser Ile Val Ser Ile Leu Pro Arg Val Arg Lys His Leu His Glu Leu
 115 120 125
 Val His Arg Ala Gly Ala Thr Asp Arg Leu Ala Ser Leu Lys Leu Pro
 130 135 140
 Asp Leu Gly Val Leu Ala Phe His Glu Asp Glu Ala Ala Ala Phe Glu

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145	150	155	160
Thr Leu Arg Arg Val Ala Gly Glu Ala Val Arg Glu Asp Gly Ala Glu			
165	170	175	
Ser Ile Val Leu Gly Cys Ala Gly Met Ala Gly Phe Ala Arg Gln Leu			
180	185	190	
Ser Glu Glu Leu Gly Val Pro Val Ile Asp Ala Val Glu Ala Ala Cys			
195	200	205	
Arg Val Ala Glu Ser Leu Val Ala Leu Gly Tyr Arg Thr Ser Lys Ala			
210	215	220	
Asn Thr Tyr Gln Ala Pro Thr Glu Lys Gln Tyr Leu			
225	230	235	

<210> SEQ ID NO 21
 <211> LENGTH: 241
 <212> TYPE: PRT
 <213> ORGANISM: Sinorhizobium meliloti
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: 1..241
 <223> OTHER INFORMATION: Sinorhizobium meliloti strain CECT 4114 (strain)
 <220> FEATURE:
 <221> NAME/KEY: gene
 <222> LOCATION: 1..241
 <223> OTHER INFORMATION: HyuA

<400> SEQUENCE: 21

Met His Ile His Leu Ile Asn Pro Asn Ser Thr Ala Ser Met Thr Ala			
1	5	10	15
Gln Ala Leu Glu Ser Ala Leu Leu Val Lys His Ala His Thr His Val			
20	25	30	
Ser Ala Ser Asn Pro Thr Asp Thr Pro Ala Ser Ile Glu Gly Gly Ala			
35	40	45	
Asp Glu Ala Met Ser Val Pro Gly Met Leu Ala Glu Ile Arg Gln Gly			
50	55	60	
Glu Ala Gln Gly Val Asp Ala Tyr Val Ile Ala Cys Phe Asp Asp Pro			
65	70	75	80
Gly Leu His Ala Ala Arg Glu Val Ala Lys Gly Pro Val Ile Gly Ile			
85	90	95	
Cys Gln Ala Ala Val Gln Val Ala Met Thr Ile Ser Arg Arg Phe Ser			
100	105	110	
Val Ile Thr Thr Leu Pro Arg Ser Val Pro Ile Ile Glu Asp Leu Val			
115	120	125	
Ser Asp Tyr Gly Ala Glu Arg His Cys Arg Lys Val Arg Ala Ile Asp			
130	135	140	
Leu Pro Val Leu Ala Leu Glu Glu Asp Pro Gln Arg Ala Glu Arg Leu			
145	150	155	160
Leu Leu Lys Glu Ile Glu Ile Ala Lys Ala Glu Asp Gly Ala Glu Ala			
165	170	175	
Ile Val Leu Gly Cys Ala Gly Met Ser Ser Leu Cys Asp Arg Leu Gln			
180	185	190	
Lys Ala Thr Gly Val Pro Val Ile Asp Gly Val Thr Ala Ala Val Lys			
195	200	205	
Met Ala Glu Ala Leu Leu Gly Ala Gly Tyr Ala Thr Ser Lys Val Asn			
210	215	220	
Thr Tyr Ala Tyr Pro Arg Ile Lys Ala Ala Ala Gly His Lys Val Cys			
225	230	235	240

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Ala

<210> SEQ ID NO 22
 <211> LENGTH: 243
 <212> TYPE: PRT
 <213> ORGANISM: Flavobacterium sp.
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: 1..243
 <223> OTHER INFORMATION: Flavobacterium sp. AJ 11199 (strain)
 <220> FEATURE:
 <221> NAME/KEY: gene
 <222> LOCATION: 1..243
 <223> OTHER INFORMATION: HRase

<400> SEQUENCE: 22

Met Lys Ile Lys Val Ile Asn Pro Asn Thr Thr Leu Thr Met Thr Ala
 1 5 10 15
 Lys Ile Gly Glu Ala Ala Ala Val Ala Ser Ala Gly Thr Glu Val
 20 25 30
 Val Ala Val Ser Pro Ala Met Gly Pro Ala Ser Ile Glu Gly His Tyr
 35 40 45
 Asp Glu Ala Val Ser Ala Leu Gly Val Leu Asp Glu Val Arg Lys Gly
 50 55 60
 Lys Ala Glu Gly Cys Asp Gly Tyr Leu Ile Ala Cys Phe Asp Asp Pro
 65 70 75 80
 Gly Leu Gln Ala Ala Arg Glu Ile Ala Asp Gly Pro Val Val Gly Ile
 85 90 95
 Ala Glu Ala Ala Met His Met Ala Ser Phe Val Ser Glu Gly Phe Ser
 100 105 110
 Val Val Ala Thr Gly His Arg Ser Arg Ile Ile Leu Glu His Leu Ala
 115 120 125
 Arg Ser Tyr Gly Met Glu His Lys Cys Arg Lys Val Arg Thr Thr Glu
 130 135 140
 Leu Ala Val Leu Asp Leu Glu Val Glu Gly Ser Asp Ala Arg Gly Ile
 145 150 155 160
 Ile Leu Glu Glu Cys Arg Arg Ala Ile Val Glu Asp His Ser Asp Cys
 165 170 175
 Ile Val Leu Gly Cys Ala Gly Met Ala Asp Leu Ala Asp Tyr Ile Ser
 180 185 190
 Lys Glu Leu Gly Val Pro Val Val Asp Gly Val Ala Ala Gly Val Lys
 195 200 205
 Val Leu Glu Gly Leu Ile Gly Leu Arg Leu Ser Thr Ser Arg Ala Cys
 210 215 220
 Gly Tyr Ala Tyr Pro Asn Pro Lys Thr Tyr Ser Gly Glu Met Ala Arg
 225 230 235 240
 Phe Gln Pro

<210> SEQ ID NO 23
 <211> LENGTH: 247
 <212> TYPE: PRT
 <213> ORGANISM: Agrobacterium sp.
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: 1..247
 <223> OTHER INFORMATION: Agrobacterium sp. IP I-671 (strain)
 <220> FEATURE:
 <221> NAME/KEY: gene
 <222> LOCATION: 1..247
 <223> OTHER INFORMATION: HyuA

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<400> SEQUENCE: 23

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Met Lys Ile Lys Val Ile Asn Pro Asn Thr Thr Trp Thr Met Thr Asp
1      5      10      15
Lys Ile Ala Gly Ala Ala Arg Ala Ala Ala Ala Pro Gly Thr Glu Ile
      20      25      30
Val Ala Val Ser Pro Asp Met Gly Pro Val Ser Ile Glu Gly Tyr Tyr
      35      40      45
Asp Glu Val Phe Ala Ala Val Gly Val Val Asp Glu Val Arg Lys Gly
      50      55      60
Glu Leu Glu Gly Cys Asp Gly Tyr Val Ile Ala Cys Phe Gly Asp Pro
      65      70      75      80
Gly Leu Asn Ala Ala Arg Glu Val Ala Arg Gly Pro Val Ile Gly Ile
      85      90      95
Ala Glu Ala Ala Met His Ala Ala Ser Leu Ile Gly Gly Ser Phe Ser
      100     105     110
Ile Ile Ser Met Leu Gly Arg Ser Arg Gly Val Leu Glu His Leu Val
      115     120     125
His Ser Tyr Gly Met Ala His Lys Cys Arg Ser Val Arg Met Thr Asp
      130     135     140
Leu Pro Val Leu Glu Phe Glu Glu Glu Gly Ser Asp Ala Arg Arg Ile
      145     150     155     160
Val Val Glu Glu Cys Arg Arg Ala Ile Glu Gln Asp His Ala Asp Ser
      165     170     175
Val Leu Leu Gly Cys Gly Gly Met Ser Asp Leu Met Ala Tyr Val Ser
      180     185     190
Gln Glu Ile Gly Ala Pro Ala Ile Asp Gly Val Ser Ala Gly Val Lys
      195     200     205
Leu Val Glu Ala Leu Val Gly Met Gly Leu Gly Thr Ser Lys Arg Ala
      210     215     220
Gly Leu Arg Leu Ser Asp Arg Glu Ser Leu His Cys Ser Phe Ser Ser
      225     230     235     240
Phe Ala Pro Ser Arg Pro Arg
      245

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<210> SEQ ID NO 24

<211> LENGTH: 247

<212> TYPE: PRT

<213> ORGANISM: Pasteurella sp.

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: 1..247

<223> OTHER INFORMATION: Pasteurella sp. AJ 11221 (strain)

<400> SEQUENCE: 24

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Met Lys Ile Lys Val Ile Asn Pro Asn Thr Thr Trp Ser Met Thr Glu
1      5      10      15
Lys Ile Gly Glu Ala Ala Arg Arg Val Ala Ala Pro Gly Thr Glu Ile
      20      25      30
Val Ala Val Ser Pro Ala Met Gly Pro Val Ser Ile Glu Gly Phe Tyr
      35      40      45
Asp Glu Ala Phe Ala Ala Ile Gly Val Ile Asp Glu Val Arg Lys Gly
      50      55      60
Glu Glu Glu Gly Cys Asp Gly Tyr Val Ile Ala Cys Phe Gly Asp Pro
      65      70      75      80
Gly Leu Leu Ala Ala Arg Glu Ile Ala Arg Gly Pro Val Val Ile Ala
      85      90      95

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Glu Ala Ala Met His Ala Ala Ser Leu Ile Gly Asn Gly Phe Thr Ile
 100 105 110
 Val Ser Met Leu Glu Arg Thr Arg Ala Thr Met Glu His Leu Val His
 115 120 125
 Ala Tyr Gly Met Ser His Lys Cys Arg Asn Ile Arg Met Thr Asp Leu
 130 135 140
 Pro Val Leu Glu Leu Glu Lys Glu Gly Ser Asn Ala Gln Ala Ile Ile
 145 150 155 160
 Met Glu Glu Cys Arg Arg Ala Leu Glu Glu Asp His Ser Asp Ala Val
 165 170 175
 Leu Leu Gly Cys Gly Gly Met Ser Asp Leu Met Ala Leu Ile Thr Arg
 180 185 190
 Glu Ile Gly Ala Pro Ala Ile Asp Gly Val Ser Ser Gly Val Lys Leu
 195 200 205
 Val Glu Ala Leu Val Ser Leu Gly Leu Gly Thr Ser Lys Arg Gln Ala
 210 215 220
 Tyr Ala Tyr Pro Val Glu Lys Thr Tyr Thr Gly Ser Phe Ser Gln Phe
 225 230 235 240
 Ser Val Pro Ala Ala Asn Val
 245

<210> SEQ ID NO 25
 <211> LENGTH: 228
 <212> TYPE: PRT
 <213> ORGANISM: Pyrococcus horikoshii
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: 1..228
 <223> OTHER INFORMATION: Pyrococcus horikoshii OT3 (strain)
 <220> FEATURE:
 <221> NAME/KEY: gene
 <222> LOCATION: 1..228
 <223> OTHER INFORMATION: PH1054

<400> SEQUENCE: 25

Met Tyr Arg Met Asp Lys Tyr Thr Ile Gly Leu Ile Arg Val Ile Thr
 1 5 10 15
 Leu Glu Asp Lys Glu Ile Leu Asn Leu His Gly Arg Ile Ile Glu Ser
 20 25 30
 Ala Phe Pro Glu Leu Lys Val Val Ser Arg Cys Ile Glu Asp Gln Pro
 35 40 45
 Lys Gly Ile Tyr Asn Glu Glu Thr Glu Arg Glu Ala Glu Pro Lys Ile
 50 55 60
 Ile Arg Leu Ala Lys Glu Phe Glu Arg Glu Gly Val Asp Ala Ile Ile
 65 70 75 80
 Ile Ser Cys Ala Ala Asp Pro Ala Val Glu Lys Val Arg Lys Leu Leu
 85 90 95
 Ser Ile Pro Val Ile Gly Ala Gly Ser Ser Val Ser Ala Leu Ala Leu
 100 105 110
 Ala Tyr Gly Arg Arg Val Gly Val Leu Asn Leu Thr Glu Glu Thr Pro
 115 120 125
 Lys Val Ile Arg Ser Ile Leu Gly Asn Asn Leu Ile Ala Glu Asp His
 130 135 140
 Pro Ser Gly Val Ser Asn Thr Leu Asp Leu Leu Thr Asp Trp Gly Arg
 145 150 155 160
 Arg Glu Val Ile Asn Ala Ala Lys Arg Leu Lys Glu Lys Gly Val Glu
 165 170 175

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Val Ile Ala Leu Gly Cys Thr Gly Met Ser Thr Ile Gly Ile Ala Pro
180 185 190

Val Leu Glu Glu Glu Val Gly Ile Pro Val Ile Asp Pro Val Ile Ala
195 200 205

Ser Gly Ala Val Ala Leu His Ala Leu Lys Arg Arg Glu Val Lys Arg
210 215 220

Phe Glu Gly Arg
225

<210> SEQ ID NO 26
<211> LENGTH: 236
<212> TYPE: PRT
<213> ORGANISM: *Arthrobacter* sp.
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: 1..236
<223> OTHER INFORMATION: *Arthrobacter* sp. BT801 (strain)
<220> FEATURE:
<221> NAME/KEY: gene
<222> LOCATION: 1..236
<223> OTHER INFORMATION: HyuA

<400> SEQUENCE: 26

Met Arg Ile His Ile Ile Asn Pro Asn Ser Ser Ser Glu Leu Thr Glu
1 5 10 15

Ala Val Ala Glu Thr Ala Arg Ser Val Val Ser Ala Gly Thr Ala Val
20 25 30

Thr Ala Val Asn Pro Ser Ser Gly Pro Ala Val Ile Glu Gly Ser Tyr
35 40 45

Asp Glu Ala Leu Ala Thr Tyr His Leu Val Gln Glu Val Met Arg Ala
50 55 60

Glu Arg Glu Asp Arg Pro Asp Ala Tyr Val Ile Ala Cys Phe Gly Asp
65 70 75 80

Pro Gly Leu Asn Ala Val Arg Glu Leu Thr Glu Lys Pro Val Val Gly
85 90 95

Ile Ala Glu Ala Ala Ile Gln Leu Ser Ser Phe Ile Gly Ala Thr Phe
100 105 110

Ser Ile Val Ser Thr Leu Pro Arg Val Arg Ser His Leu His Gly Leu
115 120 125

Val Arg Arg Ala Gly Ala Thr Asn Arg Leu Ala Ser Val Lys Thr Pro
130 135 140

Asp Leu Gly Val Met Ala Phe His Glu Asp Ala Gly Ser Ala Gln Ala
145 150 155 160

Thr Leu Glu Gln Ala Ala Lys Glu Ala Val Gln Glu Asp Gly Ala Glu
165 170 175

Val Val Val Leu Gly Cys Ala Gly Met Ala Gly Leu Ala Arg Arg Leu
180 185 190

Ser Glu Glu Leu Gln Val Pro Val Ile Asp Pro Val Glu Ala Ala Cys
195 200 205

Arg Val Ala Glu Ser Leu Ser Ala Leu Gly Tyr Gly Thr Ser Lys Ala
210 215 220

Asn Thr Tyr Glu Lys Pro Thr Glu Lys Val Tyr Asn
225 230 235

<210> SEQ ID NO 27
<211> LENGTH: 1263
<212> TYPE: DNA
<213> ORGANISM: *Arthrobacter* *aurescens*

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<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: 1..1263
<223> OTHER INFORMATION: Arthrobacter aurescens DSM 3747 (strain)
<220> FEATURE:
<221> NAME/KEY: gene
<222> LOCATION: 1..1263
<223> OTHER INFORMATION: hyuC

<400> SEQUENCE: 27

atgaccctgc agaaagcgca agcggagcgc attgagaaag agatctggga gctctcccg 60
ttctcggcgg aaggccccgg tgttaccggg ctgacctaca ctccagagca tgcgcgcgg 120
cgggaaacgc tcattgctgc tatggaagcg gccgctttga gcgttcgtga agacgctctc 180
gggaacatca tcggccgacg tgaaggcact gatccgcagc tccctgcgat cgcggtcggt 240
tcacacttcg attctgtcgc aaacggcggg atgttcgatg gcactgcagg cgtggtgtgc 300
gcccttgagg ctgcccgggt gatgctggag agcggctacg tgaatcgga tccatttgag 360
ttcatcgcga tcgtggagga ggaaggggcc cgcttcagca gtggcatgtt gggcgccgg 420
gccattgcag gtttggtcgc cgacagggaa ctggactctt tggttgatga ggatggagt 480
tcogttaggc aggcggctac tgccttcggc ttgaagccgg gcgaactgca ggctgcagcc 540
cgctccggcg cggacctgcg tgcttttata gaactacaca ttgaacaagg accgatcctc 600
gagcaggagc aaatagagat cggagttgtg acctccatcg ttggcgctcg cgcattgcgg 660
gttgctgtca aaggcagaag cgcacacgcc ggcacaaccc ccatgcacct gcgccaggat 720
gcgctggtac ccgcgcctct catggtgcgg gaggtcaacc ggctcgtcaa cgagatcgcc 780
gatggcacag tggctaccgt tggccacctc acagtggccc ccggtggcgg caaccaggtc 840
ccgggggagg tggagttcac actggacctg cgttctccgc atgaggagtc gctccgggtg 900
ttgatcaacc gcactctcgg catggtcggc gaggtcgccg cgcaggccgg tgtggtctgc 960
gatgtggatg aatttttcaa tctcagcccg gtgcagctgg ctctaccat ggtggacgcc 1020
gttcgcgaag cggcctcggc cctgcagttc acgcaccggg atatacagc tggggcgggc 1080
cacgactcga tgttcacgc ccaggtcacg gacgtcgga tggttttcgt tccaagccgt 1140
gttgcccgga gccacgttcc cgaagaatgg accgatttcg atgaccttcg caagggaaact 1200
gaggttgtcc tccgggtaat gaaggcactt gaccggggat cccatcatca tcatcatcat 1260
tga 1263

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<210> SEQ ID NO 28
<211> LENGTH: 1377
<212> TYPE: DNA
<213> ORGANISM: Arthrobacter sp.
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: 1..1377
<223> OTHER INFORMATION: Arthrobacter sp. DSM 9771 (strain)
<220> FEATURE:
<221> NAME/KEY: gene
<222> LOCATION: 1..1377
<223> OTHER INFORMATION: hyuH

<400> SEQUENCE: 28

atgtttgacg taatagttaa gaactgccgt atggtgtcca gcgacggaat caccgaggca 60
gacattctgg tgaaagacgg caaagtcgcc gcaatcagct cggacacaag tgatgttgag 120
gcgagccgaa ccattgacgc ggggtggcaag ttcgtgatgc cgggcgtggg cgatgaacat 180
gtgcataatc tcgacatgga tctgaagaac cggtatggcc gcttcgaact cgattccgag 240

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tctgcggccg tgggagggcat caccaccatc tttgagatgc cgtttacctt cccgcccacc	300
accacttttg acgccttctc cgaagaag aagcaggcgg ggcagcggtt gaaagttgac	360
ttcgcgctct atggcggttg agtgccggga aacctgccc agatccgcaa aatgcacgac	420
gcggcgccag tgggcttcaa gtcaatgatg gcagcctcag ttccgggcat gttegacgcc	480
gtcagcgacg gcgaactgtt cgaatcttc caggagatcg cagcctgttg ttcatgcgc	540
gtggtccatg ccgagaatga aacgatcatt caagcgctcc agaagcagat caaagccgct	600
ggtcgcaagg acatggccgc ctacgaggca tcccaaccag tttccagga gaacgaggcc	660
attcagcgtg cgttactact gcagaaagaa gccggctgtc gactgattgt gcttcacgtg	720
agcaaccctg acggggtcga gctgatacat cgggcgcaat ccgagggccca ggacgtccac	780
tgcgagtcgg gtccgcagta tctgaatatc accacggacg acgcccgaacg aatcggaaccg	840
tatatgaagg tcgcgccgcc cgtccgctca gccgagatga acgtcagatt atgggaacaa	900
cttgagaacg ggctcatcga cacccttggg tcagaccacg gcggacatcc tgtcgaggac	960
aaagaacccg gctggaagga cgtgtggaaa gccggcaacg gtgcgctggg ccttgagaca	1020
tccctgccta tgatgctgac caacggagtg aataaaggca ggctatcctt ggaacgcctc	1080
gtcgaggtga tgtgcgagaa acctgcgaag ctctttggca tctatccgca gaagggcacg	1140
ctacaggttg gttccgacgc cgatctgtc atcctcgatc tggatattga caccaaagtg	1200
gatgcctcgc agttccgatc cctgcataag tacagcccg tgcacgggat gcccgtcacg	1260
ggtgacccgg ttctgacgat ggtgcgcgga acgggtgttg cagagaaggg agaagtctg	1320
gtcgagcagg gattcggcc gttcgtcacc cgtcacgact acgaggcgtc gaagtga	1377

<210> SEQ ID NO 29
 <211> LENGTH: 711
 <212> TYPE: DNA
 <213> ORGANISM: *Arthrobacter aurescens*
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: 1..711
 <223> OTHER INFORMATION: *Arthrobacter aurescens* DSM 3745 (strain)
 <220> FEATURE:
 <221> NAME/KEY: gene
 <222> LOCATION: 1..711
 <223> OTHER INFORMATION: *hyuR*

<400> SEQUENCE: 29

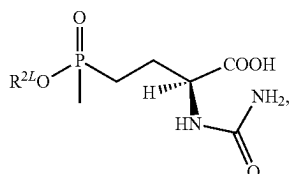
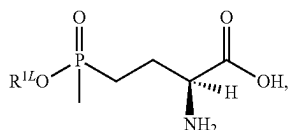
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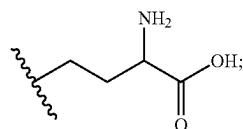
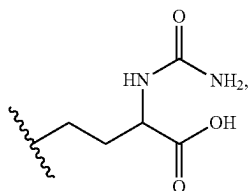
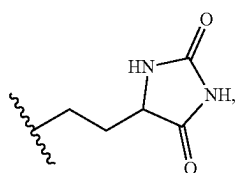
The invention claimed is:

1. A method for the production of an L-glufosinate P-ester according to formula L-(I), the method comprising:

(c) reacting a compound according to formula L-(II) to give a compound according to formula L-(I)



wherein the reaction according to (c) is catalyzed by an L-carbamoylase E_1 categorized in EC class 3.5.1.87, wherein R^{1L} and R^{2L} are radicals selected from the group consisting of formulae (VIII), (IX), and (X)



and

wherein $R^{1L}=R^{2L}$.

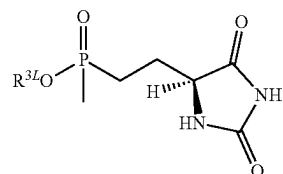
2. The method according to claim 1, wherein R^{1L} and R^{2L} are radicals that are each independently selected from the group consisting of formulae (IX) and (X).

3. The method according to claim 1, wherein a polypeptide sequence of the L-carbamoylase E_1 is selected from the group consisting of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, and variant amino acid sequences having at least 90% sequence identity to one of SEQ ID NOS: 1-9.

4. The method according to claim 1, wherein the compound according to formula L-(II) is obtained by (b) in

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which a compound according to formula L-(III) is reacted to give a compound according to formula L-(II):



L-(III)

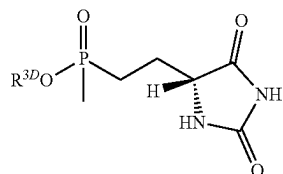
wherein the reaction according to (b) is catalyzed by a hydantoinase E_2 categorized in EC class 3.5.2.2, and wherein R^{3L} is a radical that is selected from the group consisting of formulae (VIII), (IX), and (X).

5. The method according to claim 4, wherein R^{3L} is a radical of the formula (VIII).

6. The method according to claim 4, wherein a polypeptide sequence of the hydantoinase E_2 is selected from the group consisting of SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 16, and variant amino acid sequences having at least 90% sequence identity to one of SEQ ID NOS: 10-16.

7. The method according to claim 4, wherein the hydantoinase E_2 is an L-hydantoinase E_2 .

8. The method according to claim 4, wherein the compound according to formula L-(III) is obtained by (a) in which a compound according to formula D-(III) is reacted to give a compound according to formula L-(III):



D-(III)

and

wherein R^{3D} is a radical that is selected from the group consisting of formulae (VIII), (IX), and (X).

9. The method according to claim 8, wherein R^{3D} is a radical of the formula (VIII).

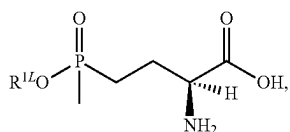
10. The method according to claim 8, wherein the reaction according to (a) is catalyzed by a hydantoin racemase E_3 categorized in EC class 5.1.99.5.

11. The method according to claim 10, wherein a polypeptide sequence of the hydantoin racemase E_3 is selected from the group consisting of SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26, and variant amino acid sequences having at least 90% sequence identity to one of SEQ ID NOS: 18-26.

12. The method according to claim 1, wherein the compound according to formula L-(I) is saponified to give L-glufosinate.

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13. A method for the production of an L-glufosinate P-ester according to formula L-(I):

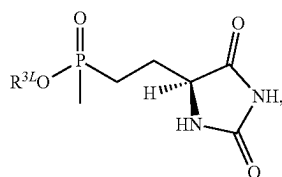


L-(I) 5

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the method comprising:

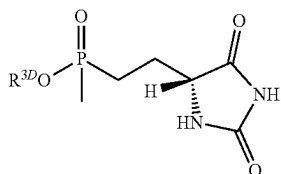
(i-A) providing a mixture M_{IIA} comprising at least one compound L-(III) and at least one compound D-(III), wherein L-(III) and D-(III) have the following formulae: 15



L-(III)

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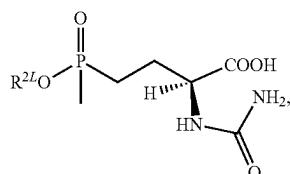


D-(III)

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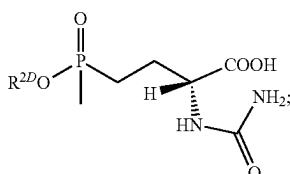
(i-B) (a) reacting at least a part of the compounds according to formula D-(III) comprised by the mixture M_{IIIA} into the compound according to formula L-(III) to form a composition M_{IIIB} comprising compounds according to formula L-(III) and compounds according to formula D-(III), and 40

(ii) subjecting composition M_{IIIB} to a reaction according to (b), giving a composition Mu comprising at least one compound according to formula L-(II) and at least one compound according to formula D-(II), wherein the reaction according to (b) is catalyzed by a hydantoinase E₂ categorized in EC class 3.5.2.2, wherein L-(II) and D-(II) have the following formulae: 45



L-(II) 50

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D-(II)

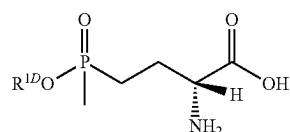
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and

(iii) subjecting composition Mu to a reaction according to (c), wherein the reaction according to (c) is catalyzed by an L-carbamoylase E₁, giving a composition M_I comprising at least one compound according to formula L-(I) and at least one compound according to formula D-(I),

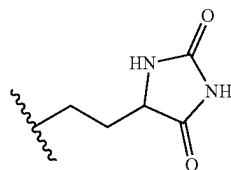
wherein a molar ratio of all compounds according to formula L-(I) in composition M_I to all compounds according to formula D-(I) in composition M_I is greater than a molar ratio of all compounds according to formula L-(III) in M_{IIIA} to all compounds of formula D-(III) in M_{IIIA}, wherein D-(I) has the following formula:



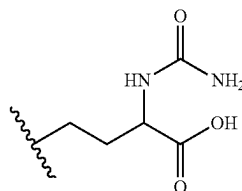
D-(I)

and

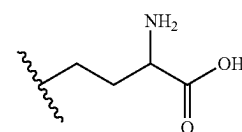
wherein R^{1L}, R^{1D}, R^{2L}, R^{2D}, R^{3L} and R^{3D} are each the same radical selected from the group consisting of formulae (VIII), (IX), and (X) having the following formulae:



(VIII)



(IX)



(X)

* * * * *